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Multimodal Teleoperation of a Wearable Dual-Arm Supernumerary Robotic System

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Abstract

A wearable robotic arm has a person as its base. This project built a teleoperation system for two seven-joint robot arms carried on a backpack frame and controlled by a *second* person, gave it a safety layer suited to a robot standing beside somebody who is not commanding it, and ran a first pilot comparing direct control against shared-autonomy assistance.

As mounted, the two arms could reach no common point; a costed mount change fixed it and raised the clearance to the wearer from 0.1610 m to 0.2202 m. Holding the wrist at a fixed orientation, not the wearer's body, is what limits reach – by a factor of five at the working position. The task set solves 13 927 joint-angle problems with no failures; a new motion generator removed 51.3 mm of unrequested hand excursion; four real cameras find the table surface to 1.43 mm; and across every recorded session the automatic clearance floor never engaged while the manual stop was used twelve times.

Under ethical approval (SETREC 7111986), five operators completed twenty-three sessions comparing direct VR control against assistance: assisted sessions were shorter and needed fewer re-grips, but the robot's path was 21 % longer while the operators' own hand paths were within 2 % of each other – the extra distance comes from the assistance, not the operator. The pilot has no separate wearer; nothing was validated on a loaded physical arm; and the two-person study this system was built for remains the target.

Acknowledgements

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I thank my co-supervisor, Dr. Etienne Burdet, for the framing that changed the shape of the project: treating the wearer and the operator as two people with different stakes, rather than as one role, is what turned a control problem into a question worth asking.

I am grateful to the Human Robotics Group for access to the Dr. Octopus platform, and to everyone who kept two Kinova arms and a backpack frame available and working. Finally, I thank my family and friends for their support over the course of the year.

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Chapter 1

Introduction

A supernumerary robotic limb is an extra arm worn on the body. It works alongside the wearer’s own limbs rather than replacing or rehabilitating them [YHC⁺21, P⁺21], and the same idea serves an assistant at a workbench, a surgeon who needs a third hand, or a person with limited mobility. What makes such a device different from a fixed industrial robot is simple but consequential: the base of the robot is a person. It accelerates, sways and steps, and it cannot be told to hold still.

This thesis reports work on Imperial College London’s Dr. Octopus platform (the Multi-limb Virtual Environment [Imp]), which mounts two Kinova Gen3 robot arms [Kin22], each with seven joints, on a backpack frame. The arms are controlled by a *second* person: an operator who cannot feel the arms, steering them for a wearer who cannot command them. No previous study has measured, in one system and in the same trial, both what automated assistance gives such an operator and what it costs the person carrying the arms (Section 1.2). Closing that gap is what this project was built for.

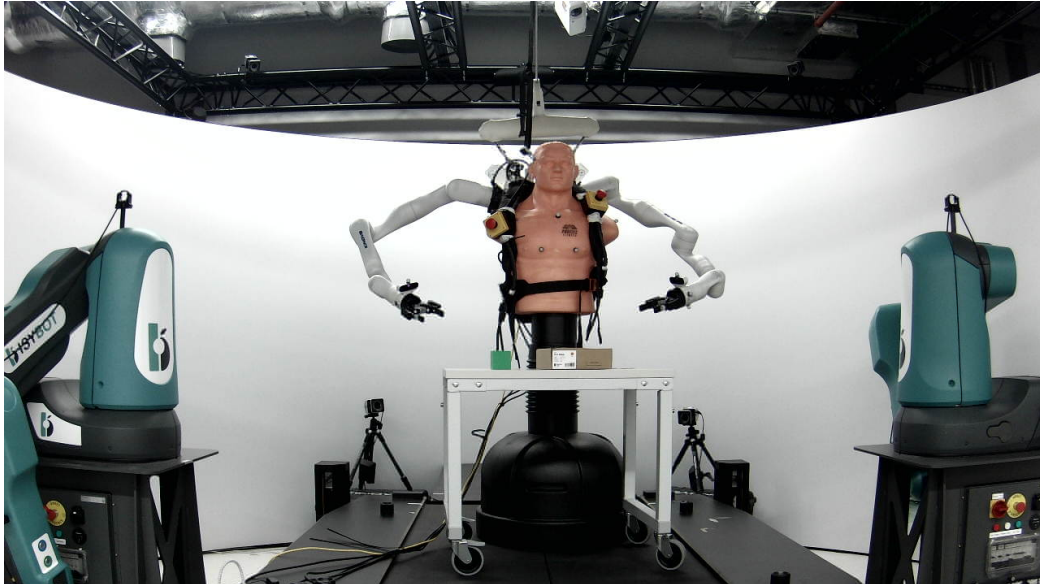


Figure 1.1: The platform as built: two Kinova Gen3 arms on a backpack frame, worn by a torso mannequin, photographed by the room camera. The two teal columns are the laboratory’s fixed robots and are not part of this system.

1.1 Background

Supernumerary robotic limbs. Parietti and Asada’s founding demonstrations brace an extra limb against the environment to steady the wearer [PA14, PA13]. Later reviews map the design choices along consistent axes – where the limb is mounted trades reach against leverage on the wearer, how it is driven trades speed against carried weight – and note that the field reports how the limb *feels* far more often than how well it *performs* [YHC⁺21, P⁺21, TL21, Z⁺23, Z⁺26]. Weight is the tightest limit: each Gen3 weighs 8.2 kg, the pair with harness exceeds 17 kg, and simulated loads suggest the cost to the wearer’s body is not negligible [MK24].

Operator and wearer as different people. Fusion mounts a robotic head and two arms on a worn frame driven by a remote operator [SSM⁺18], but its modes are degrees of *physical* coupling, not machine autonomy, and no user study is reported. Work on robots close to the body found that raising *apparent* autonomy – a hidden human operator, a wearer with nothing to do – raised physiological arousal and lowered trust [ZGD⁺26], so the wearer’s experience matters. The two-person case also differs mechanically: compensating a single wearer’s own body motion reaches (1.37 ± 0.58) mm of hand error [Z⁺24], but there the disturbance comes from the operator’s own nervous system, which knows in advance what it is about to do; here it comes from somebody else’s.

Controllers shaped unlike the robot. Guggenheim and Asada argue that a miniature controller of similar shape gives the operator an intuitive feel for the robot [GA21]; the argument covers *shape*, not *size*, and this controller is a third the robot’s size. Copying joint angles needs every controller joint measured accurately; copying hand position does not, but leaves the robot’s spare seventh joint undirected.

Shared autonomy. Dragan and Srinivasa formalise shared control as a blend of what the operator commands and what the robot predicts, weighted by the robot’s confidence [DS13]; Javdani et al. treat it as a decision under uncertainty about the operator’s goal and show that assisting before the robot is certain still helps operators finish faster [JSB15]. Neither was developed for a robot on a second person, where a wrong guess sends a manipulator toward someone who did not ask.

Measuring embodiment, workload and trust. A person’s sense that an artificial limb is theirs is measurable and separates into ownership, agency and location [BC98, LSK⁺08]; workload is measured by the NASA Task Load Index [HS88], exertion by a Borg scale [Bor82]. Feedback that improves control does not reliably improve ownership [PLF⁺23], so performance and embodiment must be measured separately; the pilot here measures the first and claims nothing about the second.

1.2 The gap

No study measures the operator and the wearer together, under the same autonomy, in the same trial. Shared autonomy is known to help an operator when the robot is attached to nobody; apparent autonomy is known to lower a wearer’s trust when the autonomy was a hidden human. Until both are measured in one system, the exchange rate between what assistance gives the operator and what it costs the person carrying the arms is unknown. The full review is in [Chapter M](#).

1.3 Aim, objectives and hypotheses

The aim was to build and verify, on the platform of [Fig. 1.1](#), a two-person teleoperation system on which that exchange rate can be measured, and to begin measuring it. Six objectives:

1. Design and build an instrumented mannequin controller (the *master*) with the robot's joint layout at reduced scale, and characterise it against recorded data.
2. Derive a mapping from master pose to robot command that degrades predictably as sensors are lost, and quantify what each loss costs.
3. Implement a safety architecture suited to a robot mounted beside a person, validated by deliberately injecting faults.
4. Measure the reachable workspace of the mounted arms and which manipulation tasks it can support.
5. Establish a verification method that produces traceable evidence for every claim, and apply it throughout.
6. Run a pilot of direct control against shared-autonomy assistance with real operators, and identify what stands between it and the full two-person study.

The study the system was built for states five hypotheses, each with a predicted direction, before any data ([Chapter J](#)): task completion time is shorter under shared autonomy (H1); operator workload is lower (H2); wearer-reported trust is *lower* under shared autonomy (H3), opposite to H1 and H2; the wearer's sense of agency falls while ownership does not (H4); and interference between the two arms falls while the delay between them does not (H5). The pilot tests H1 directly; no workload questionnaire was administered, so H2 is untested, and H3–H5 need the separate wearer the pilot did not have.

Each objective is reported in [Chapter 3](#) with the strength of its evidence – implemented, verified in simulation, measured on hardware, or measured in the pilot. The planning report's force-feedback master was not delivered ([Chapters B and 4](#)). **No result below the pilot was validated on a physical arm carrying a load.**

Chapter 2

Methods

This chapter summarises what was designed, built and run; the engineering detail needed to reproduce it is in [Chapters A and C to E](#).

2.1 The master arm

The master is a miniature mannequin arm with the Gen3's joint layout at roughly a third of its size ([Fig. A.1, Chapter A](#)), moved with the fingers. A Teensy 4.1 microcontroller reads fourteen rotary position sensors (potentiometers), two inertial sensors, two force-sensitive pads and two buttons, and reports at 100 Hz ([Table A.1](#)). Each reading r_n is smoothed on the microcontroller by an integer running average before conversion to degrees,

$$e_{n+1} = \frac{\alpha r_n + (256 - \alpha) e_n}{256}, \quad \alpha = 64. \quad (2.1)$$

Each sensor carries a health verdict from its own recent history (*dead*, *incoherent*, *intermittent* or *alive*; [Section H.2](#)); only the sensors the active mode uses can reject a reading, and when no trustworthy sensor remains the system reports the position as *unavailable* rather than guessing.

2.2 From master pose to arm command

The commanded hand position is a direction and a distance from the master's shoulder rather than a computation from all seven joint angles. With $\mathbf{g}$ the measured gravity vector, $\hat{\mathbf{a}}$ the upper arm's own axis, θ_i the joint angles and θ_1^0 the shoulder's rest angle,

$$\phi = \arcsin\left(\hat{\mathbf{a}} \cdot \frac{\mathbf{g}}{\|\mathbf{g}\|}\right), \quad \psi = \theta_1 - \theta_1^0, \quad r = \|\text{FK}(\theta_1, \theta_2, \theta_3, \theta_4, 0, 0, 0)\|, \quad (2.2)$$

so a dead rotation sensor zeroes ψ without corrupting r . The robot's target is a fixed reference plus a scaled displacement from where the master tip $\mathbf{d}$ was when the operator pressed the *clutch*, a button held to make the robot follow:

$$\mathbf{p} = \mathbf{p}_{\text{anchor}} + s (\mathbf{d} - \mathbf{d}_{\text{ref}}), \quad (2.3)$$

so releasing and re-pressing the clutch works like lifting and repositioning a mouse.

Five modes are delivered ([Table 2.1](#)); one safety layer serves all of them and no mode may switch any part of it off. Shared autonomy decides by a threshold rather than by continuous blending [[DS13](#)] ([Fig. 2.2](#)). With P the

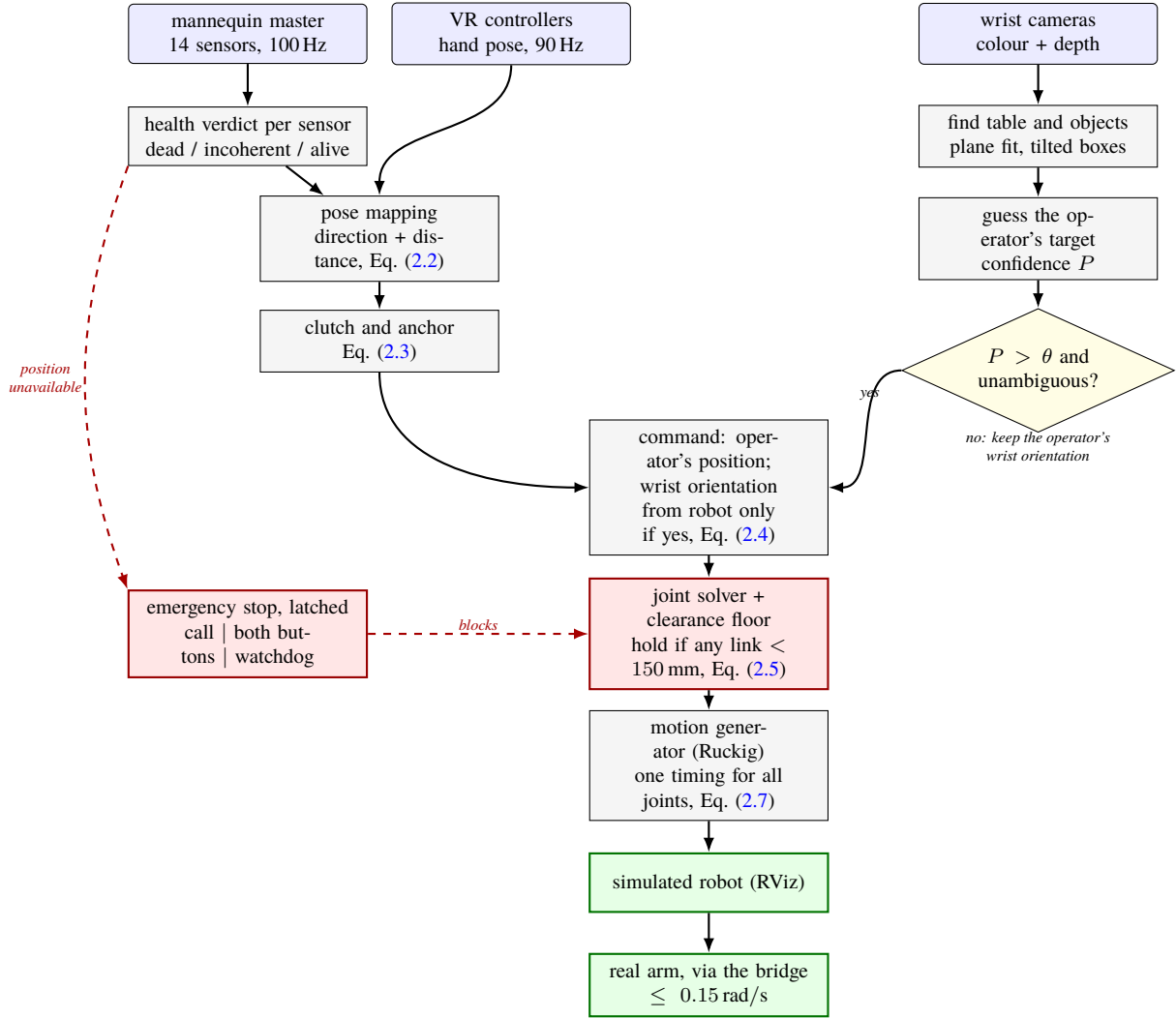


Figure 2.1: The signal path from the operator’s input to the real arm, with every point at which something can stop the arm in red. The operator’s position always passes straight through; the robot may supply the wrist orientation only.

Table 2.1: Operating modes, condensed from Table C.1. $v_{\max}$ is the simulation default; the hardware bridge caps every mode at 0.15 rad/s. Autonomy runs at a quarter of teleoperation’s speed by design: nobody is holding a controller, so the only limit on a wrong motion is how long it takes.

Mode	Input	Robot decides	$v_{\max}$
01/02 direct	mannequin or VR controllers	nothing	0.60 rad/s
03/04 shared autonomy	master or controllers, vision	wrist orientation	0.60 rad/s
06 full autonomy	typed or spoken goal	target, approach, grasp, place	0.15 rad/s

estimator’s confidence in its guess of the operator’s target and $\mathbf{q}$ a wrist orientation,

$$\mathbf{p}_{\text{cmd}} = \mathbf{p}_{\text{op}} \text{ always; } \quad \mathbf{q}_{\text{cmd}} = \begin{cases} \mathbf{q}_{\text{op}} & P \leq \theta \text{ or ambiguous,} \\ \text{slerp}(\mathbf{q}_{\text{op}}, \mathbf{q}_{\text{grasp}}, \sigma(t)) & P > \theta, \end{cases} \quad (2.4)$$

blended in over 0.5 s. Position is never blended, because a blend toward the wrong goal here moves the arm toward somebody who did not ask. In the laboratory the operator sits across the room, holding the VR controllers as a hand-position input and watching the robot directly (Section C.2.1); the operator-to-robot alignment is a measured rotation, never a mirror image, since copying someone you face reverses every orientation.

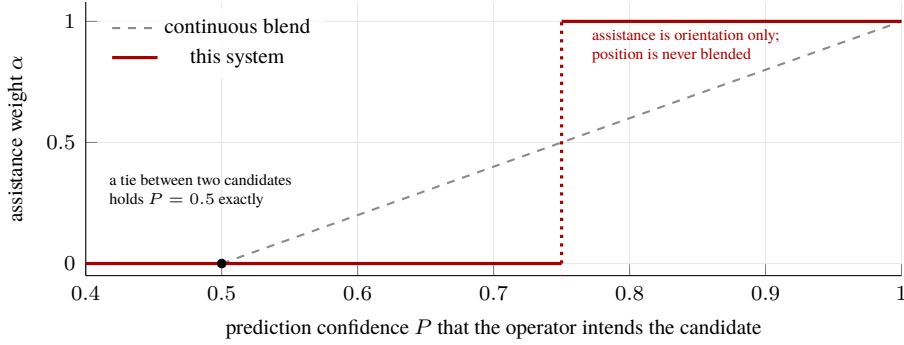


Figure 2.2: The rule this system uses to decide when the robot assists, plotted from its own settings, beside the continuous blend it deliberately does not use.

2.2.1 Safety architecture

Collision avoidance has three stages (Section C.1.1): the joint solver rejects colliding poses, the spare joint is re-seeded and the solve retried, and a clearance floor is enforced from where the arm actually is. With ℓ ranging over the arm’s links (modelled as tubes of radius r_{tube}) and w over the wearer’s body parts, the arm holds unless

$$\min_{\ell, w} [\text{dist}(\ell, w) - r_{\text{tube}}] \geq 0.150 \text{ m}, \quad (2.5)$$

computed on the robot’s own link model, not the planner’s collision list, which omits the shoulder-adjacent pairs. Every mechanism able to stop the arm registers by name, states how it clears, and falls into an explicit *unknown* state – still blocking – if its process stops. Three independent triggers latch the emergency stop until reset: a software call, both master buttons held together, and a watchdog on the master data’s *source* timestamp, because a failing master keeps sending stale values rather than going silent.

2.3 Perception, planning and motion

A calibrate-then-plan stage (Chapter D) replaced typed-in object and table positions with measured ones: a wrist-camera sweep fits the table surface, picks the objects out of the image and hands the result to a motion planner (RRT-Connect [KL00]) checked against the safety layer’s own clearance. Detection uses an open-vocabulary model (YOLO-World [CSG⁺24] with FastSAM [ZDA⁺23]) with a colour-and-depth filter as fallback; the fallback carries the load, since on real frames (432 object instances) the learned detector found 9.0 %. The table is the plane $(\mathbf{n}, d)$ on which most depth points $\mathbf{P}_k$ lie, $|\mathbf{n} \cdot \mathbf{P}_k + d| < 6 \text{ mm}$, kept from 400 random draws [FB81]. Grasps

come from each object’s principal axes: with $\mathbf{u}_3$ the direction of its least extent,

$$w = \max_k \mathbf{u}_3^\top \mathbf{P}_k - \min_k \mathbf{u}_3^\top \mathbf{P}_k \leq w_{\max} = 85 \text{ mm}, \quad (2.6)$$

the jaw closes across $\mathbf{u}_3$ and a wider object is refused by name (Eq. (D.1)) – which a learned model cannot do.

Motion is generated by Ruckig [BK21], which limits acceleration and its rate of change and drives all seven joints on one shared timing $\sigma(t) \in [0, 1]$,

$$\mathbf{q}(t) = \mathbf{q}^{(k)} + \sigma(t)(\mathbf{q}^* - \mathbf{q}^{(k)}), \quad (2.7)$$

so the commanded pose stays on the straight line between endpoints; it replaced a per-joint step limiter (Section E.1). Replaying recorded commands on the physical arm tested two models of the simulation-to-real gap – a fixed stop-short, $q_{\text{real}} = q_{\text{cmd}} - \varepsilon \text{sign}(\Delta q)$, or an error growing with distance, $q_{\text{real}} = (1 - k) q_{\text{cmd}}$ – which differ only on short moves, so joint errors were propagated to the hand, $\Delta \mathbf{p} \approx \mathbf{J} \Delta \mathbf{q}$, to choose (Section E.3). The positioning error budget combines its terms b_i as

$$\sigma_{\text{RSS}} = \sqrt{\sum_i b_i^2}, \quad \sigma_{\text{worst}} = \sum_i b_i, \quad (2.8)$$

the second being the honest figure against the 30 mm gate, since the two largest terms are fixed offsets.

2.4 Verification methodology

Twenty-six times a result that looked like a system failure was a defect in the measuring tool (Chapter G, Section 4.3). So every analysis script carries a test with a known correct answer; synthetic data is used only where the truth is *constructed*, never rendered; and a result that contradicts an earlier measurement is a suspect instrument until reconciled. Because the joint solver (TRAC-IK [BA15]) starts from a random guess, every reachability claim is ten repeats along a path sampled every 20 mm, and a pose that fails any repeat is not reported reachable. Each of the perception pipeline’s thirty-three stages likewise returns a result, a refusal with a reason, or “not applicable” – never a silent zero (Fig. O.34).

2.5 Ethics

This study was approved by Imperial College London’s Science, Engineering and Technology Research Ethics Committee, approval number 7111986. All participants gave informed consent to taking part and to the use of their data. Chapter P analyses the work under the Engineering Council’s four pillars and the department’s six domains. *Human subjects*: the wearer bears a risk they cannot control, so consent is taken from operator and wearer separately, in separate rooms; the wearer holds their own emergency stop and rehearses it before any data; a stop is data, never a lost participant; worn operation was not sought for approval until the mount clash of Section 4.6 is fixed. *Scientific integrity*: every number carries its evidence class and twenty-six instrument defects are reported. *Institutional integrity*: identities are held on encrypted College storage outside the repository, and `write_manifest()` refuses to record a name. *Social responsibility*: assistance that helps an operator while eroding a wearer’s trust shifts harm onto the person least able to refuse it – why H3 and H4 exist.

2.6 The pilot study

Five operators (P1–P5) each completed two to seven recorded sessions – following a held object, reaching a target, matching a pose – once under direct VR control and once with assistance, from the desk arrangement of [Section C.2.1](#) ([Fig. 2.3](#)). Positions, clutch state and the robot’s lag were logged twenty times a second with an event log of every clutch press and emergency stop, so completion time, robot path, re-grips, steering fraction and lag came from the record, not a stopwatch. Three further operators (M1–M3) attempted the same tasks from the mannequin master. The pilot has no separate wearer, so H3–H5 remain for [Chapter J](#).

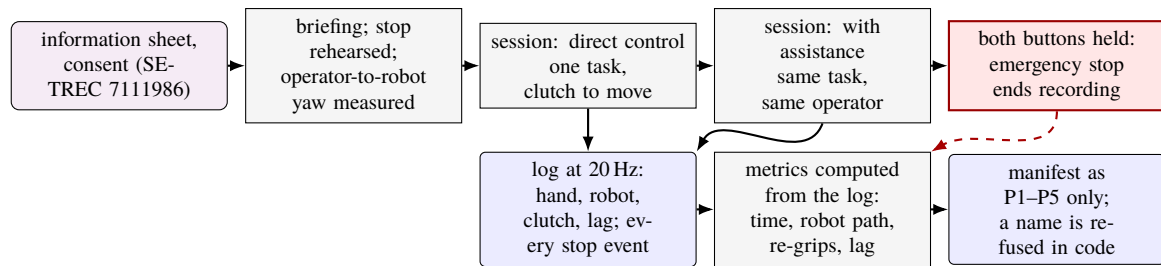


Figure 2.3: One pilot session as run, and the record every metric came from; nothing was timed by hand, and the stop that ends a recording is itself in the log (twelve holds, eleven sessions).

Chapter 3

Results

Every number below states the strength of its evidence, and **no result before the pilot in Section 3.4 was validated on a physical arm carrying a load**: most of this project is simulated against mock hardware, plus recorded data from the physical master and four cameras (Chapter H: full derivations).

3.1 The command mapping and its safety layer

Six of fourteen master sensors are alive (Section H.2; the graceful-degradation design is discussed in Section N.1). The safety layer handled fourteen of fourteen injected faults (Section H.2).

The pilot (Section 3.4) is the only *live* evidence for this layer (Fig. 3.1): across every recorded session the automatic clearance floor – the rule that holds an arm 150 mm short of the wearer – never activated. The manual emergency stop did: twelve holds across eleven of twenty-three sessions, nine of them in a session’s final 15 % as the operator ended the recording, and three not – one assisted session stopped at 67 % and 83 % of its length, with steering resuming after each.

3.2 Workspace and motion

Over a 63-cell grid across the working volume in front of the wearer, **no cell was reachable by both arms** (Fig. 3.2).

Moving the mounts inboard, forward and down with more forward tilt raises shared cells to 22 of 63 within a buildable 0.20 m separation; applied, it raised clearance at the rest pose from 0.1610 to 0.2202 m and moved the pick-and-place targets from 0.530 to 0.290 m off centre. The tightest limit on reach is the fixed wrist orientation: from the working position, the hand can travel a mean 0.065 m with the wrist held fixed against 0.363 m with it free – over five times – because a fixed orientation uses up the seventh joint (Table I.2).

The per-joint step limiter that was the whole motion generator is compared with its replacement in Table 3.1: it let the hand leave the planned straight line by nearly twice the 30 mm grasp gate, at over twelve times the declared joint speed, because it never read the speed-limit file.

Replaying recorded commands on the physical arm shows every joint stops 0.305° short of its target on the side it came from – a fixed stop-short, not an error growing with distance ($r = 0.923$ when propagated to hand position). One correction per arm removes 74 to 76 % of it, taking the combined positioning error from 18.64 to 12.91 mm (worst case 33.39 to 19.99 mm), inside the 30 mm requirement *provided* the camera sits where the model says on the wrist. Measured, it does not: 8.45° and 34 to 59 mm of error across the working range

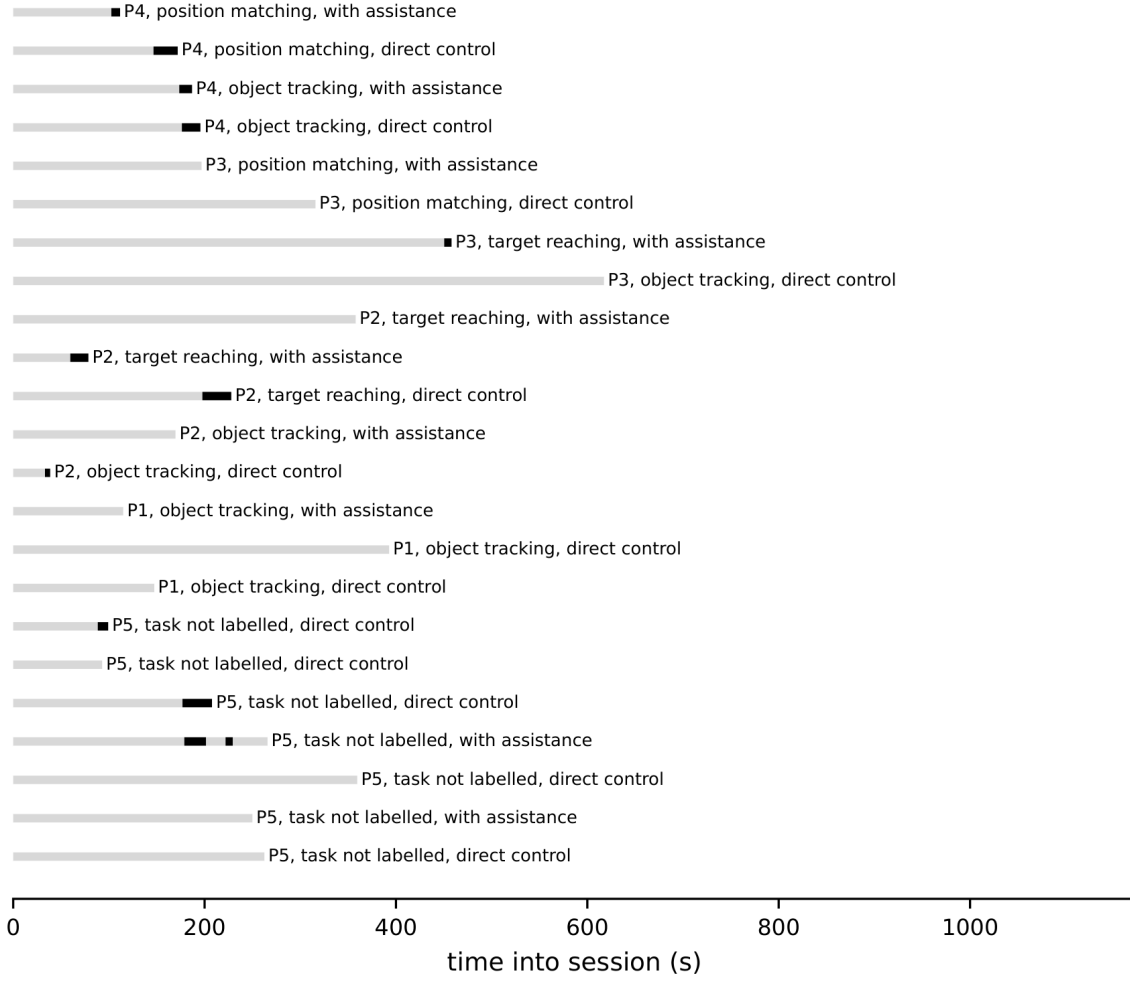


Figure 3.1: Every VR pilot session as a bar, emergency-stop holds in black: twelve holds in eleven sessions, median 16.1 s, longest 30.9 s; the automatic clearance floor never fired.

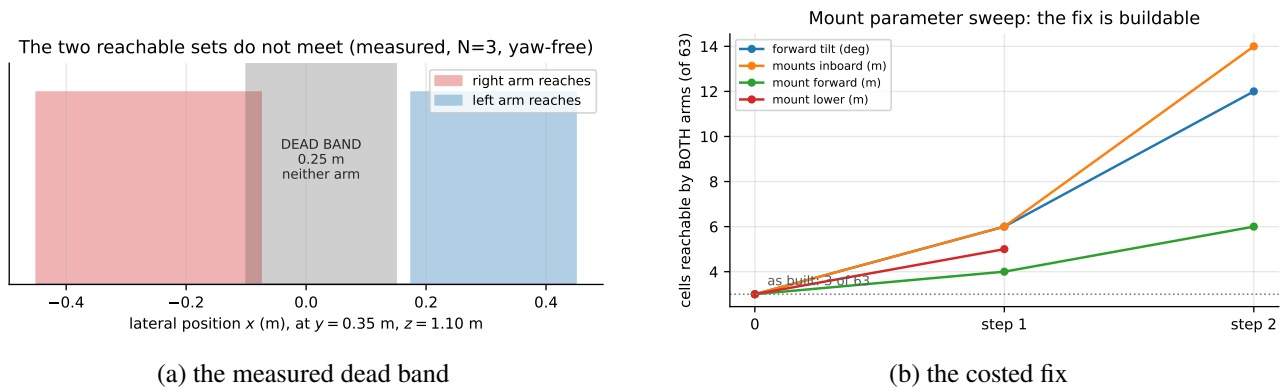


Figure 3.2: (a) The two arms' reachable regions do not meet. (b) Sweeping the mount position: tilting forward and moving inboard help, splaying outward *hurts*. Geometry only, so these are upper bounds pending a clearance check.

Table 3.1: The same commanded move, both arms, at the master’s own 50 Hz. Off-path is the furthest the hand leaves the planned straight line.

Generator	Off-path	Peak speed vs limit	Arrival
Per-joint clamp, as shipped	51.3 mm	12.53×	2 cycles
Ruckig, delivered	0.0 mm	1.00×	1 cycle

(Section H.7).

3.3 Perception: what the cameras actually detected

Pointed at four physical cameras for the first time (Chapter F), the wrist cameras find the table surface to 1.43 mm with 74 % of their depth points on it (plane fitting by random sample consensus [FB81]). Segmentation is scoped to the task’s two objects (Fig. 3.3), recovered as oriented boxes so the planner sees a turned object’s true width.

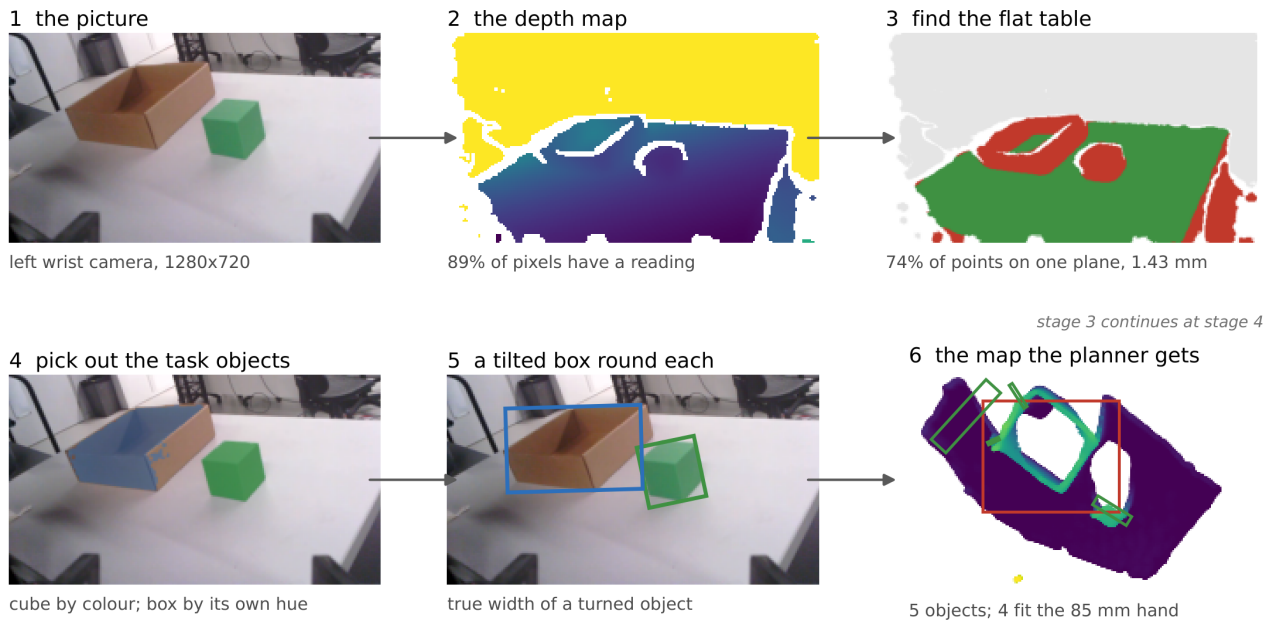


Figure 3.3: The pipeline on one real left-wrist-camera frame, stage by stage: picture; depth map; table plane (74 % of points on it, 1.43 mm); the two task objects picked out; a tilted box round each; and the map handed to the planner – five objects, four fit the 85 mm hand, the widest is 341 mm.

Depth, not colour, is what a box is drawn from (Fig. 3.4): the same two objects with their measured distance. Table 3.2 collects this frame’s numbers; its two distance methods measure different points on a tilted object.

No grasp closed on any camera in that session: the nearest object’s narrowest side measured 144 mm against an 85 mm gripper opening. Figure F.1 shows the whole scene the safety layer reasons about.

In simulation the full task set solves 13 927 joint-angle problems with no failures and no clearance breaches, and every scripted grasp closes on its object in every mode (Fig. 3.5); the cubes are set down 56 mm from the centre of their pad, past the 30 mm gate (Fig. 3.6). The intent estimator – the part of shared autonomy that guesses which object the operator wants – shows why measuring matters: from a typed-in table position it guesses the wrong object with confidence 35 % of the time at one cube spacing (60 mm) and 17.5 % at 40 mm, past the 16 %

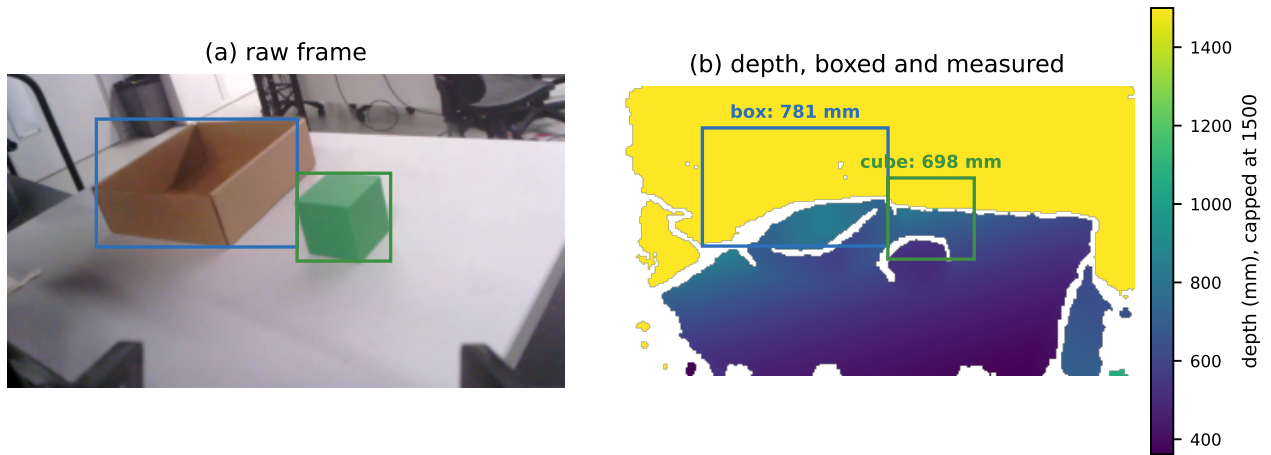


Figure 3.4: Figure 3.3’s frame (a) beside its depth map (b), each object boxed and labelled with its median measured distance.

Table 3.2: Every value from the one real frame of Figs. 3.3 and 3.4.

Quantity	Value	Method
Support-plane RMS	1.43 mm	RANSAC, this frame
Support-plane inlier fraction	74.0 %	RANSAC, this frame
Cube range	525.5 mm	plane-fit centre
Cube range	698 mm	depth-box median
Box footprint	341×278 mm	oriented 3D box
Box range	677.6 mm	plane-fit centre
Box range	781 mm	depth-box median

error rate above which aggressive assistance is a net loss [DS12]; from the camera’s measured position, 0.0 %.

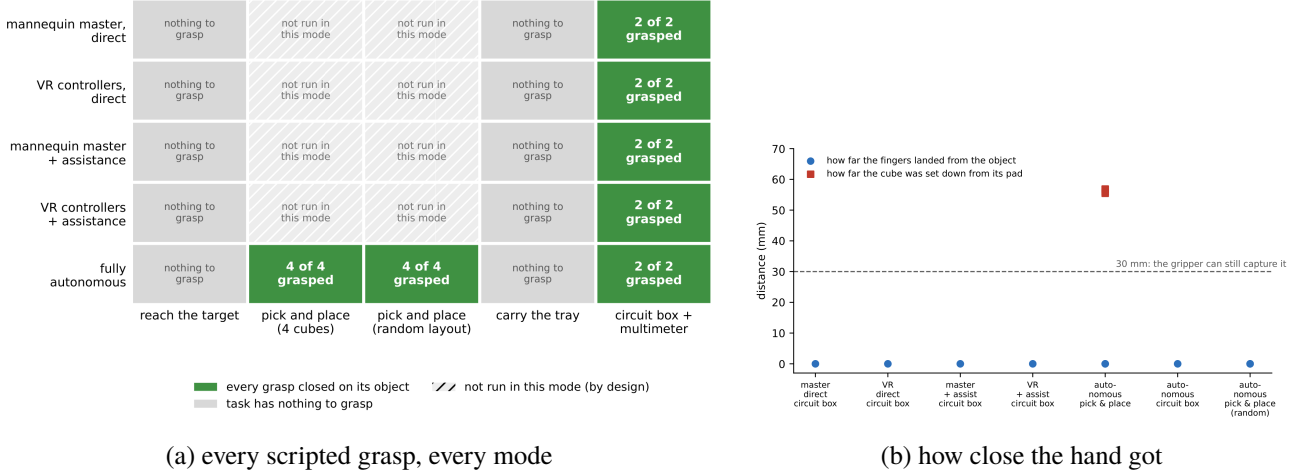


Figure 3.5: (a) Twelve of twelve grasps closed; hatched cells are tasks not run in that mode by design, grey ones have nothing to grasp. (b) The fingers met every object where planned; set-down lands 55.5 to 56.9 mm from the pad centre. Simulation.

3.4 A pilot of direct control against shared autonomy

Five operators (P1–P5) completed twenty-three recorded sessions under the protocol of Section 2.5, each on one of three tasks – *object tracking* (following a held, moving object with the robot’s hand), *target reaching* (a reach to a marked position) and *position matching* (moving to and holding a target pose) – once under direct control and once with assistance. P5’s seven unlabelled sessions are pooled, their condition read from the logged flag. A fourth planned task, using a multimeter, was never recorded with a participant on any of three tries (Chapters L and O). Coverage is uneven, so the pooled comparison mixes tasks and the paired one isolates the condition.

Table 3.3: Twenty-three pilot sessions, pooled by condition. Duration includes pausing and hesitation, so it measures the session rather than pure movement. Emergency-stop holds (Section 3.1) are excluded here as engagement time, not as a safety verdict.

Measure	Direct ($n = 13$)	Shared autonomy ($n = 10$)
Time to finish, mean (s)	241	219
Robot’s hand path, mean (m)	2.30	2.78
Re-grips per session, mean	25.8	23.3
Share of session actively steering, mean	55 %	61 %
Robot’s lag behind the operator’s hand, mean (mm)	2.6	2.8

Pooled across all twenty-three sessions (Table 3.3, Fig. 3.7a), assisted sessions averaged shorter completion times (219 s against 241 s) and fewer re-grips (23.3 against 25.8), with the robot lagging the hand by under 2.9 mm in both conditions – but a *longer* robot path (2.78 m against 2.30 m), a direction that holds in the paired comparison too (Fig. 3.9, Section 4.4). Figure 3.8 isolates the seven cases in which the *same* operator repeated a task under both conditions: the robot travels further with assistance in six of seven, the operator’s own hand in only two of seven.

Every session also recorded the real arm’s own joint encoders, so the gap between the simulated and the real hand is known throughout (Figs. 3.10 and 3.11): typically 11.8 mm per VR session and 24.8 mm for the

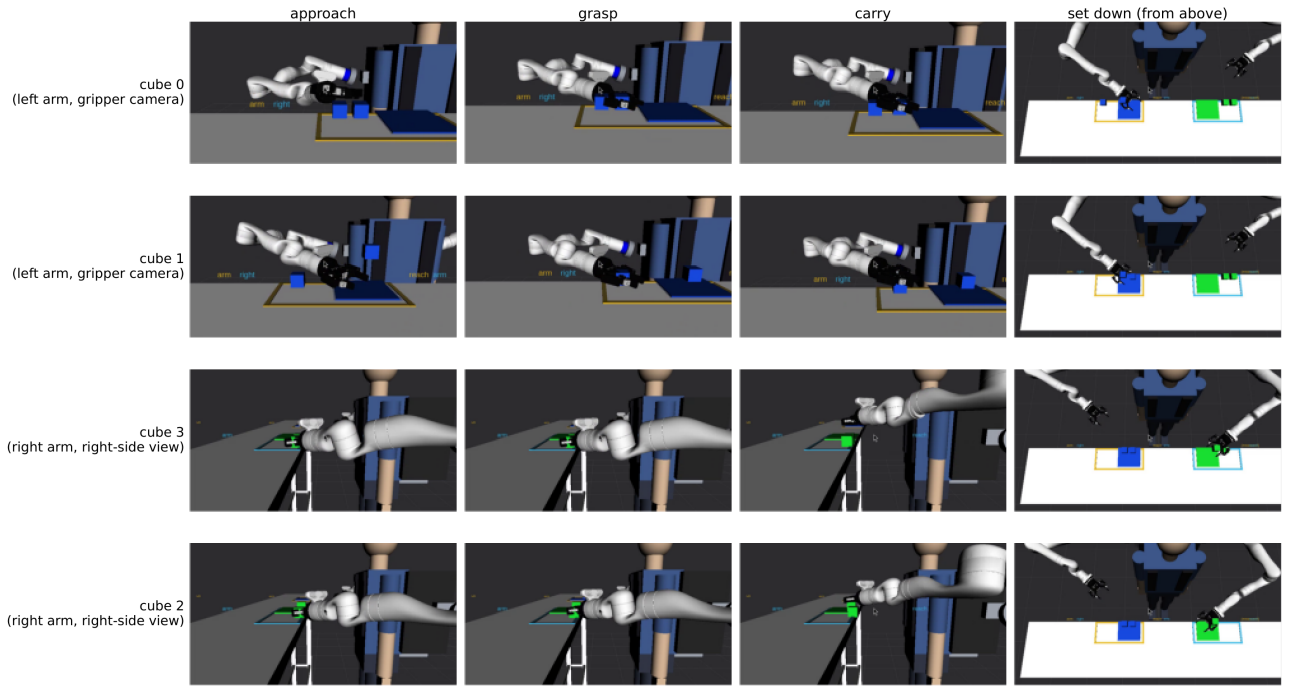
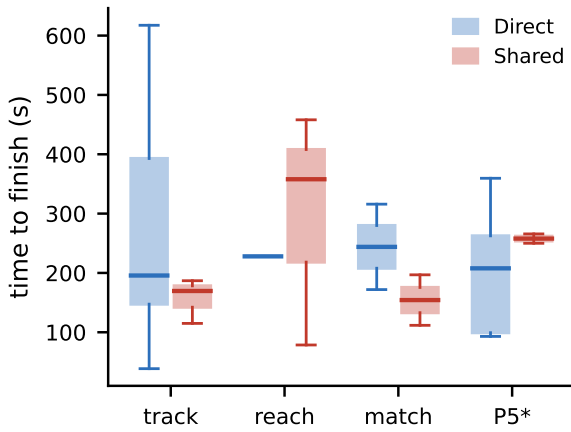
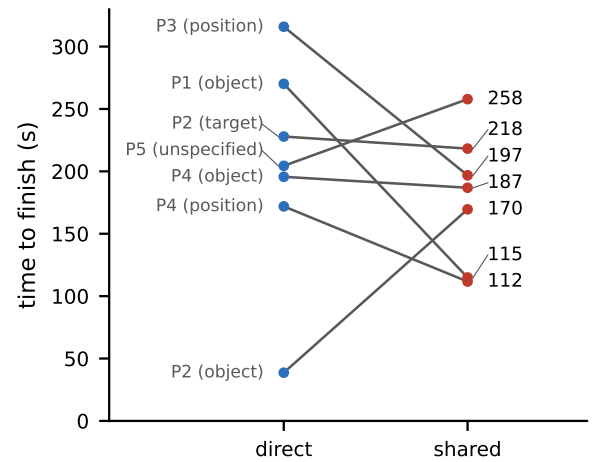


Figure 3.6: The autonomous pick-and-place in simulation, one row per cube: approach, grasp, carry and set-down, at the logged event times; both arms are used.



(a) pooled by task (P5's task was unrecorded, *P5**)



(b) paired within operator, both conditions

Figure 3.7: Time to finish, pooled as box plots – median, quartiles, whiskers and outliers (a) – and paired within operator (b, the fairest comparison here). Duration includes setup and hesitation, so it is the least precise pilot measure.

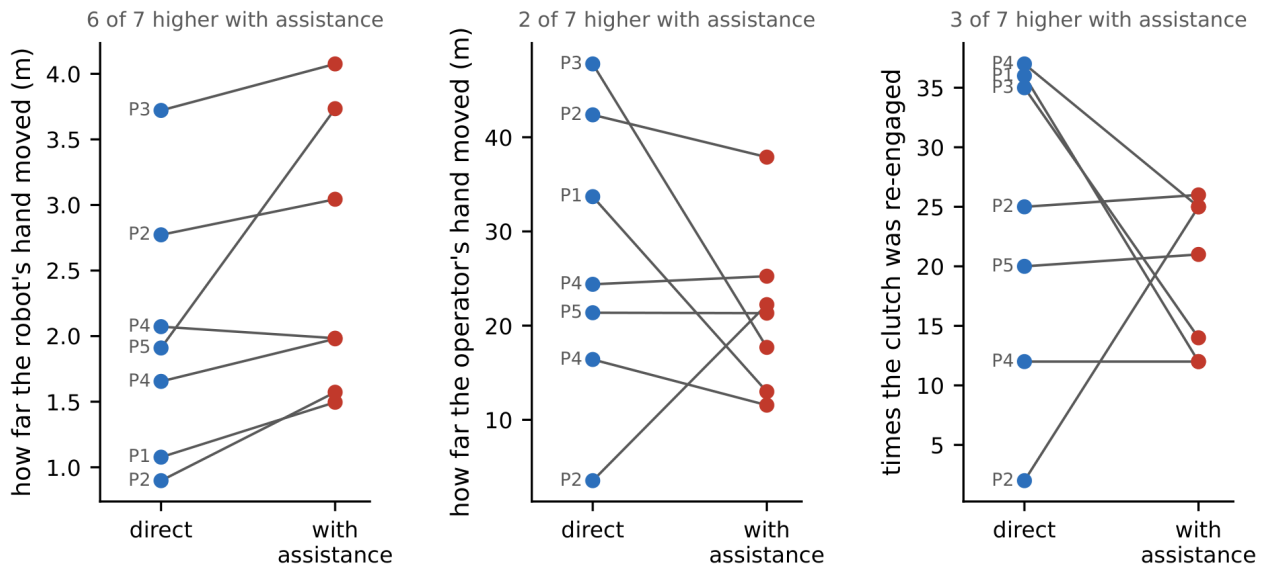


Figure 3.8: The seven paired cases: the robot's path rises with assistance in six of seven pairs while the operator's own hand path does not (two of seven) – the extra distance comes from the assistance.

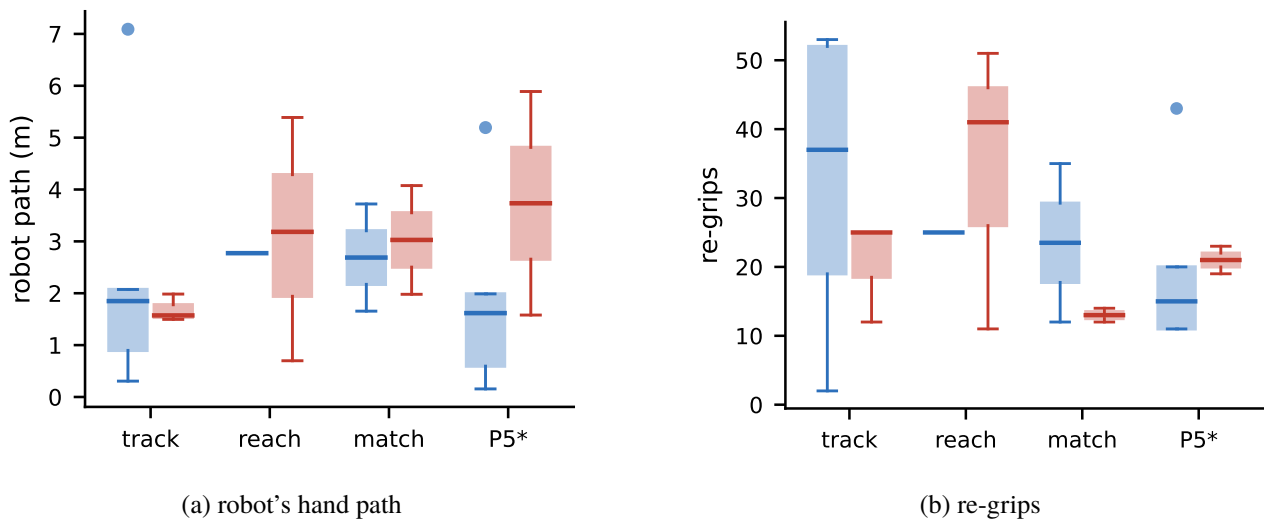


Figure 3.9: The counter-intuitive half of Table 3.3: with assistance the robot's path is *longer* in three of four task groups, while re-grips fall.

mannequin cohort, most sessions inside the 30 mm grasp gate.

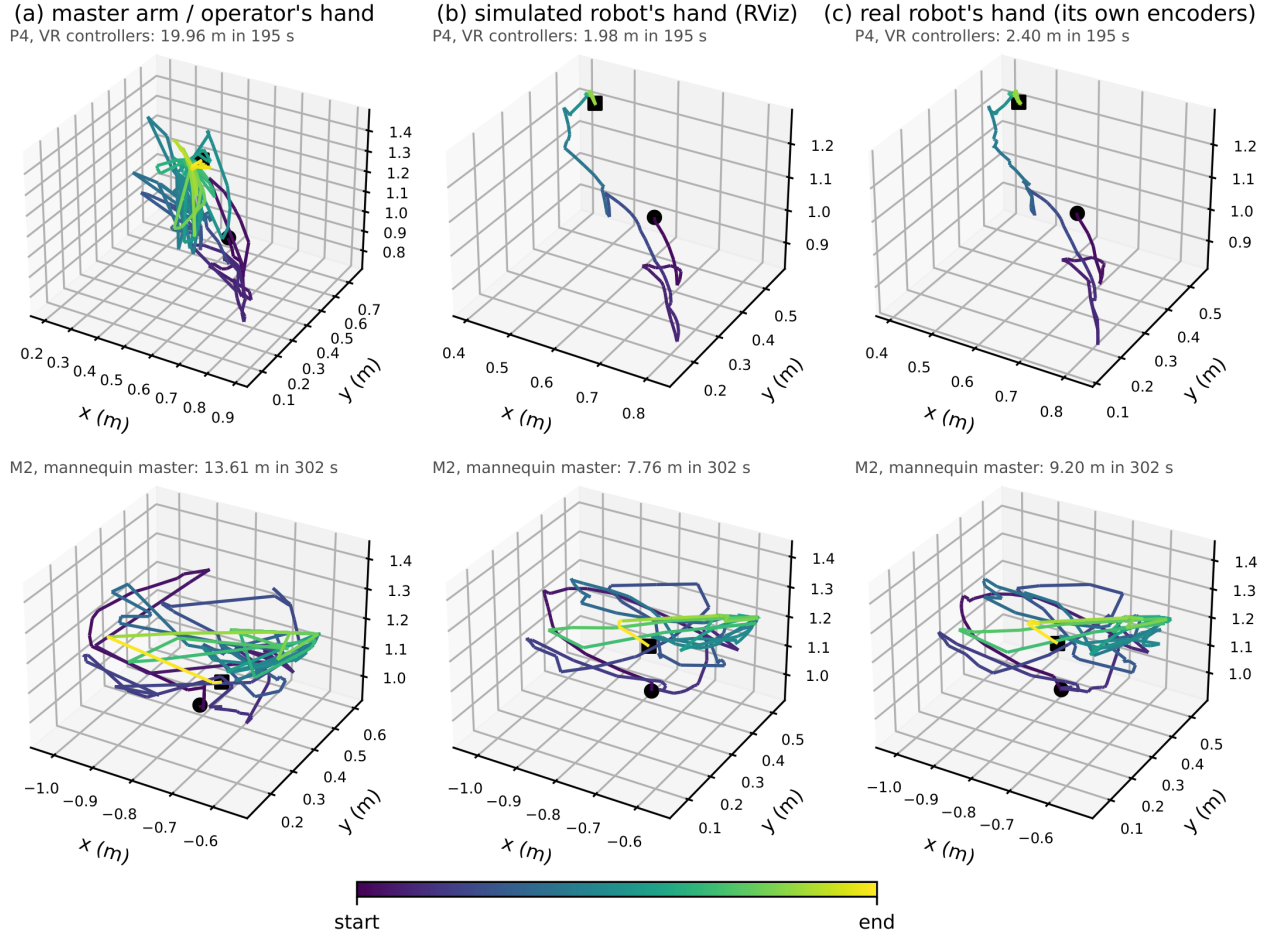


Figure 3.10: One session per cohort as three separate paths, coloured from start to end: (a) the master arm or operator's hand, (b) the simulated robot's hand as RViz shows it, (c) the real robot's hand from its own joint encoders. Top: VR (P4, object tracking, direct; simulated-to-real gap 11.1 mm). Bottom: mannequin master (M2, target reaching, direct; gap 21.8 mm). The hand covers far more distance than the robot because the mapping scales and filters it. Every session: [Chapter O](#).

This pilot measures operator performance only; it has no separate wearer, so H3–H5 are untested ([Section 4.4](#)).

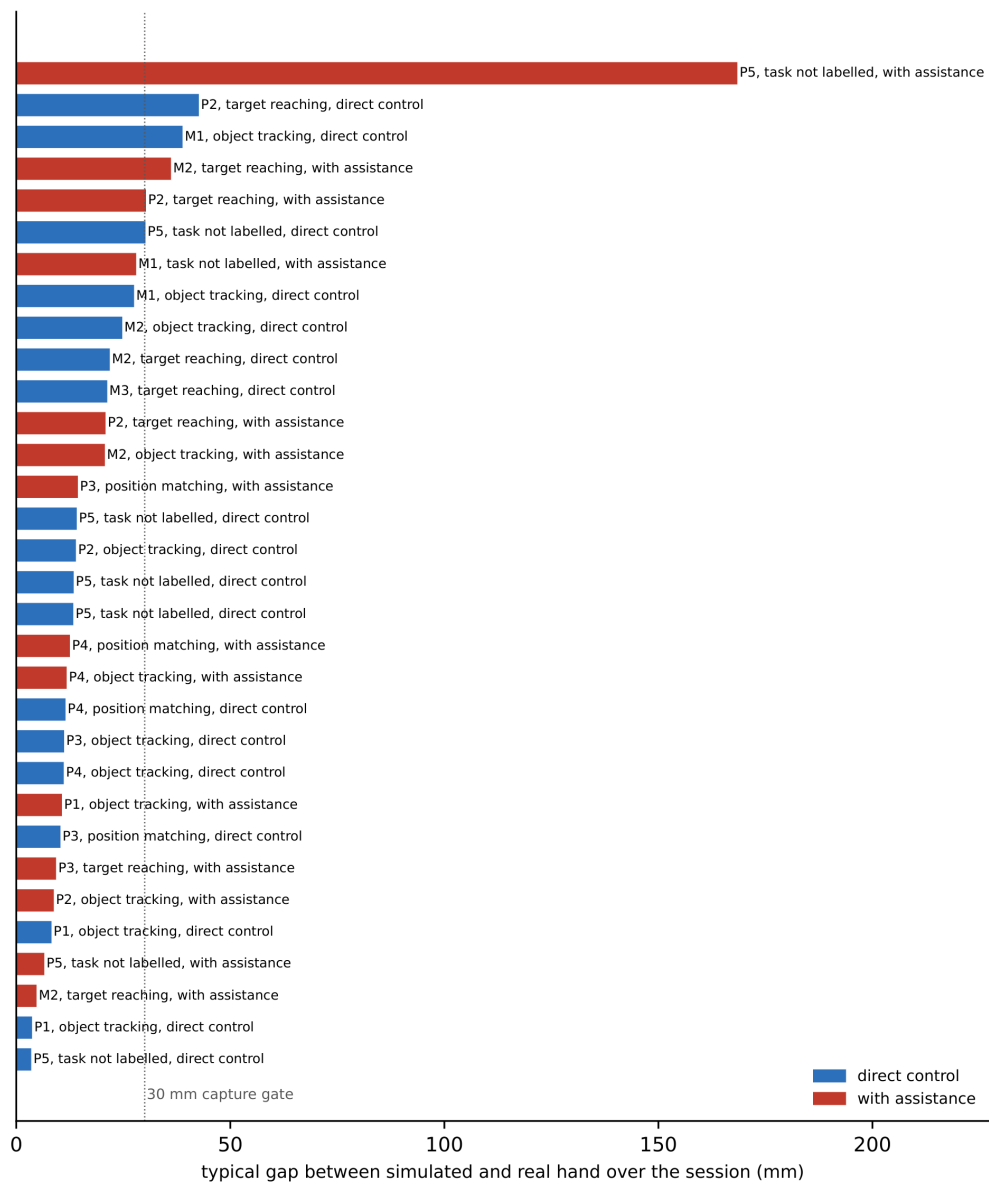


Figure 3.11: Every session of both cohorts ranked by the typical gap between the simulated and the real robot hand, against the 30 mm grasp gate. The one large outlier is the P5 session with the two mid-session emergency stops.

Chapter 4

Discussion

This project built and verified a teleoperation system for two robot arms worn by one person and controlled by another, and ran a first pilot of direct control against shared-autonomy assistance with real operators. Its main findings: the two arms as mounted could reach no common point, and a costed mount change fixed it; the safety layer’s own pilot record shows the automatic clearance floor was never needed; and assistance shortened sessions and reduced re-grips while lengthening the robot’s path.

4.1 The workspace result, and what it costs

The finding that the arms shared no reachable point is severe because it is geometric: no planner or wrist policy can produce a point neither arm can reach, which ruled out hand-to-hand transfer. The cause was not the mount’s rotation but the arms’ asymmetric rest angles, which are fixed facts because the simulation-to-hardware bridge replays them onto the real arm. Fixing it needed a mechanical change, costed, applied and re-verified in one pass.

4.2 The safety layer is thorough about distance and silent about time

The clearance floor is a *distance*, and the wearer can move. Combining the measured delay of the wearer-tracking camera with limb speeds from the literature gives [Table 4.1](#): in most modelled limb motions the limb crosses the whole 150 mm margin before the robot knows anything changed. It is this report’s principal open safety question, and three responses exist, none yet implemented: enlarge the floor by the delay times an assumed limb speed, a one-parameter change that makes the margin mean what it claims; cut the delay, since most of the 562 ms tail is a configured publishing rate, not the 28 to 34 ms the tracker itself takes; and scale commanded speed with measured clearance, two quantities the system already computes separately. The fallback makes it worse: when the wearer estimate is stale the guard reverts to a mannequin with its arms down and believes there is 0.3170 m of clearance where a wearer with arms held out has 0.0933 m – optimistic by 224 mm, mitigated only by instructing the wearer.

Table 4.1: How far a limb travels during the measured tracking delay, against the 150 mm clearance floor; full table in [Chapter E](#).

Delay	Deliberate reach	Quick reach or startle
62 ms (median)	62 % of the floor	exceeds the floor
562 ms (worst)	exceeds the floor	exceeds it several times over

4.3 Validity of the measurements

Twenty-six times in this project a measurement was wrong and the system was not ([Chapter G](#)). Fifteen were caught because an independently known quantity disagreed with the reading, and fourteen could have been caught in advance from something already in hand – the case for building a cross-check into every instrument rather than trusting it harder. Every result other than the pilot and the sensor measurements comes from simulation with mock hardware, whose force readings are exactly zero, so no force-based quantity is defined here; and every delay reported excludes operator, display and physical arm, so it is a lower bound.

4.4 What the pilot shows, and its limits

The pilot points the same way as the shared-autonomy literature for robots attached to nobody [[DS13](#), [JSB15](#)]: shorter completion times and fewer re-grips with assistance, and no measurable cost in responsiveness. The robot's path is *longer* under assistance, in the paired comparison too (six of seven cases) – but the operators' own controller paths are not: 29.9 m against 30.5 m, within 2 %. The extra distance therefore belongs to the assistance, and the platform shows the same with no operator at all: identical scripted waypoints travel 4 % further under VR shared autonomy than direct VR, and 45 % further under mannequin-master shared autonomy than mannequin direct ([Fig. O.14](#)). The literature predicts the cost this pilot could not see: higher autonomy improves performance but lowers the sense of agency [[CNA25](#)], cuts effort without raising satisfaction [[KHKCG⁺12](#)], and raises safety while lowering perceived control [[YH11](#)]; no agency or workload questionnaire was administered here, which is why H2 is untested and H4 is in the protocol. Five operators with uneven coverage is a pilot, not a powered comparison: the full study's pre-registered analysis, run on 20 000 constructed data sets ([Chapter K](#)), detects a medium effect ($d_z = 0.73$) with 97.3 % power across all tasks at sixteen pairs, but on any one *named* task with only 63.8 %; 80 % needs twenty pairs, the sample the next study should recruit. Most importantly, the pilot has no wearer: it measures what the operator experienced, not what carrying 17 kg of assisted motion costs a second person – the comparison [Chapter J](#) is for.

4.5 Against the planning report

The planning report specified physical springs at the master and a virtual spring-damper at the robot; neither survives the measured system, whose usable command path runs at 18.4 to 18.7 Hz, too slowly for a spring-damper law to behave as one. The compliance sought would have to come from the mechanism – a spring in series with each motor [[TGT⁺22](#), [LKK19](#)] or current-based control [[dWKB24](#)] – both hardware changes. The safety, sensor-degradation, workspace and pilot work were not in the plan; each followed from a measurement the plan did not anticipate.

4.6 Against the objectives, and what remains

Of the six objectives of [Section 1.3](#), five were met and one partly: the master built and characterised, its repair narrowed to two joints (1); a mapping that degrades predictably (2); a safety layer that handled fourteen of fourteen injected faults and whose live record shows the automatic floor was never needed (3); a workspace characterisation that found no shared reachable point and costed, applied and re-verified the fix (4); ten-repeat, known-answer verification that caught twenty-six instrument defects before any became a finding (5); and a pilot under approval

that supports H1 at pilot scale but, with no workload questionnaire and no wearer, leaves H2–H5 untested, so objective 6 is met in its first half only.

What remains, in priority order and with the number that sets it: repair the two named master sensors (54.5 mm of unrequested motion); measure the camera’s true position on the wrist (34 to 59 mm of error, the largest unmeasured term and the reason no camera-guided grasp has closed on hardware); correct the clearance check that samples link origins rather than link geometry, the one safety defect here rather than an incompleteness ([Chapter N](#)); and run the protocol of [Chapter J](#) with a separate wearer, now that approval and a working input path both exist.

Chapter 5

Conclusion

A teleoperation system for two seven-joint robot arms worn by one person and controlled by another was built, verified, and taken to a first pilot with real operators under ethical approval. The arms as mounted shared no reachable point and a costed mount change fixed it; the safety layer stopped the arm through fourteen of fourteen injected faults, and its live record shows the manual stop used twelve times while the automatic floor was never needed; and across twenty-three sessions by five operators, shared autonomy shortened completion and cut re-grips at the cost of a longer robot path – the assistance’s doing, not the operators’. Nothing was validated on a loaded physical arm and the pilot had no wearer, so the exchange rate this system exists to measure – what assistance gives the operator against what it costs the person carrying the arms – remains to be measured, and the means to measure it now exist. The habit behind every number here – twenty-six times an apparent system failure was a measuring-tool defect, and establishing which is in hand before acting – is the one most worth carrying into that study.

Appendix A

The master arm

The master is a miniature mannequin reproducing the Gen3's joint sequence at roughly a third of its scale, moved with the fingers of one or both hands. Source: `firmware/master_arm/teensy_final.ino`.

A.1 Mechanical chain and electronics

Each arm carries seven potentiometer-instrumented joints, alternating a roll about the preceding link's own axis with a bend perpendicular to it; link lengths sum to 0.272 m. A roll joint moves the tip only through a downstream bend (rotating a straight arm about its own axis translates the tip by zero), which is why the loss of J2 or J4 costs reach while J3 or J5 costs almost nothing.

Master arm, from CAD — zero configuration

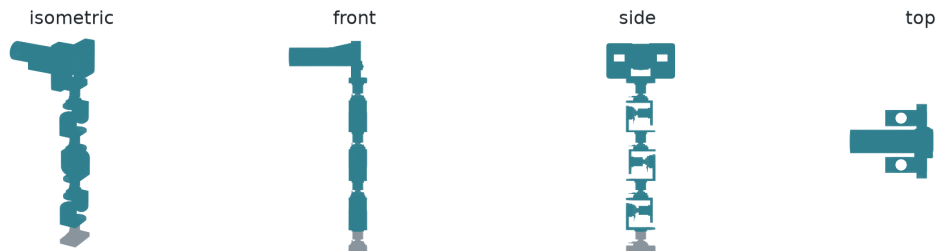


Figure A.1: Four orthographic views of the assembly at the zero configuration. The mechanical model carries seven potentiometer housings, a shaft coupling at each joint, and a wrist platform for the inertial sensor and force-sensitive resistor.

A Teensy 4.1 reads fourteen potentiometers, two MPU-6050 inertial sensors, two force-sensitive resistors and two buttons, emitting one ASCII text frame per cycle at 100 Hz ([Table A.1](#)). An earlier firmware revision placed two potentiometer channels on A4/A5, which are physically the default Wire bus both inertial sensors use; calling `analogRead()` on a pin the I²C peripheral is driving corrupts the bus intermittently and returns meaningless potentiometer values, with no symptom that looks like a pin conflict from the outside. The fix excludes both pins from analog use: Teensy 4.1 exposes eighteen analog-capable pins, minus two leaves sixteen, exactly what fourteen potentiometers plus two force sensors require – no spare.

Table A.1: Serial frame, in transmission order.

Field	Count	Format	Meaning
k1j1..k1j7	7	%.1f	left arm joints, 0 to 360°
k2j1..k2j7	7	%.1f	right arm joints
K1IMU, K2IMU	6+6	%d	$a_x a_y a_z g_x g_y g_z$, raw LSB
fsr1, fsr2	2	%d	grip force, raw counts
btn1, btn2	2	%d	latched toggle state

A.2 Signal conditioning

Each potentiometer channel is read at 12-bit resolution and passed through an integer exponential moving average,

$$e_{n+1} = \frac{\alpha r_n + (256 - \alpha)e_n}{256}, \quad \alpha = 64, \quad (\text{A.1})$$

a smoothing factor of 0.25 per sample. The filtered count maps to degrees by a clamped linear transform that discards the outer 50 counts at each end of the ADC range, where a potentiometer wiper is least linear.

Recorded captures show each potentiometer producing a new *distinct* value at only 14.7 to 17 Hz against the 100 Hz frame rate, so 82 to 88 % of consecutive rows repeat – every per-row statistic computed before this was noticed was invalidated by it ([Chapter G](#)). At the power-on default of ± 2 g full scale, accelerometer sensitivity is 16384 LSB/g; the widely copied value of 8192 LSB/g corresponds to the ± 4 g range and, used here, halves every reported acceleration – invisible in the elevation estimate (a uniform scale error cancels in a normalised vector) but fatal to the accelerometer gate, which then rejects every frame and looks like a dead sensor. Force-sensitive resistors are read as raw integers with no filtering; the mapping applies its own deadband and hysteresis downstream, so the discrepancy between the design (which specifies a filtered value) and the firmware in hand is recorded rather than resolved.

A.3 The CAD is not the kinematic source

Neither STEP file contains a single kinematic-joint entity; they are solid geometry for manufacture, describing one fixed configuration. No URDF or xacro anywhere in the workspace references either file – **the master arm has no URDF at all**. What the software uses is seven measured constants in `srl_teleop/master_calibration.py`, taken directly from the physical arm. The CAD’s joint-module pitch agrees with those constants to 1.2 %, which is a cross-check, not a licence to treat the CAD as the source: it cannot distinguish the 43 mm roll from the 37 mm bend within one measured pitch.

Appendix B

The route the design took

A record of process, not of result: the routes tried and abandoned before [Chapter C](#) took its delivered form. Recovered prior programs are archived at `thesis_report/_source/prior_work/`, dated 7 June to 27 July 2026; the full divergence analysis against the planning report is at `thesis_report/DIVERGENCE.md`.

The first master-to-slave mapping sent master joint angles directly to slave joint angles. It failed for a structural reason rather than a tuning one: forward kinematics compounds every joint angle in the chain, so a 9 mm/3° disagreement between the master’s CAD and its measured link lengths ([Section A.3](#)) propagates to a materially wrong slave pose at the wrist, and there is no reconstruction path once a channel carrying an intermediate joint fails. Task-space mapping, sending master *tip displacement* to slave end-effector displacement, decouples the two chains and is what shipped – at the cost of leaving the slave’s redundant degree of freedom unspecified, which is what [Chapter I](#) later shows to be expensive near the wearer.

Two further redirections are worth recording because each was forced by a measurement rather than chosen for elegance. The planning report’s haptic objective – physical springs at the master, virtual impedance at the slave – was abandoned once the arm’s cyclic control interface was found unusable over the available network path ([Chapter 3](#)): an impedance law running at 18 Hz is not an impedance law. And validating the master-slave mapping in simulation before touching hardware, as planned, used MuJoCo [[TET12](#)] rather than Isaac Lab [[NVI26](#)] – it starts in under a second, its model format is one readable file, and for a kinematic-mapping question its contact solver is irrelevant, so the heavier alternative bought nothing.

Sensor faults turned out to be the dominant failure mode, not control faults: injecting a stuck analog reading sent a wrong angle to the corresponding slave joint with no reconstruction available under the joint-space mapping, which is the finding that motivated the graceful degradation architecture of [Chapter C](#) and, with it, the reduced azimuth/elevation/reach decomposition that the task-space mapping made possible.

Appendix C

System as built

Chapter 2 gives the command mapping and safety architecture in summary, including the package architecture (Fig. C.1). This appendix covers what is not repeated there: the full mode registry, the desk arrangement the laboratory actually operates, calibration, the operating window, and the verification audit.

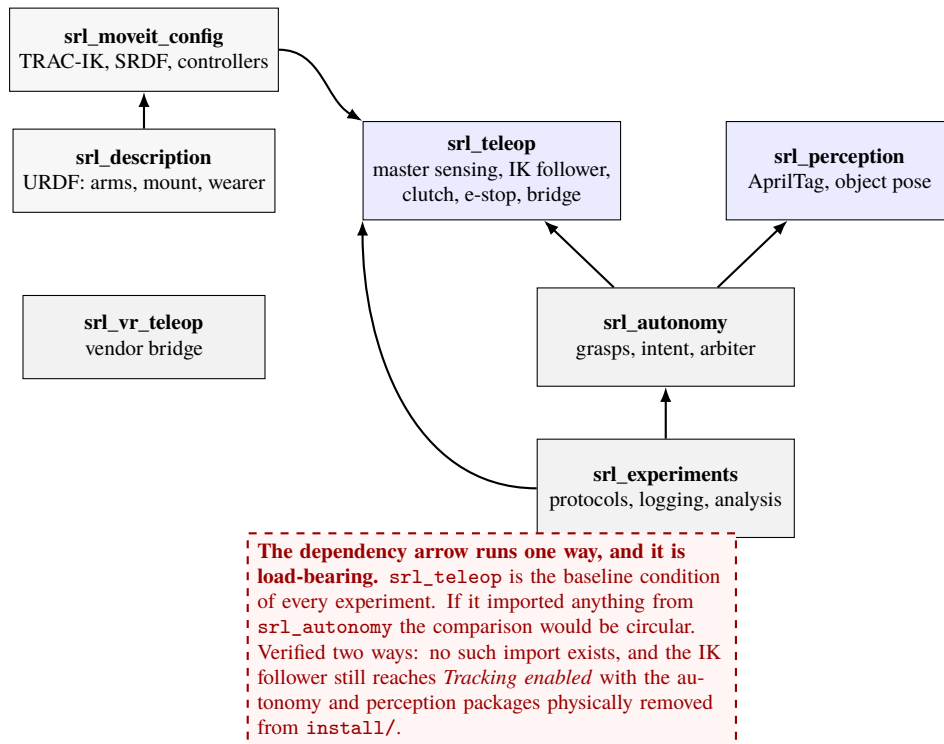


Figure C.1: The software packages and their one-way dependencies: the teleoperation package, the baseline condition of every planned experiment, imports nothing from the autonomy package – verified by inspecting the source and by running it with the autonomy and perception packages deleted.

C.1 Command mapping, in full

The commanded position is built in spherical form about the master’s shoulder rather than by forward kinematics from seven joint angles, each component taken from whichever sensor observes it most directly:

$$\phi = \arcsin\left(\hat{a} \cdot \frac{\mathbf{g}}{\|\mathbf{g}\|}\right) \quad \text{elevation, from gravity} \quad (\text{C.1})$$

$$\psi = \theta_1 - \theta_1^0 \quad \text{azimuth, from the shoulder roll} \quad (\text{C.2})$$

$$r = \|\text{FK}(\theta_1, \theta_2, \theta_3, \theta_4, 0, 0, 0)\| \quad \text{radius, magnitude only} \quad (\text{C.3})$$

with $\hat{a}$ the arm’s long axis in the sensor’s own frame, captured once at rest, and $\mathbf{g}$ the measured acceleration. Elevation is mount-independent and drift-free; radius is dominated by the bend joints and tolerates a dead roll passed through as zero. The commanded end-effector position is

$$\mathbf{p} = \mathbf{p}_{\text{anchor}} + s(\mathbf{d} - \mathbf{d}_{\text{ref}}), \quad (\text{C.4})$$

with $\mathbf{d}$ the master tip in the robot frame and $\mathbf{d}_{\text{ref}}$ the reference latched at clutch engagement, from the first *valid* frame after boot at the master’s actual pose. An earlier version initialised the reference to the origin of the master’s own frame, so the first command was a step of the master’s entire tip vector (roughly 0.22 m) in whatever direction the master happened to be lying.

C.1.1 Collision avoidance in three stages

The wearer is in the collision model: thirteen collision elements were added, paired with thirty-six exclusions for the mount-adjacent proximal links (the arm bases are bolted to the harness and permanently adjacent, so without exclusions every pose is trivially in collision – everything distal of the first arm link stays enabled, which is what actually protects the wearer). Avoidance is graduated: collision-aware inverse kinematics rejects a configuration; on rejection the redundant degree of freedom is re-seeded and the solve retried; a geometric clearance floor measured from the transform tree provides the final margin. The floor is measured against the *fused* body of [Section D.4](#), not the semantic collision description, which excludes forty-four proximal pairs and would report a pose valid with the upper-arm tube inside the person; and `base_link` sits 0.1610 m from the torso and is moved by no joint, so a clearance reading of exactly that value is a measurement of the mount, not the arm ([Chapter I](#)). Driving the commanded pose directly at the wearer’s head and torso stopped the arm at 0.072 to 0.079 m with zero clearance-floor blocks, so the earlier stages did the work and the floor remained a backstop.

C.2 Control modes, in full

Orientation assist is a stub: it would supply wrist orientation under an orientation constraint, and the inverse-kinematics plugin in use ignores the constraint field entirely (270 solver calls, success identical with and without it). It needs a constraint-aware plugin or explicit yaw sampling.

C.2.1 The desk arrangement

The laboratory’s actual arrangement seats the operator across the room, facing the wearer, holding the VR controllers as a six-degree-of-freedom input device and watching the physical robot directly. **Nobody wears the**

Table C.1: Every delivered operating mode. The numbering is 01, 02, 03, 04 and 06: there is no 05, and the gap is preserved because recorded data carries these labels.

Mode	Input	Robot decides	$v_{\max}$ (rad/s)
01 direct, mannequin	master arm	nothing	0.60
02 direct, controllers	tracked controllers	nothing	0.60
03 shared autonomy	master, vision	wrist orientation	0.60
04 shared autonomy, controllers	controllers, vision	wrist orientation	0.60
06 full autonomy	typed or spoken goal	target, approach, grasp, place	0.15

headset; it stands on a shelf as the tracking reference for the controllers. The alignment between the operator’s frame and the robot’s is a measured yaw and can never be a mirror: facing someone and copying their motion is a reflection (determinant -1), which looks correct in position alone but reverses every commanded orientation. The angle is measured by having the operator move 0.4 m toward the robot, with a perpendicular cross-check; calibration refuses to write a value if the two motions are not perpendicular and horizontal.

Bringing the WebXR transport up for the first time (2026-08-19) exposed two defects no unit test would have found: the bridge node had never started at all, because an attribute named `clients` shadowed a read-only property of the base class; and its axis map had been written for the conventional robotics frame rather than this project’s, so a hand moved 3 m forward commanded 3 m to the wearer’s right. The delivered route serves the client page and the wss socket from one origin so a single certificate exception covers both, needs no adb or sideloading, and was verified end to end on 2026-08-19.

C.3 Calibration

C.4 The operating window

Every capability reaches one window: a feature reachable only from a terminal does not exist for the person running a session with a wearer standing in the rig. `scripts/verify_gui_buttons.py` presses every control and checks what the press did (257 checks). The first complete run found four defects: five session controls raised an exception inside a Qt slot, which the framework swallows, so they did nothing and said nothing; and the stray-process preflight for the one-stack constraint had never executed, because its module was never imported.

C.5 Verification methodology

The rule – a surprising failure is evidence about the instrument until the instrument has been cleared – and the ten-repeats-over-a-densified-path practice are stated in [Chapter 2](#); [Chapter G](#) is the full case-by-case record.

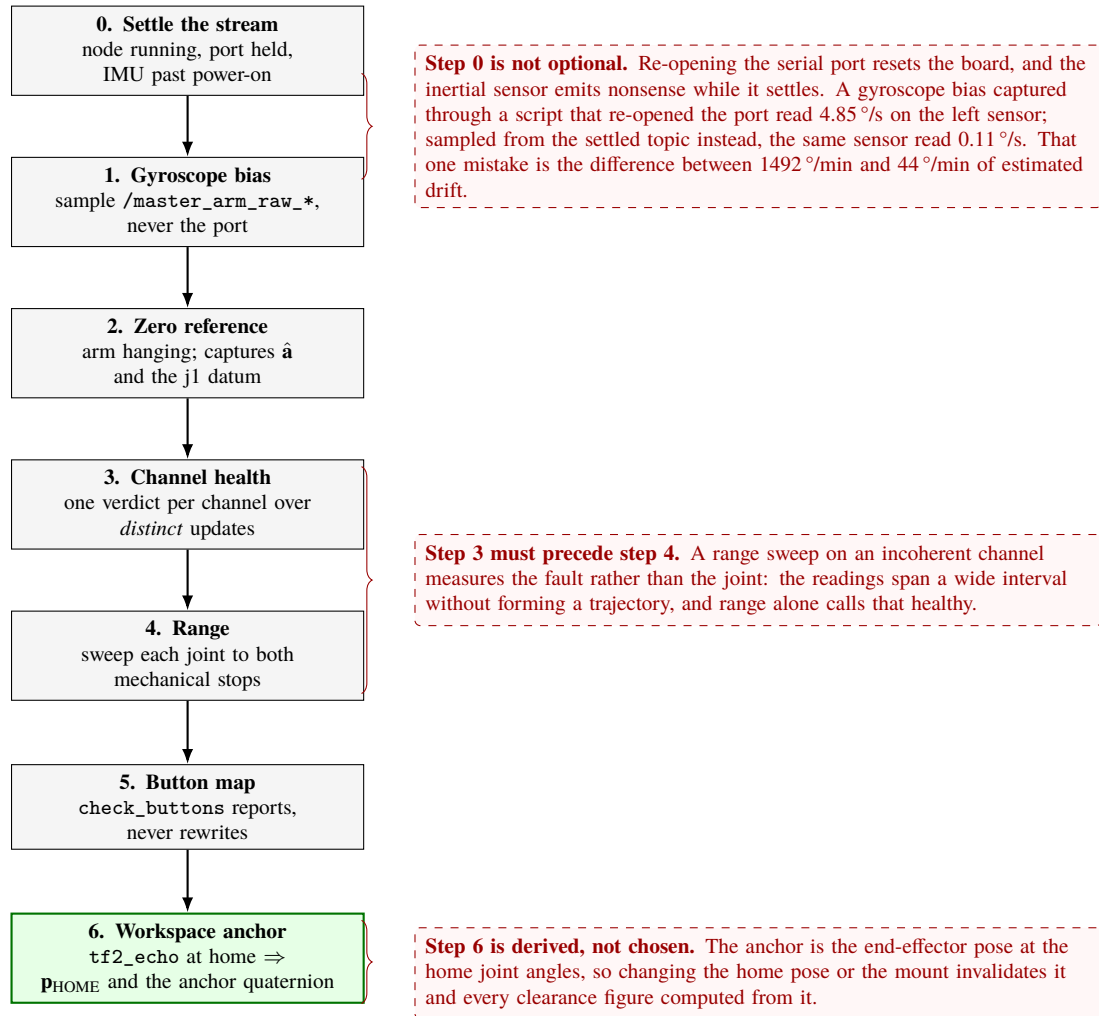


Figure C.2: The calibration procedure, in order. A calibration routine measures whatever the instrument was actually doing at the time, so establishing that the instrument is in a valid state is part of the measurement and not a preliminary to it.

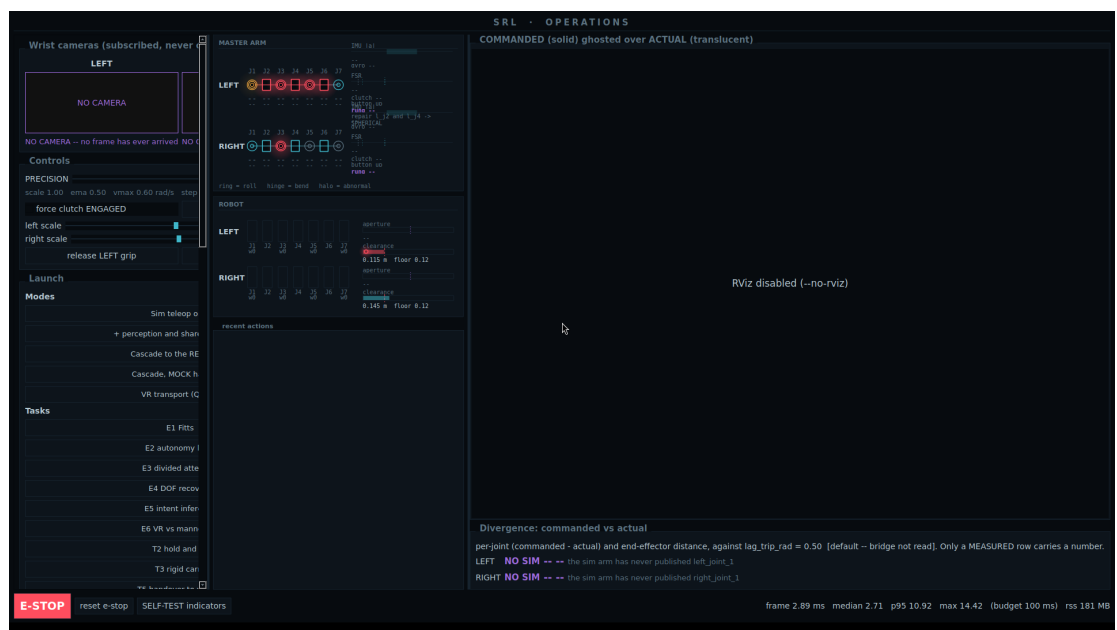


Figure C.3: One command opens it. Left: set-up, run, status. Centre: mode registry and per-mode controls. Right: the three-dimensional commanded view beside a panel drawn from the arms' own reported joint states, with the divergence between them read out beneath both, so a mismatch between commanded and reported state is visible without reading a log.

Appendix D

Perception, environment mapping and autonomous manipulation

Chapter 2 summarises this stage. Almost everything in the delivered task layer previously ran on *declared* coordinates verified at ten repeats over densified paths, none of which survives a different table, a different object, or the same object displaced by 50 mm. Three components capable of sensing the same quantities already existed and had never been composed: a support-plane finder used only to print a description, a voxel-world planner constructed nowhere outside its own self-test, and a work-surface module whose depth-driven height setter nothing called.

D.1 Detection and grasp synthesis

Detection is open-vocabulary (YOLO-World [CSG⁺24] with FastSAM [ZDA⁺23]), falling back to a hue-and-depth filter; the backend that produced a detection is reported with it. Range is used as a discriminator where hue cannot be: the laboratory’s teal robots and the target green block differ in hue by only two units, well inside noise, but sit 2.8 m away against a 1.6 m working range, so range separates them at zero extra cost – the same size-at-range expression ($w = sZ/f$) also rejects a 210 mm pad presented as a 40 mm cube. Grasp synthesis takes the principal axes of the object’s depth cloud: writing the mean-centred points as $\mathbf{P}_k$, the axes are the eigenvectors of the scatter matrix

$$\mathbf{C} = \frac{1}{N} \sum_{k=1}^N (\mathbf{P}_k - \bar{\mathbf{P}})(\mathbf{P}_k - \bar{\mathbf{P}})^\top, \quad \mathbf{C}\mathbf{u}_j = \lambda_j \mathbf{u}_j, \quad \lambda_1 \geq \lambda_2 \geq \lambda_3, \quad (\text{D.1})$$

and the jaw closes across $\mathbf{u}_3$, the direction of least extent, capped at

$$w = \max_k \mathbf{u}_3^\top \mathbf{P}_k - \min_k \mathbf{u}_3^\top \mathbf{P}_k \leq w_{\max} = 85 \text{ mm}. \quad (\text{D.2})$$

A geometric method can *name* the cause of a refusal (too wide, no depth, too few points), which a learned alternative cannot and which also would not fit beside the detector in 4 GiB of video memory.

A reachability gate checking candidate grasps for reach and wearer clearance was inoperative for the life of the module: it scored the clearance from the whole kinematic chain including the immobile mount stub, which reads a constant 0.2202 m in every pose and so could never fail against the 0.15 m floor. It now reads the moving-chain clearance (0.4031 m at home) and a regression test requires a named refusal when aimed at the wearer’s torso.

D.2 Measuring the working area

The sweep holds one wrist attitude per pass and translates, printer-bed-probe style, rather than re-aiming at each grid point: re-aiming would make a residual camera-to-wrist error a *different* error in every frame, which can neither cancel across views nor be estimated from them. A frame is accepted only once no joint has moved by more than 0.0015 rad over 400 ms – a tenth of the arrival tolerance.

Table D.1: The fused map after one full sweep, against the scene it was measuring.

Quantity	Measured
Surface height	1.250 26 m against a declared 1.250 m
Objects recovered	6 of 6, every centre within 1 mm
Object extents	every dimension within 3 mm
Coverage of reachable cells	98.65 %, 0 stillness refusals
Elapsed	13 min 15 s

An earlier surface error of -5.2 mm was itself an instrument defect (Chapter G): the reported height was the fitted plane’s intercept at the *origin*, a point inside the wearer, not its height over the working area. Reachability (a fact about the robot, computed once) and the scene map (a fact about the room, rebuilt from scratch every time) are separate products with separate lifetimes; `scripts/pick_from_map.py` refuses to move from a map older than thirty minutes, because a stale map is indistinguishable from a fresh one by inspection.

D.3 Planning and natural language

`joint_planner.py` implements RRT-Connect [KL00] in joint space (the binding constraint is the whole arm’s distance to the wearer, not the hand), evaluated against the same moving-chain clearance the guard enforces, with every edge densified at 0.05 rad to match the path checker. Without a planner, a pick whose pre-grasp and grasp poses were each individually clear passed within 0.1462 m of the wearer mid-path.

Table D.2: Seventy-five phrasings, delivered grammar against an earlier revision checked out of version control.

Outcome	Earlier	Delivered
Correct	21	34
Asked a clarifying question	14	14
Refused	37	27
Misunderstood	3	0

The three earlier misunderstandings shared one mechanism: a compound instruction (*put the blue ones on the blue mat and the green ones on the green mat*) was cut at the first destination cue, so the two-target check counted targets only in the head clause, planned the first, and silently discarded the second while reporting success.

D.4 Observing the wearer

The clearance floor of Chapter 2 is enforced against a body. `wearer_tracker_node.py` replaces the mannequin in the robot description, part by part, with one built from markerless pose estimation and metric landmarks lifted by perspective-n-point [LTN⁺19]: 28 to 34 ms inference, 61.6 ms median frame-to-estimate, 62 to 562 ms end to

end (the spread is a configured publication rate limit, not the estimator). The safety argument is one monotonicity rule enforced in one function: **the camera may only ever make the wearer larger** – per body part, `fuse()` keeps whichever of the tracked and mannequin primitive is *nearer* the robot. A body injected 0.10, 0.30, 1.00, 5.00 m further away moved the enforced margin by nothing; six injected faults (dark frame, blank frame, two people, empty frame, occluded arms, a control) were refused seven of seven, each naming the fault. The honest limitation is on the fallback side: if the wearer raises their arms and tracking is then lost, the guard reverts to arms-down and believes 0.317 m of clearance where there is 0.093 m.

Before this was connected, the tracker reached the planning scene and *not* `mount_guard_node`, which builds its wearer at import time – the body the planner avoided and the body the floor enforced against were two different bodies. A second camera observing the workspace from outside the arms is implemented, refuses to republish stale frames, and both calibration tools pass self-tests to 0.43 % focal length and 0.0000 m/0.000° extrinsic error – but **no camera has ever been attached to it**: the device is the host’s own webcam, and forwarding it into the Linux environment needs an administrator-level bind that was refused.

Appendix E

Motion generation and simulation-to-hardware transfer

[Chapter 3](#) reports the headline numbers. This appendix gives the mechanism.

E.1 Why a per-joint clamp cannot stay on the line

The limiter it replaced applied, per cycle and independently to each joint i , $q_i^{(k+1)} = q_i^{(k)} + \text{clip}(q_i^* - q_i^{(k)}, -\Delta, +\Delta)$ with one step bound Δ shared by all seven joints. Writing the required displacement as $\mathbf{e} = \mathbf{q}^* - \mathbf{q}^{(k)}$, staying on the straight joint-space line requires the applied step be *parallel* to $\mathbf{e}$; the clamp applies $\text{clip}(\mathbf{e}, -\Delta, +\Delta)$ component-wise, which is parallel to $\mathbf{e}$ only when every component is inside the bound or every component saturates equally. In every other case the direction of travel rotates and the end effector leaves the path. This is why sweeping Δ over a factor of 35 does not drive the excursion to zero – it settles at 68 mm – and why the excursion is a property of the algorithm, not its tuning.

E.2 What replaced it

Ruckig [\[BK21\]](#): time-optimal, jerk-limited, per-degree-of-freedom limits, accepting a non-zero target velocity. Phase synchronisation drives every joint from one scalar phase $\sigma(t) \in [0, 1]$:

$$\mathbf{q}(t) = \mathbf{q}^{(k)} + \sigma(t)(\mathbf{q}^* - \mathbf{q}^{(k)}), \quad \sigma(0) = 0, \sigma(T) = 1, \quad (\text{E.1})$$

so the commanded configuration is on the segment between endpoints by construction for every t – unlike merely stretching each joint’s own profile to the slowest joint’s duration (time synchronisation), which measures 9.5 mm of path deviation on the same case against phase’s 0.02 mm. The acceleration and jerk limits are *assumed* and labelled so: `joint_limits.yaml` declares zero acceleration limits for all fourteen joints, so there is nothing to read; only the velocity limits are the robot’s own.

Five defects were caught in the replacement, four of which would have passed a conventional test suite: re-planning every cycle destroyed phase synchronisation (a mid-profile re-plan cannot reproduce the profile it is half through); floating-point noise forced a re-plan every cycle even after fixing that, because the target was recomputed via a wrapped subtraction not bit-identical to itself; the synchronised-clamp fallback reported zero acceleration (because it does not model one) and so passed a continuity check by fabricating the quantity being checked; the off-path metric had a 0.1 mm grid-resolution floor and was reporting the instrument’s resolution as

the robot’s behaviour; and an arrival test used an absolute tolerance on travels spanning a factor of twenty-five.

E.3 The simulation-to-hardware gap

Of 252 recorded per-joint errors, 211 lie between 0.25° and 0.30° with the sign following the direction of travel – a *bound*, not a spread, which identifies a terminal offset rather than noise. Two candidate models, a terminal offset and a proportional gain error, coincide at one travel distance and differ by a factor of five at 20 mm; propagating the recorded errors through the manipulator Jacobian reproduces the measured Cartesian error to 90 % ($r = 0.923$) under the offset model and not the gain model. The data had been recorded for weeks in `recordings/baselines/arm_directional_calibration.json` before anything read it.

An earlier report claimed the terminal error is speed-independent across a fivefold range; the velocity limit was read once at bridge start-up and seven of thirty-six runs moved *faster* than commanded, so the three speeds were three replicates of one condition and the speed dependence has never actually been measured. The bridge deadband was 1.0° against a measured 0.305° park – a factor of 3.3 – and is 0.10° now, with an opt-in overshoot correction; neither has been tried on hardware.

E.4 The positioning error budget

$$\sigma_{\text{RSS}} = \sqrt{\sum_i b_i^2}, \quad \sigma_{\text{worst}} = \sum_i b_i, \quad (\text{E.2})$$

with σ_{worst} the honest figure against the 30 mm capture gate, since the two largest terms are declared constants rather than noise. Corrections – the pad offset re-derived from forward kinematics (13.45 to 0.05 mm), the shell bias removed via the measured support plane, and the terminal-offset overshoot (joint term 7.24 to 1.88 mm) – take the budget from 18.64 to 12.91 mm RSS and 33.39 to 19.99 mm worst case, *provided* the camera-to-wrist transform is small. [Section H.7](#) measured it directly and it is not.

E.5 The reaction budget

Table E.1: Limb travel during the measured wearer-tracking latency against the 150 mm clearance floor. Speeds are literature ranges, labelled as assumptions; latencies are measured.

Limb motion	Latency	Travel	Against the floor
Deliberate reach	62 ms	50 to 93 mm	62 % of it
Deliberate reach	562 ms	450 to 843 mm	exceeds it
Quick reach	62 ms	93 to 155 mm	exceeds it
Startle	62 ms	155 to 248 mm	exceeds it

In five of six modelled limb motions the limb travels further than the entire margin before the guard is aware anything has changed – [Chapter 4](#) treats this as the principal open safety question.

Appendix F

The vision pipeline, measured on the real cameras

Measured on four physical cameras, 2026-08-30 – together with [Section H.7](#), the only hardware numbers in this report.

The pipeline is instrumented at thirty-three stages per camera; every stage draws itself on the frame it was given, states its formula, and reports its quantities with units, so a claim can be checked rather than taken on trust. A stage has three possible outcomes, not two: it succeeds, it *refuses* and names its cause and remedy, or it is *absent by design* (the webcam’s fifteen depth stages, because it has no depth sensor) – rendering an absence identically to a defect has cost this project findings before.

F.1 The orientation defect

Three places in the software independently decided whether the scene depth camera is mounted upside down, and disagreed. Settled by correlating each frame’s grey signature against the HD webcam (rotated by nothing, on the same side of the room) in both orientations: the un-rotated frame agrees at $r = 0.66$ to 0.70 over four captures, the rotated one at $r = 0.27$ to 0.34 – the same verdict every time, a factor of 2.4 apart. **The camera is mounted the right way up**, and every earlier scene figure was inverted. Because the wrong rotation had been applied *consistently* to the image and the intrinsics together, the recovered point cloud was an exact reflection of the true one and every distance, extent and residual was preserved – the defect was invisible to every metric the pipeline itself computes and was found only by comparing one camera’s picture against another’s.

F.2 What the cameras measured

Table F.1: One frame per camera. The inlier fraction decides whether the row below it means anything.

	left wrist	right wrist	scene depth
pixels returning a depth	89.4 %	80.5 %	14.5 %
plane inliers	74.0 %	52.8 %	3.8 %
plane residual, RMS	1.43 mm	1.59 mm	3.41 mm
objects posed in 3D	5	12	12
graspable within the jaw	0	0	0

The wrist cameras measure a work surface to about 1.5 mm – within a factor of three of the same code against a renderer, the first evidence the surface-fitting stage transfers to hardware. **The scene camera does not identify a support plane**: its best plane holds 3.8 % of the points, so every quantity that depends on it (height,

object boxes) rests on a plane one point in twenty-six agrees with. **No object in any frame could be grasped:** the nearest object's narrowest axis measures 144 mm against an 85 mm jaw. **The open-vocabulary detector returned nothing on any camera** for the prompt *green cube*; the classical colour-and-geometry path found the same objects it did not.

Two defects in the first runs were software, not hardware. The pick path's own depth working band, correct for a wrist camera, applied unchanged to a camera across the room retained **0 of 85 061 returns** – the band is now declared per source. And two stages disagreed about the same frame because depth at 480×270 forward-projected into colour at 1280×720 lands one sample in roughly every seven pixels, so the above-plane mask was a dot pattern that failed a connectivity check while a per-mask stage saw it fine; splatting each projected sample raised coverage to 64.4 % and 58.8 %.

The arms were to be parked at the pick pose for this session; measured rather than assumed, the left arm was 10.67° and the right 13.53° from the reference configuration, against a 2° tolerance, so no figure from this session is captioned "at the pick pose."

F.3 The scene the safety layer reasons about

Figure F.1 names four classes in one frame rather than isolated detections: the mannequin the clearance floor of Section 3.1 is enforced against, the table, the two wearable arms, and the task objects between them. No detector in this project covers the first three classes – only the task-object palette does – so all six boxes here are hand-identified against a pixel grid on this exact frame, stated as such rather than presented as automatic detections.

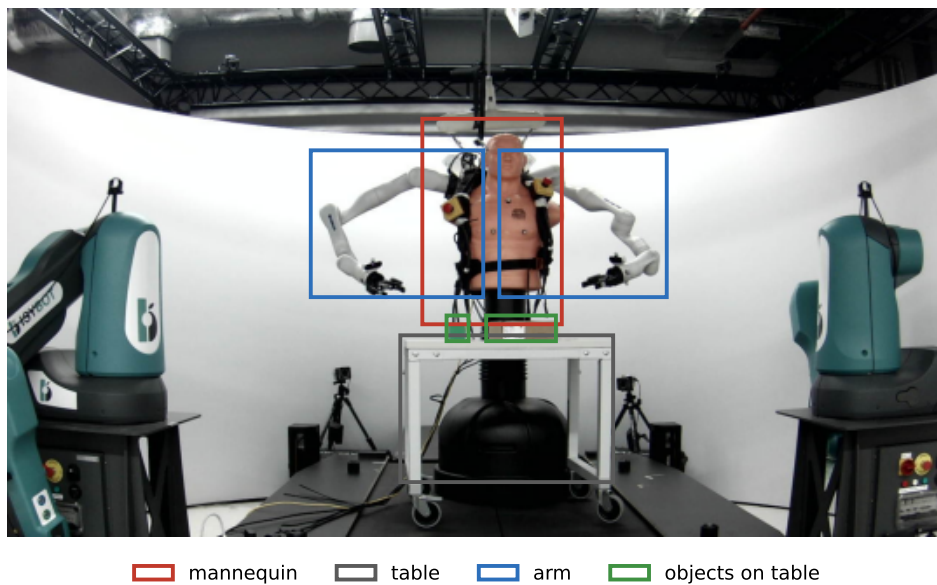


Figure F.1: The same figure referenced from Chapter 3.

Appendix G

Measurement validity: the full record

Chapter 4 states the validity argument and the two numbers that carry it. This appendix is the evidence: what each of the twenty-six cases actually was, grouped by mechanism.

A surprising failure is evidence about the instrument until the instrument has been cleared.

Twenty-six times a measurement in this project was wrong and the system was not. In every case the wrong reading was *plausible* – none announced itself – and two would have removed results now central to this thesis. Sorted by mechanism, they fall into five groups.

An absence read as a value (7 cases). Zero variance from oversampling a slower sensor ([Section H.2](#)); a detection rate of 0 to 4 % from a renderer producing images unlike anything the detector was trained on; a surface height reporting the fitted plane’s intercept at the origin rather than over the working area.

A test harness constructing the condition it tests (6 cases). A reachability control comparing a search-cap lower bound against a boundary, which was never going to pass regardless of the arm’s true reach; a path-verification failure from the verifier’s own test path running in the one direction that leaves the feasible region.

State left over from a previous run (5 cases). A reachability sweep measured with the arms parked 0.87 and 0.59 rad from home, reporting zero failures under the same name as the genuine measurement.

Order of operations inside the measurement (5 cases). An isotropy statistic taken as the minimum over directions of a single sweep, dominated by rare early solver terminations and varying by a factor of two between runs; stable once repeated ten times and read as a median.

A constant asserted in more than one place, measured in none (3 cases). The scene camera’s mounting orientation ([Section F.1](#)); the wrist-to-pad offset, treated as a scalar when it is a function of gripper opening.

Two numbers carry the argument. **Fifteen of the twenty-six were caught by an independently known quantity contradicting the reading** – five by inspecting the code, six by looking at the raw artefact, usually an image. Care was not the defence that worked; redundancy was. **Fourteen of the twenty-six were already answerable from something in hand** – a published stroke, an arithmetic prediction, a quantity exact by construction – which is the recommendation this project’s experience supports: build the cross-check in rather than trying harder.

G.1 What follows from it

Every reachability figure in this report carries its repeat count, because one solver call is one coin flip. Every stage of the simulation campaign carries controls that must *fail*. Where a quantity has never been measured, the text says so rather than offering an estimate – the camera mounting transform in the positioning budget is one such term, and it is not small ([Section H.7](#)). Any unreproduced single measurement in this report should be read as provisional.

G.2 Three standing caveats

Simulation is not the arm. Every quantitative result except the channel measurements and the pilot comes from simulation with mock hardware, whose force channels are identically zero – any quantity derived from force is structurally undefined, not merely unmeasured. **Latency figures are lower bounds**, measured between software stages, excluding the operator, the display and the physical arm’s response. **The one-stack constraint:** two instances of the master node once split the serial stream between them and invalidated a full day of measurements before the cause was found; this is now prevented at the file-descriptor level, but any measurement taken before that prevention carries a risk that cannot be retrospectively excluded.

Appendix H

Results, in full

Chapter 3 reports the headline numbers with their evidence class. This appendix is the supporting record: every table the simulation campaign, the hardware sessions and the safety and instruction-handling measurements produced, with only enough text to say what each is.

H.1 The simulation campaign

`scripts/run_simulation_campaign.py` runs every simulation stage by one command and records what each produced with its exit status and duration; `scripts/make_sim_figures.py` refuses to emit a table whose stage did not run. Sixteen stages, all of which ran; 13 927 inverse-kinematics calls at zero failures.

Table H.1: The simulation campaign inventory.

Stage	Needs	Ran	Time
<i>computer vision</i>			
cv_arithmetic	standalone	✓	12 s
cv_orientation	standalone	✓	< 1 s
cv_replay	standalone	✓	88 s
<i>teleoperation</i>			
teleop_budget	standalone	✓	< 1 s
teleop_motion	standalone	✓	5 s
teleop_orientation_cost	simulated platform	✓	20.6 min
teleop_replay	recorded data	✓	< 1 s
teleop_smoothing	standalone	✓	< 1 s
<i>robot tasks</i>			
tasks_accuracy	standalone	✓	< 1 s
tasks_dance	simulated platform	✓	38 s
tasks_msc	simulated platform	✓	48 s
tasks_status	standalone	✓	< 1 s
tasks_t1	simulated platform	✓	2.2 min
<i>autonomy</i>			
autonomy_arbitration	standalone	✓	4 s
autonomy_instructions	standalone	✓	< 1 s
<i>study design</i>			
study_power	standalone	✓	17 s
<i>transfer to hardware</i>			
sim_to_real	recorded data	✓	< 1 s

H.1.1 Computer vision against a constructed scene

Twenty-one known-answer checks on a frame whose plane, object pose and size were chosen. The fitted plane normal lands 0.0001° from the constructed one; a 40 mm cube measures $40.8 \times 40.7 \times 40.7$ mm and centres to 0.41 mm. Two naive shortcuts are measured wrong by name: an axis-aligned bounding box reads 41 % too wide, and taking the object centre as the mean of its visible points is out by 16.15 mm along the surface normal (a single view sees a shell, not a solid) – corrected to 0.35 mm by assuming the object rests on the fitted plane.

H.1.2 Shared autonomy: what the intent estimator is sensitive to

Sweeping the declared-position error over the range a real table spans, a declared-coordinate estimator becomes *confidently wrong* 50 % of the time once the declaration is one object-pitch out; a measured-coordinate one stays at 0.0 % across the same sweep, with the correct rate flat at 71.7 to 75.7 %. The correct rate never exceeds 76 % in any condition, including perfect information – the remaining quarter is ambiguity, and the arbiter is specified to stay in direct teleoperation whenever ambiguous.

H.1.3 Full autonomy: seventy-five phrasings

Correct interpretations rose from 21 to 34 of 75 against an earlier grammar revision, refusals fell from 37 to 27, misunderstandings fell from 3 to 0. Ten of fourteen clarifying questions carry a scored reply; all ten resolve correctly.

H.2 Channel health

Measured on recorded hardware data, 2026-08-06. Six of fourteen channels met the alive criterion.

Table H.2: Per-channel verdicts, over distinct sensor updates.

Left arm		Right arm	
Channel	Verdict	Channel	Verdict
j1	intermittent, 2.2 %	j1	alive, 195°
j2	incoherent, 5 %	j2	alive, 261°
j3	incoherent, 8 %	j3	incoherent, 14 %
j4	incoherent, 6 %	j4	alive, 291°
j5	incoherent, 12 %	j5	dead
j6	incoherent, 8 %	j6	alive, 250°
j7	alive, 26°	j7	dead

Left j7 had been recorded as railed and dead: the evidence was a spread of exactly zero over forty consecutive *transmitted rows*, which is what an oversampled sensor (82 to 88 % repeated rows) produces regardless of health. Regressing true reach on the surviving channels plus the inertial sensor gives $R^2 = 0.133$ (left) and $R^2 = 0.986$ (right), naming the repair as exactly left j2 and j4.

H.3 Axis correctness

Regressing commanded pose against the master's physical spherical position (6443 samples, $R^2 = 0.999$) found a gain matrix of $\text{diag}(+1, -1, -1)$ against the intended identity: raising the master drove the arm down. Corrected. The lateral axis remains wrong and cannot be fixed by any axis map: measured directions meant to be 90° apart are 137.7° apart, and a rotation preserves angles, so no rotation maps one triad onto the other. Gyroscope-derived azimuth halves the residual on the left arm (25.6° to 13.5°) but is not enabled: the right sensor drifts at $44^\circ/\text{min}$ stationary.

H.4 Safety: fault injection and timing

Fourteen of fourteen injected faults handled: channel dropout, board disconnect/reconnect, emergency stop mid-motion, wrong-object latching, ambiguous-target flicker, stale object, detection failure, logging failure, identifying data in a manifest, non-monotonic clock, session loss, zero detections, network dropout. Dead-man timing against a 1.00 s deadline: healthy master, 243 frames, no trip; **degraded (publishing at full rate, frozen source timestamp): trip at 0.94 to 0.99 s** – the case that could not have passed before keying the dead-man on the source timestamp; clean unplug: trip at 0.97 s. A separate emergency-stop defect: the trip handler called a blocking real-driver shutdown first, so decision at 0.97 s produced trip *logged* at 4.98 s – fixed by parking first, from local publishes only.

H.5 Task-set verification

Every coordinate of the five-task set verified at ten repeats over paths densified to 20 mm, both arms confirmed at 0.0000 rad from home before any sample: zero failures, and the result reproduces field for field on re-run.

Table H.3: Five-task verification.

Task	Coverage	Result
Positioning	3 targets per arm	3/3, 3/3
Pick and place	3 picks per arm, approach and place	3/3, 3/3
Rigid carry	3 paths, 25 densified waypoints	all verified
Compliant carry	as rigid carry	all verified
Dual pursuit	26-direction shell per arm	26/26, 26/26

H.6 Performance figures

Control-path latency; a lower bound on end-to-end latency.

The Kortex figure constrains the architecture: the cyclic control interface needs a 1 kHz loop on a dedicated link, and over this machine's virtualised network each cyclic write costs about 10 ms – at 100 Hz the write alone consumes the cycle budget and no command reaches the arm while telemetry keeps arriving. The high-level velocity interface avoids this because it holds a commanded motion between sends.

Table H.4: Measured rates and latencies, against mock hardware.

Quantity	Value	Note
Master frame rate	100 Hz	firmware loop
Distinct sensor updates	14.7 to 17 Hz	Chapter A
VR packet rate	68.5 Hz	822 packets / 12.0 s
VR round trip	8.66 ms median, 16.5 ms p95	
Voice stop to /estop	2 ms	
Kortex high-level loop	18.4 to 18.7 Hz	two round trips/cycle

H.7 The first hardware session

Measured on a physical arm – the only section of this chapter whose numbers come from a moving Kinova arm. A single arm was driven from its own wrist camera for the first time. Three findings survive as results. **The camera-to-wrist transform is wrong by roughly 8.5° and rotates with the wrist:** the surface normal measured from four poses of a stationary table read $9.74, 15.42, 16.26, 18.14^\circ$ from vertical, a worst pairwise disagreement of 8.45° – 34 mm at 0.23 m, 59 mm at 0.40 m. **The depth sensor itself is excellent:** the support plane fits at 94.8 % inliers, 0.46 mm residual, and a 40 mm calibration cube measures $39.2 \times 38.9 \times 40.9$ mm – only the transform out of the camera frame is wrong. **The middleware was a first-class hazard:** shared memory in a state where no new process could join while running nodes continued normally, and an unset middleware variable caused silent discovery failure. Both present as an unresponsive robot.

H.8 Verification, in totals

Table H.5: What is checked automatically, as of 2026-08-31.

Measure	Count
Automated tests passing	1505
Test modules	136
Checks on the operating window	257
Faults injected and handled	14 of 14
Simulation stages, all of which ran	16
Inverse-kinematics calls, failures	13 927, 0
Known-answer checks on the vision arithmetic	21
Simulated studies of the analysis procedure	20 000
Recorded demonstration cells	17, at 8 camera angles
Vision stages instrumented on real cameras	132

H.9 What is not measured

Anything derived from a force reading (the mock returns identically zero); object detection rate at working distance (the renderer used to measure it is out of the detector’s training distribution); grasp success against a physical object; end-to-end latency including the arm and operator; how the terminal joint error varies with commanded speed (retracted in [Chapter E](#): three speeds were three replicates of one condition); speech recognition against a human speaker (no audio input device on this host; every voice figure characterises the grammar, not a recog-

niser); two arms driven simultaneously on hardware (measured at 8.1 to 12.2 Hz with send latency spiking to 519 ms, past the bridge's own watchdog).

Appendix I

Workspace characterisation and its consequences

Chapter 3 gives the headline finding and the mount fix it produced. This appendix is the chain of measurements behind it.

I.1 The two arms share no reachable workspace

Over a 63-cell grid spanning the frontal working volume, arms verified at home, no cell was reachable by both arms (Fig. 3.2 in Chapter 3). The result survives freeing the wrist yaw (raising each arm's individual reach from 16/18 to 21/21 cells, still 0/63 shared) and additional solver retries. The right arm reaches $x \leq -0.10$ m, the left $x \geq 0.15$ m: a 0.25 m dead band, measured at $y = 0.35$ m, $z = 1.10$ m, yaw free, three repeats per point.

This holds within $x \in [-0.6, 0.6]$, $y \in [0, 0.4]$, $z \in [0.90, 1.30]$ m, with the as-built mount and the legacy home joint angles – the arms share no workspace *as mounted and as parked*, and the cause is not the mount's rotation (proved invariant) but the arms' asymmetric home joint angles, which are ground truth because the simulation-to-hardware bridge replays them onto the real arm.

I.1.1 The fix is buildable

A parametric sweep moving the mounts inboard, forward and down, with additional forward tilt, raises shared cells from 3 to **22 of 63** within a buildable 0.20 m mount separation (splaying the mounts outward *hurts*). Recorded and deliberately not applied at the time (changing the mount invalidates the home pose, the anchor and every clearance figure), then applied later as one deliberate pass with the whole task set re-measured:

Table I.1

Quantity	Before	After
Clearance at home	0.1610 m	0.2202 m
Nearest robot part to wearer at home	the mount (no joint moves it)	the moving chain
Task objects resting on the surface	not achievable	achieved
Pick-and-place targets off centre	0.530/0.350 m	0.290 m either side

I.2 One boundary, two independent constraints

Wearer clearance and top-down grasp feasibility, measured for different reasons, land on the same lateral boundary: clearance rises monotonically and crosses the 150 mm floor at 0.25 m; top-down grasp candidates fail entirely (0/9) at $|x| = 0.25$ m and succeed entirely (9/9) at $|x| = 0.30$ m, with collision checking disabled too – kinematic infeasibility of the wrist orientation, not obstruction.

I.3 What the pinned wrist costs at the work point

Reach under each orientation policy, walked from home, understates the cost badly because home is in free space. Walked instead from (0.4, 0.175, 1.12 m) on the working surface, with the 150 mm floor enforced throughout:

Table I.2: Mean reach under each orientation policy. The two columns disagree by a factor of seven, because reach is meaningless without the point it was measured from.

Orientation policy	from home (m)	from work point (m)
pinned to the anchor (delivered)	0.460	0.065
within a 15° cone	0.492	0.304
within a 45° cone	0.530	0.325
position only	0.547	0.363

At the point where the work happens, the delivered policy leaves the arm 65 mm of mean reach against 363 mm with the wrist free – a factor of 5.6. The mechanism is redundancy, not the wearer: no walk from the work point stops on the clearance floor, they stop because there is no solution – a fixed six-degree-of-freedom pose on a seven-degree-of-freedom arm spends the entire redundancy, and pinning the wrist removes the joint that would otherwise move the elbow clear while the hand stays put. Freeing the roll alone buys 0 mm at the work point; almost all recovery comes from the first 15° (80 % of the pinned-to-free gap), which the follower's own 15° solving cone already supplies – the reason the task set solves at all.

I.4 The centre of the working space, asked four ways

Table I.3: Four attempts to place usable work within 100 mm of the body centre line, at ten repeats.

Question	Answer
Innermost workable column	$ x = 0.325$ m (L), 0.450 m (R)
Same, wearer's arms deleted entirely	0.300/0.375 m; every column 0.125 to 0.300 m binds on the torso
Objects resting on a surface, 3360 configurations	no configuration within 100 mm of centre
Same, top-down approach	0 of 840 cells, any height 0.70-1.10 m

The constraint is the torso, not the wearer's limbs: sweeping five arm postures moves the boundary by at most 75 mm, and two postures (arms clasped behind the back, arms out to the sides) place the wearer's own limbs inside the clearance floor at home, before the robot does anything. Neither arm crosses the centre line at all – zero of ten solutions, both arms, every cross-side slot – which is why the pick-and-place task draws the *side* and lets colour follow it.

I.5 Consequences for the task set

Inter-arm handover is impossible: no control, planning or autonomy change can produce a point neither arm reaches. A bimanual task must span the dead band or exploit simultaneity. The coupled carrier's original 310 mm width placed each grip inside the dead band (10 of 25 waypoints failed once the arms were verified at home); widened to 500 mm it satisfies both boundaries, and the carrier's sag tolerance was re-derived to keep the failure threshold inside the reachable band rather than outside it, where it would measure nothing.

Appendix J

Protocol for the two-person study

This is the study [Chapter 2](#) says the system was built for. The pilot in [Chapter L](#) is a smaller, operator-only precursor; this protocol – with a separate wearer – is still the target.

J.1 The question and hypotheses

Does supplying part of the operator’s control with autonomy improve performance, and what does it cost the person carrying the arms? No prior study measures both under the same autonomy in the same trial ([Section 1.1](#)). Five hypotheses, stated with direction before any data exists: task completion time is shorter under shared autonomy (H1); operator workload is lower (H2); **wearer-reported trust is lower under shared autonomy** (H3), against the direction of H1/H2, taken from a concealed-operator study [[ZGD⁺26](#)] and tested here with autonomy that is genuinely a program and a wearer with a concurrent task; the wearer’s sense of agency is lower while ownership is unchanged (H4), since the two dissociate [[LSK⁺08](#)]; cross-arm interference is lower under shared autonomy while phase lag is not (H5), which would identify the benefit as attentional. **H3 and H4 are the hypotheses worth running the study for**; H1 and H2 exist to establish the platform can show an effect at all.

J.2 Design

Two modes (direct teleoperation, shared autonomy) crossed with five tasks, within subjects, sixteen dyads, roles fixed for the whole session. Two modes rather than four is costed, not a convenience: four modes across five tasks needs 260 to 320 min against a wearer exposure cap of 48 min for carrying 17 kg continuously; a between-subjects design across modes needs ninety-six participants for the same sensitivity, since between-subject variance here exceeds the effect sought. Mode order is counterbalanced with Williams squares, not a Latin square, because a Latin square balances position but not immediate sequence, and a task with a learning curve is sensitive to the sequence.

Grasping under direct teleoperation is not achievable: the default orientation mode pins the wrist to the anchor, the pick task needs a 169.7° rotation from it, and no master channel can command that. T2 (pick and place) therefore carries no mode contrast and is analysed as a single condition – retained because grasp success under autonomy is worth measuring, and because it is the one task where autonomy’s advantage over teleoperation on this platform is categorical rather than quantitative.

Table J.1

Task	Measures	Mode contrast
T1 positioning	Throughput vs. index of difficulty [Fit54]; single- vs. both-arm decrement	both modes
T2 pick and place	Grasp success, placement accuracy	shared autonomy only
T3 rigid carry	Coordination error as object tilt	both modes
T4 compliant carry	Coordination error as gripper separation	both modes
T5 dual pursuit	Cross-arm interference under divided attention	both modes

J.3 Measures and procedure

NASA-TLX [HS88] after every block, both participants separately (the two roles load differently, and the comparison between them is the object of the study); Borg CR10 [Bor82] exertion from the wearer between every block, since a single end-of-session rating turns a covariate into a guess; trust and embodiment [LSK⁺08] at the end of session, wearer only for embodiment. A safety observer, present throughout, holds an emergency stop and watches the wearer rather than the screen; their presence is enforced in software as well as procedurally – the safety node treats two seconds of heartbeat silence as withdrawal of consent to move. Consent is taken separately, in separate rooms.

J.4 Ethics and risk

The dominant risk is a 17 kg manipulator pair, moving under partly autonomous control, within reach of a seated person’s head and torso. Mitigations: the 150 mm margin enforced over the whole path and measured geometrically (Section C.1.1); a stop within the wearer’s own reach and a second held by the observer; twenty minutes of carried-load exposure at a time with exertion rated between blocks; order counterbalanced against fatigue confounding condition; consent restated at every break; identifying data refused in code (`write_manifest()` raises on name, email, date of birth, address or telephone number). Video recording is not part of the protocol: a video of a participant wearing this rig is not anonymous, and no measure here requires one.

Appendix K

The analysis procedure, and its behaviour on constructed data

The analysis [Chapter J](#) would be evaluated with, measured against data whose answer is chosen. No dyad data exists yet; this is a property of the procedure, not a result.

The design is within dyads: for task j , the paired difference $d_{ij} = y_{ij}^{\text{SA}} - y_{ij}^{\text{DT}}$ and its standardised mean $d_z = \bar{d}_{\cdot j} / s_{d, j}$ is the quantity the design was powered for (distinct from the standardised difference of means, which deflates every effect by $\sqrt{2}$ or more once within-dyad correlation is accounted for). Five tasks give five comparisons, corrected with Holm’s step-down procedure, which controls the family-wise error rate under any dependence structure and is uniformly more powerful than Bonferroni.

Applied to 20 000 simulated studies at the planned sixteen dyads (a true $d_z = 0.73$ effect on all five contrasts; and separately, no effect at all), through the unmodified analysis code:

Table K.1: $n = 16, m = 5, \alpha = 0.05$.

Constructed truth	one contrast, uncorrected	one contrast, Holm-corrected	some contrast in the family
$d_z = 0.73$ on all five	0.777	0.638	0.973
no effect at all	0.052	0.010	0.050

The null row is the check that could fail: a false-positive rate of 0.050 against nominal $\alpha = 0.05$ shows the correction is implemented as specified. The design’s own 80 % power figure is confirmed *uncorrected* (77.7 %) but no contrast will be reported uncorrected; after Holm, recovering the effect on a *named* task falls to 63.8 %, and restoring 80 % on a named task needs twenty dyads, not sixteen. The study is nevertheless well powered for the question it actually asks – whether shared autonomy changes performance on this platform at all, where the family-wise power is 97.3 % – so a per-task null at $n = 16$ must be reported as an interval, not an absence. Trial validity is defined in one shared module that exits non-zero when no trials survive or more than half are excluded; before that existed, a session with every trial invalidated analysed cleanly to empty means and undefined intervals, indistinguishable from a null result on inspection.

Appendix L

The pilot session, in full

[Section 3.4](#) reports the headline pilot numbers. This appendix is the full record: coverage, every session, and the supplementary measures.

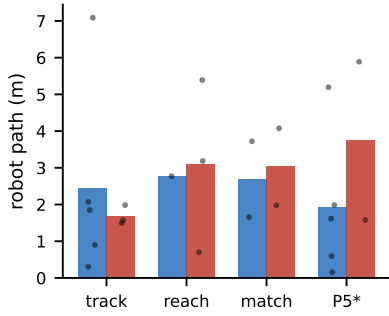
Twenty-three sessions were recorded, sixteen on 2026-09-03/04 for P1–P4 and seven on 2026-09-02 for P5 (recordings/sessions/). Duration, the robot’s end-effector path and lag come from the full-resolution position trace (20 Hz); clutch and emergency-stop counts come from each session’s event log. P5’s sessions predate the per-participant folder-naming convention (generic timestamp names, no task label) and are identified as direct or shared from the `shared_autonomy` flag each session’s own `events.jsonl` logs at start, which – unlike the folder names relied on for P1–P4 – is populated correctly for every one of P5’s sessions. Participant identity is P1–P5 throughout, including in every figure pixel, consistent with `write_manifest()` refusing to record a name ([Section 2.5](#)).

Table L.1: Which participant performed which task under which condition. Coverage is uneven because a pilot session is limited by time. P5’s task was not recorded; all seven of P5’s sessions are one undifferentiated block.

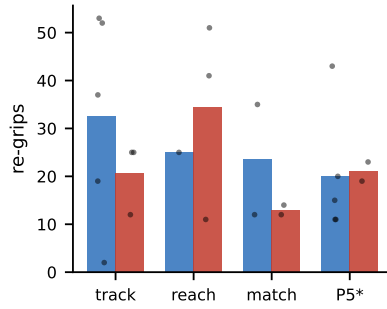
Participant	Object tracking	Target reaching	Position matching	(unlabelled)
P1	both (direct $\times 2$)	—	—	—
P2	both	both (shared $\times 2$)	—	—
P3	direct only	shared only	both	—
P4	both	—	both	—
P5	—	—	—	both (direct $\times 5$, shared $\times 2$)

L.1 Sim-to-real tracking during the pilot

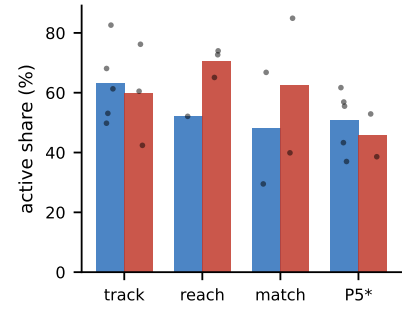
Every pilot session polled the physical Kinova arms’ own joint encoders alongside the simulated joint state, which makes a direct check possible: does the real arm actually follow the simulated one it was commanded through. [Figure L.3](#) is the best-tracking session across all twenty-three (P4, object tracking, direct control, chosen by lowest sim-to-real joint RMS): the real arm’s joint 4 angle follows the simulated one to 1.06° RMS over 3859 samples, with the largest departures during the fastest motion and near-zero error while the arm holds still. This is the same terminal-offset mechanism [Section E.3](#) measures from replayed commands, seen here live during an operator session rather than in an offline replay – **it is evidence the sim-to-real bridge was live and tracking throughout the pilot**, not a claim that any pilot result was validated against a loaded arm, which requires the arm to be carrying and moving an object rather than only mirroring a joint angle.



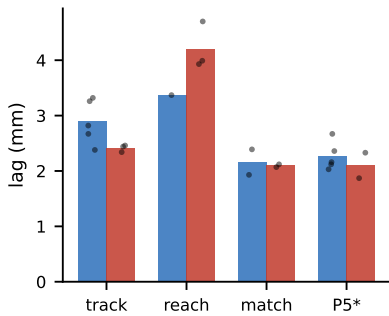
(a) robot end-effector path



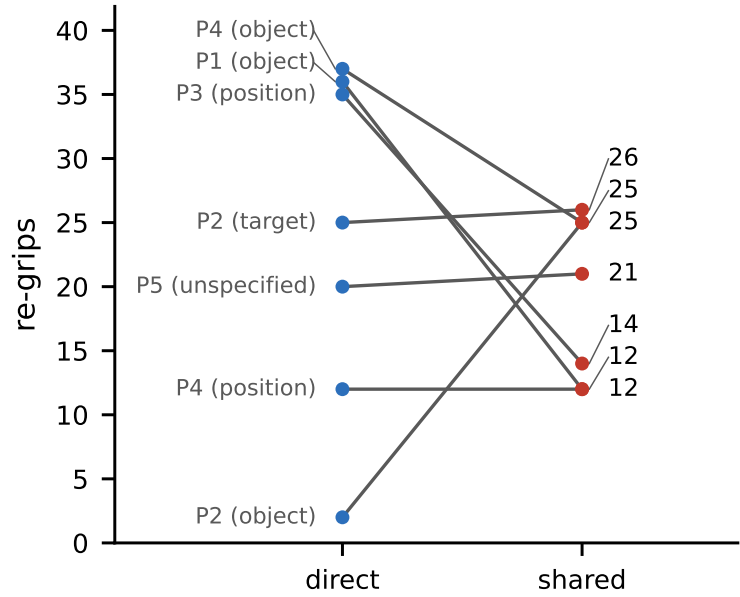
(b) re-grips



(c) active share



(d) control lag



(e) re-grips, paired within operator and task

Figure L.1: The supplementary pilot measures, by task and condition. Lag is essentially unchanged between conditions; the robot's end-effector path (a) is longer under shared autonomy in three of four task groups, consistent with the paired result in [Section 4.4](#) – and it is the *robot's* path: the operators' own controller paths differ by 2 % between conditions (29.9 m direct, 30.5 m shared). Both are drawn from the position trace rather than a stopwatch, which is why they are read as more reliable than session duration in a pilot where duration is confounded with setup time.

Table L.2: Every recorded session. Duration includes setup and hesitation. Active share is the fraction of the session spent with the grip engaged. P1’s two direct-control sessions are a short warm-up followed by the recorded attempt, both retained as separate points.

P	Task	Cond.	Duration (s)	Travel (m)	Re-grips	Active
P1	Object tracking	Direct	147.4	0.31	19	50 %
P1	Object tracking	Direct	392.8	1.85	53	53 %
P1	Object tracking	Shared	114.9	1.50	12	61 %
P2	Object tracking	Direct	38.7	0.90	2	61 %
P2	Object tracking	Shared	169.6	1.57	25	76 %
P2	Target reaching	Direct	227.9	2.77	25	52 %
P2	Target reaching	Shared	78.6	0.70	11	65 %
P2	Target reaching	Shared	357.9	5.39	41	73 %
P3	Object tracking	Direct	617.4	7.09	52	83 %
P3	Target reaching	Shared	458.1	3.18	51	74 %
P3	Position matching	Direct	315.9	3.72	35	67 %
P3	Position matching	Shared	196.8	4.08	14	85 %
P4	Object tracking	Direct	195.6	2.07	37	68 %
P4	Object tracking	Shared	186.9	1.98	25	42 %
P4	Position matching	Direct	171.9	1.65	12	30 %
P4	Position matching	Shared	111.7	1.98	12	40 %
P5	(unlabelled)	Direct	262.5	1.99	20	37 %
P5	(unlabelled)	Shared	250.0	1.58	23	39 %
P5	(unlabelled)	Direct	359.5	5.19	43	62 %
P5	(unlabelled)	Shared	265.8	5.89	19	53 %
P5	(unlabelled)	Direct	207.8	1.62	15	43 %
P5	(unlabelled)	Direct	93.0	0.16	11	56 %
P5	(unlabelled)	Direct	99.3	0.59	11	57 %

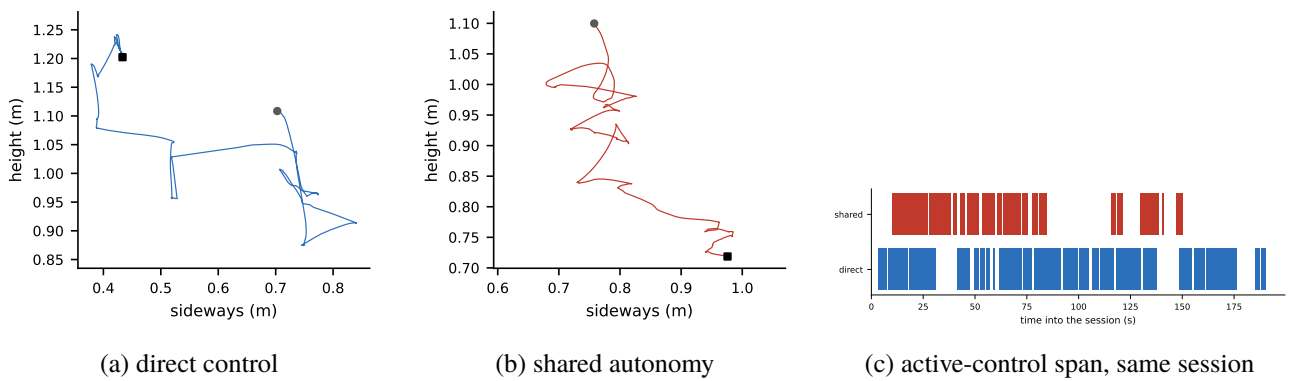


Figure L.2: The actual recorded end-effector path (`ee_left_x/y/z` in `trail.csv`) for one participant on the same task under both conditions, recorded back to back, with the corresponding span of active control alongside. Neither path is smooth; a shorter, straighter line would indicate a more direct approach to the target. Gaps in the engagement panel are the operator releasing the grip to reposition their hand before re-engaging.

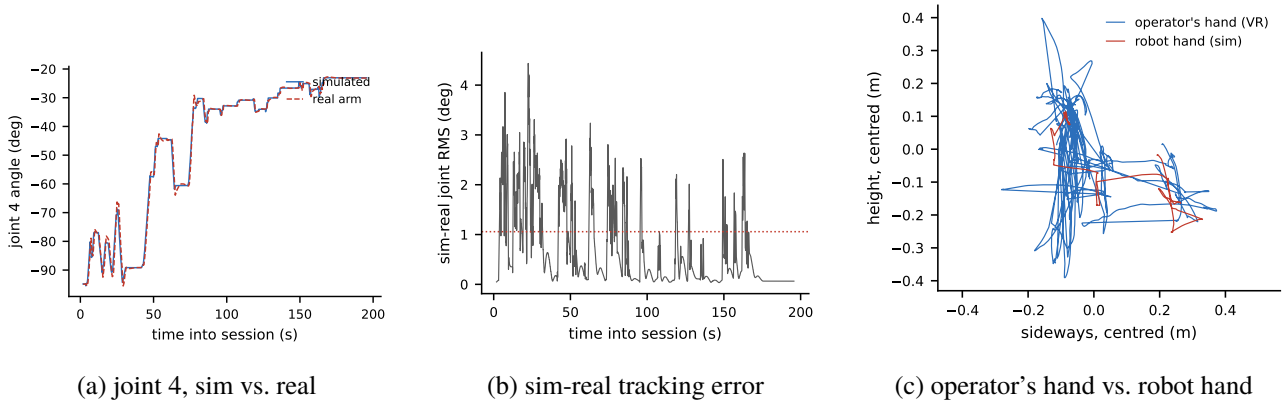


Figure L.3: The best-tracking pilot session by sim-to-real joint RMS. (a) One joint, simulated command against the real arm's own encoder, over the session. (b) The resulting tracking error, root-mean-square over all seven joints; the dotted line is the session RMS, 1.06° . (c) The operator's raw VR-hand path against the simulated robot hand's achieved path, each centred on its own mean – the two live in different coordinate frames and at different scales by design, so an absolute overlay would not be meaningful; centring each on itself is the honest way to compare their shapes.

L.2 The master-arm cohort

Three further operators (M1–M3, disjoint from P1–P4) attempted the same protocol from the instrumented mannequin master rather than VR controllers, across sixteen further recording attempts on 2026-09-02 and 2026-09-04. **Seven of those sixteen are confirmed empty:** `ros2 bag info` reports zero messages, or no bag metadata at all, for every attempt at the multimeter task and both of M1's target-reaching attempts – the same recording-pipeline failure this project's own session index for that day logged independently (eight of twenty-six recordings clean, ten with no data at all). The nine that did record are real sessions, 72 to 304 s each, tens of thousands of messages.

Table L.3: The nine recorded master-arm sessions. Joint RMS is the sim-to-real tracking error, computed the same way as Fig. L.3, over the active arm's seven joints; see Fig. L.4.

M	Task, condition	Duration (s)	Master path (m)	Joint RMS (deg)
M1	Object tracking, direct	154.1	21.44	21.2
M1	Object tracking, direct	145.8	14.11	15.9
M1	(unspecified), shared	209.8	22.02	16.3
M2	Object tracking, direct	110.9	10.85	2.2
M2	Object tracking, shared	71.7	7.57	1.9
M2	Target reaching, direct	303.8	20.80	1.8
M2	Target reaching, shared	83.6	4.19	0.4
M2	Target reaching, shared	206.4	11.60	3.0
M3	Target reaching, direct	254.1	11.48	1.8

M1's tracking error is not noise: it is consistent across all three of that operator's sessions and separated from M2 and M3 by a factor of five to fifty. The mechanism this project's own channel-health measurement predicts (Chapter H) is a session that leans on the incoherent left-arm channels rather than the coherent right-arm ones – all three of M1's sessions used the *right* arm as the active hand (Table L.3), so a simple left/right split does not explain it on its own, and which specific channels M1's sessions exercised has not been isolated. It is reported here as a finding this pilot surfaced, not a diagnosis: the master-arm modality can apparently produce sim-to-real

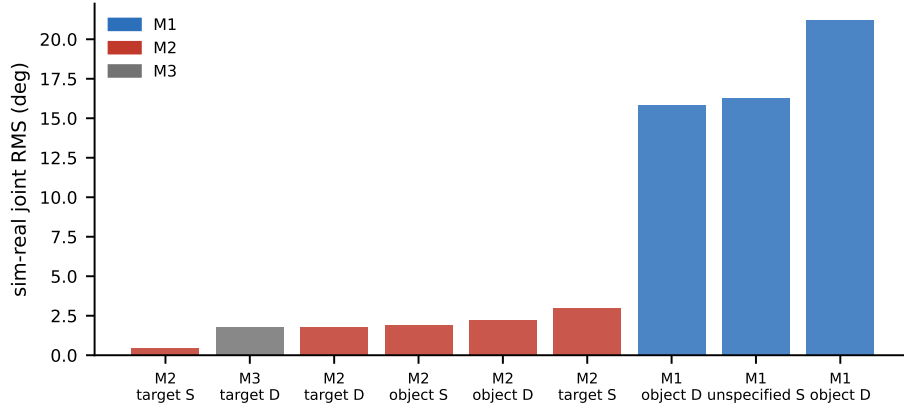


Figure L.4: Sim-to-real joint tracking error for every recorded master-arm session, sorted low to high. M2 and M3 track within 3° throughout; **every one of M1’s three sessions tracks an order of magnitude worse** (15.9 to 21.2°).

tracking an order of magnitude worse than either VR teleoperation or another operator on the same hardware, on the same day, and the pooled VR-only comparison of [Section 3.4](#) would have missed it entirely.

Limits. Five participants with uneven task coverage is a pilot-scale sample: no significance test is reported, and none of the differences above should be read as a proven effect. The pooled figures mix three tasks together; the paired panels are the part of this pilot closest to the paired design of [Chapter J](#). Session duration runs from the start of the recording to the end, including any explanation or hesitation, so it is a session-length measure rather than a pure motor-execution time. Every session’s emergency stop was engaged once, near the end, to stop the recording – a procedural marker rather than a safety incident – and is excluded throughout. This pilot has no separate wearer: every measure here is what the operator experienced, and the wearer-side hypotheses of [Chapter J](#) (H3, H4) remain untested.

Appendix M

Background, in full

[Section 1.1](#) states the gap this project addresses. This is the full review behind it, organised by theme, with each citation's scope stated precisely rather than implied.

M.1 Supernumerary robotic limbs

The field's founding demonstrations are Parietti and Asada's bracing limbs for aircraft fuselage assembly, later extended to whole-body support [PA16], which anchor against the environment so the reaction load leaves the wearer's body rather than passing through it (Fig. M.1). In the arrangement studied here the arms are free in space, so the chain closes through the wearer instead – and every reaction load is carried by a person who is not the person commanding the motion. This difference, not the control law, is what makes wearer clearance the binding constraint of [Chapter I](#).

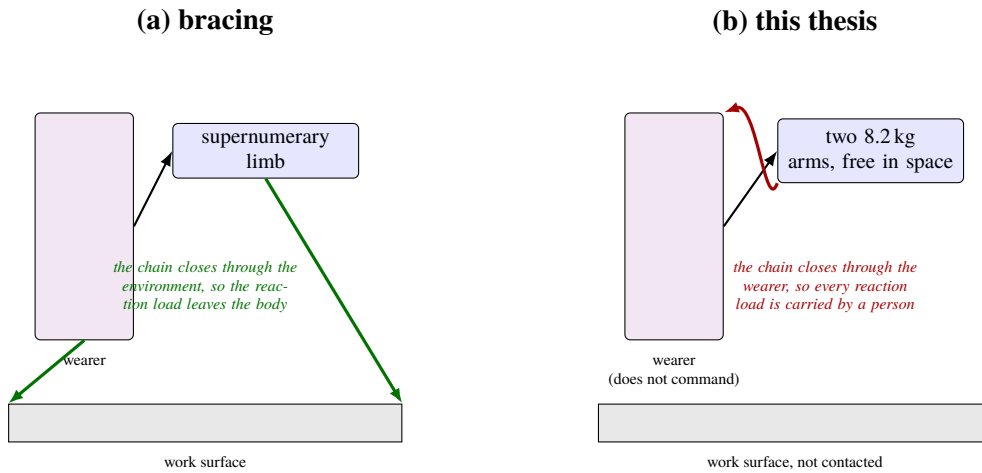


Figure M.1: Bracing (the field's founding configuration [PA14]) against free-space mounting (this platform).

Reviews since map the design space along consistent axes – mounting location trades reach against the moment arm imposed on the wearer, actuation trades bandwidth against worn mass [YHC⁺21, P⁺21, TL21, Z⁺23] – and a recent quantitative survey adds that the field reports embodiment far more often than task performance, and rarely both [Z⁺26]. Worn mass is the constraint reported least: each Gen3 masses 8.2 kg, the pair with harness exceeds 17 kg, and simulated wearer loads suggest the biomechanical cost is not negligible [MK24].

M.2 The operator and the wearer as different people

Fusion mounts a robotic head and two arms on a worn frame, driven by a remote operator [SSM⁺18]. Two qualifications travel with that citation: its three modes (Direct, Induced, Enforced) are levels of *physical coupling* between the robot arms and the wearer’s own, not levels of machine autonomy – in all three the robot decides nothing – and the publication is a two-page demonstration with no user study and no performance numbers. It establishes that the architecture existed; it cannot support a claim about how well it worked.

Work on spatial guidelines for near-body interaction ran a concealed operator, varied apparent autonomy, and measured the wearer’s physiological arousal and self-reported trust [ZGD⁺26]: higher apparent autonomy made both worse. Their autonomy was a hidden human and their wearer had no task of their own, so the result does not settle what happens when autonomy is real and the wearer is occupied – it establishes only that the wearer’s experience is not a free variable. Disturbance rejection for the two-person case is partly solved: Zhang et al. compensate human-induced base motion and report end-effector errors of (1.37 ± 0.58) mm [Z⁺24], with the human shoulder as a floating base – covering the case where wearer and operator are the same person (the base motion is partly predictable to the controller because the person commanding it also caused it), not the case studied here.

M.3 Master interfaces for dissimilar kinematics

Guggenheim and Asada argue for miniature physical masters on the grounds that geometric similarity supplies feedback inherently, without a separate haptic display [GA21]. The argument covers similarity of *form*; this system’s master-to-slave reach ratio is roughly one to three, which is where it departs. Joint-space correspondence preserves posture but needs every master joint measured accurately; task-space correspondence decouples the chains at the cost of the slave’s redundant degree of freedom (Chapter B). Pinardi et al. review supplementary feedback in movement augmentation and find that feedback improving control does not reliably improve the sense of ownership [PLF⁺23] – a caution against treating either as a proxy for the other.

M.4 Shared autonomy

Dragan and Srinivasa formalise arbitration as policy blending: a predicted user goal and the user’s actual input combined with a weight set by prediction confidence [DS13], explicit that blending aggressively on a wrong prediction is worse than not blending. Javdani et al. recast the problem as a POMDP over the user’s goal, approximated by hindsight optimisation [JSB15]: their user study found assistance that does not wait for high confidence lets operators finish faster with less input. Neither line of work was developed for a robot mounted on a second person – there, the cost of assisting toward the wrong goal is a manipulator moving toward someone who did not command it, not a slower trial.

M.5 Measuring embodiment, workload and trust

The rubber hand illusion showed ownership can be induced over an object that is plainly not one’s hand [BC98]; Longo et al. decompose the resulting experience into ownership, agency and location, which can move independently [LSK⁺08] – an operator may feel authorship of a motion without feeling the arm is part of them, and a wearer may feel neither while bearing the consequences. Workload is conventionally the NASA Task Load

Index [[HS88](#)]; for a two-person system it must be administered to both, since the two will not load the same way. Physical exertion is better tracked with a Borg category-ratio scale [[Bor82](#)]. ANSI/ISA-101.01's central principle for operator displays [[Int15](#)] – colour is a scarce alarm channel, so a normal state should be largely uncoloured – transfers directly to a robot console.

M.6 The gap, restated precisely

Fusion established the architecture and measured nothing. Shared autonomy raises operator performance where the robot is attached to nobody. Raising apparent autonomy lowers a wearer's trust where the autonomy is a concealed human and the wearer has no task. What has not been done is to measure both ends in one system, under the same autonomy, in the same trial – the engineering contribution of this thesis is a system in which that question could be asked; [Chapter 4](#) sets out what still stands between the system as built and the full session that would ask it.

Appendix N

Discussion, extended

Chapter 4 is the argument. This appendix adds the detail that did not fit there.

N.1 What the degradation architecture bought, and what it did not

The degradation architecture kept the system running and its failures visible: with eight of fourteen channels failing ([Section H.2](#)), the right arm retains full position sensing and the left retains direction, and the incoherent verdict exists because a failing potentiometer does not go quiet – range alone calls a wide-spanning nonsense signal healthy. What it did not buy is capability: freezing the left arm’s bend joints collapses its commanded set from a shell to a surface, and no health scheme restores a measurement never taken. The reduced-degree-of-freedom premise succeeded exactly where a single sensor observes the quantity directly (elevation from gravity, reach from a forward-kinematic magnitude) and failed where it does not: the spherical model assumes azimuth and elevation are independent, and they are not, because the first joint’s axis does not tilt as the arm bends – a design rule for a successor, not a defect to patch with better calibration.

N.2 The clearance model itself is not yet trustworthy

The most uncomfortable single finding of the hardware programme belongs here rather than in the results. Asked about a pose in which the arm was *physically against the mannequin*, one clearance model – the one that samples link *origins* rather than link geometry, and which runs on the physical arm during homing – returned 367.0 mm, the same value it returns for a visibly clear pose. Every clearance figure in this report computed through the guard’s capsule model stands (the majority); but the standing claim that wearer margin is enforced *at all times* was, at the time of writing, **not enforced in code on hardware**. This is the one item in [Chapter 5](#)’s further-work list that is a safety defect rather than an incompleteness, and it is placed above the mount change and above the study for that reason.

N.3 Three things the real cameras changed

Beyond the headline numbers of [Chapter F](#): the surface-fitting stage transferring to hardware within a factor of three of its simulated figure is the one component of the perception chain now known to work outside simulation, and the one the pick path leans on hardest. The scene camera is a *different* instrument from the wrist cameras, not merely a distant one – an eighth of the depth returns, a twentieth of the plane support – and no autonomous behaviour should expect a support surface from it. And the learned detector contributed nothing the classical

colour-and-geometry path did not; no result in this report rests on a learned detector, worth noting given that a language-model-free, learned-detector-optional design was argued for on cost grounds rather than accuracy grounds.

N.4 Comparison against the planning report, in full

Divergences fall into three groups. *Superseded by measurement*: the haptic objective ([Chapter 4](#)). *Deferred and still open*: coordinated pick-and-place on the real platform – the task set is specified and verified geometrically and the runner exists, but the arms were not available for the period in which it would have run. *Added*: the safety architecture, the degradation architecture, the workspace characterisation and the pilot were not in the plan; the first two follow directly from discovering that sensor faults dominate, the third from asking whether the task set was reachable before running it.

Appendix O

Additional figures from the recordings

Every figure here is drawn from data already described in [Chapters F](#), [H](#) and [L](#); this chapter exists so that a reader can see the whole record rather than the headline. Labels are in plain words. Scripts: `thesis_v3/figures/gallery/*/make*_gallery.py`.

O.1 The pilot, session by session

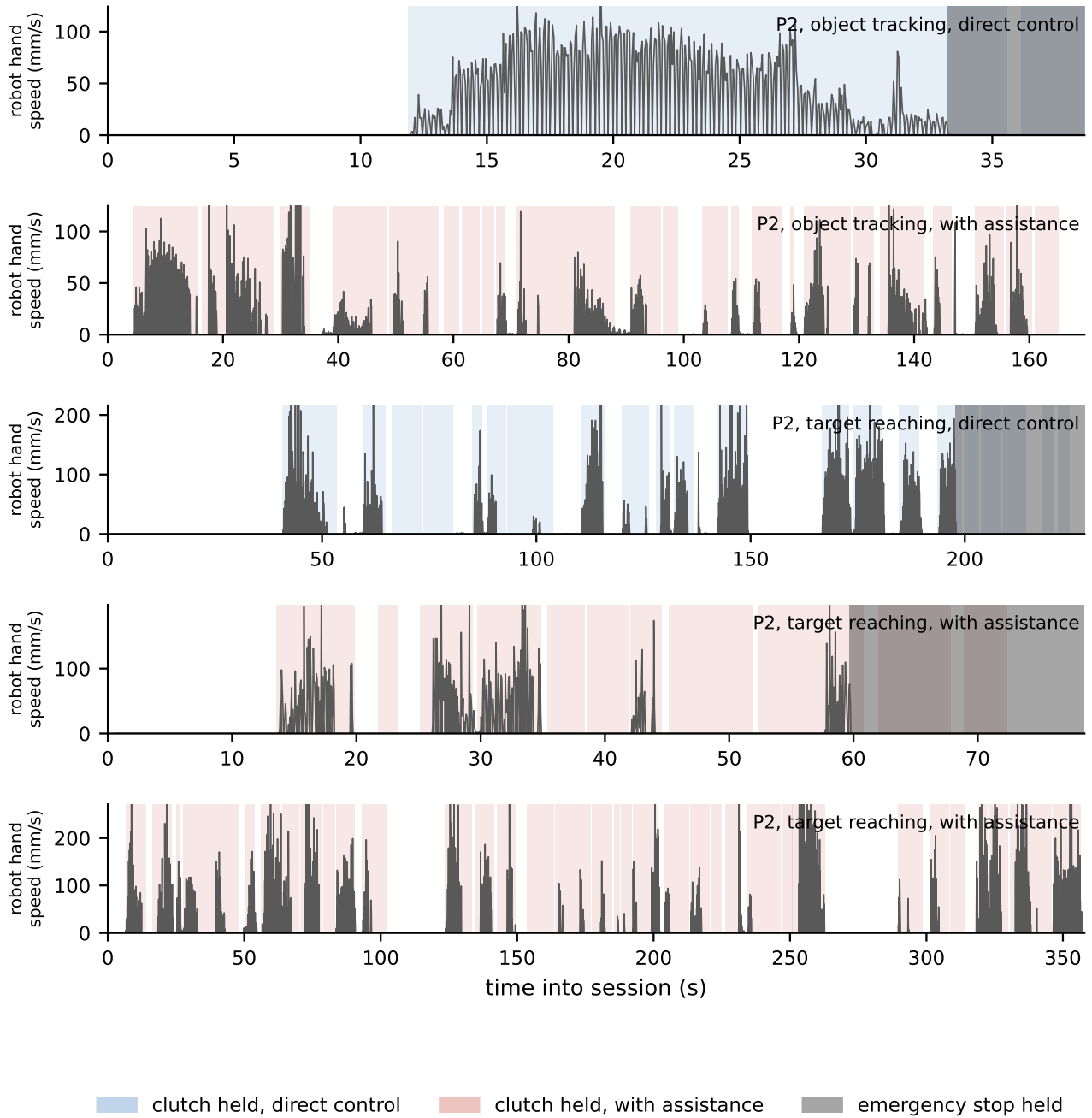


Figure O.1: One operator's five sessions (P2), second by second: shaded where the clutch was held and the operator was steering, the grey line the speed of the robot's hand, black where the emergency stop was held. The other four operators' strips are in the gallery directory.

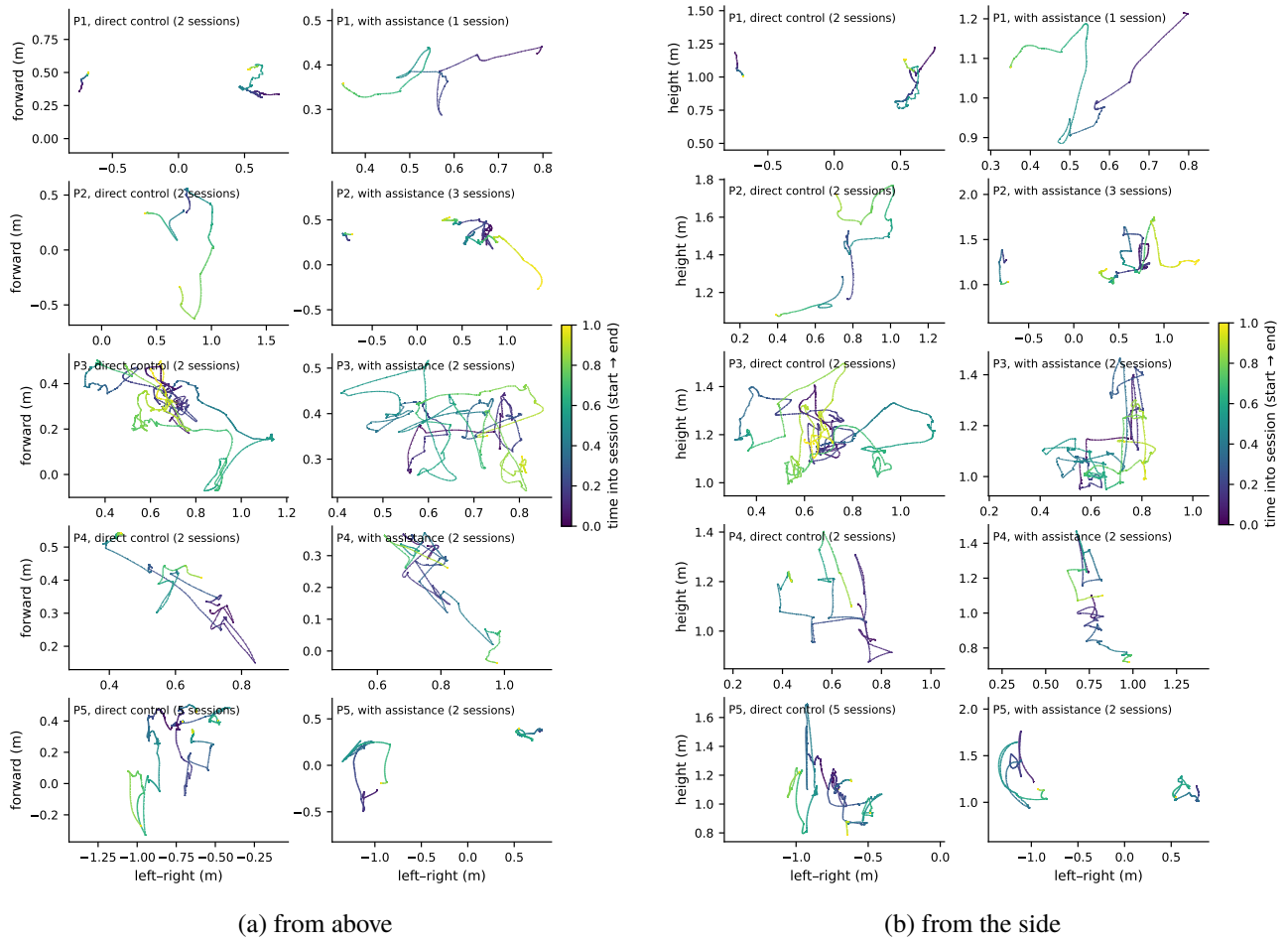


Figure O.2: Where the robot's hand went in every one of the twenty-three VR sessions, coloured from the start (dark) to the end (yellow) of each; one cell per operator and condition.

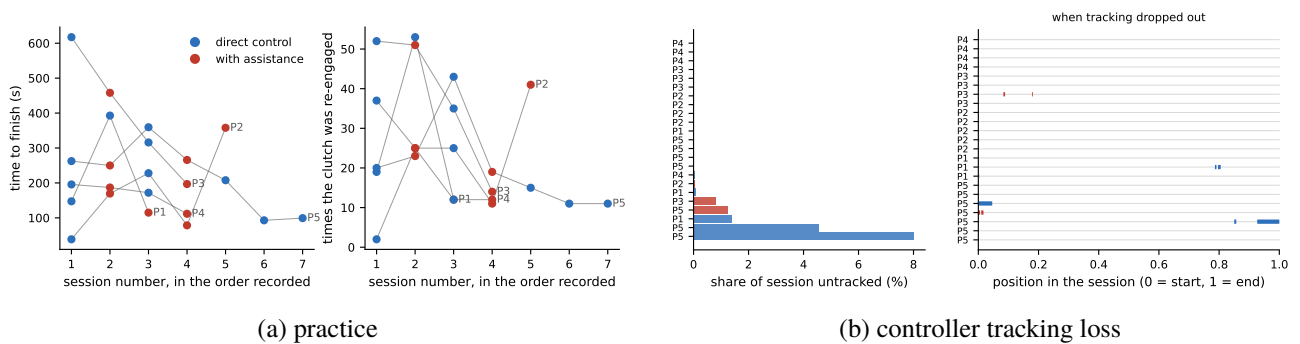


Figure O.3: Left: each operator's sessions in the order recorded; no consistent drift with practice, so order is not obviously driving the direct/assisted difference. Right: the headset lost sight of the controller in a median 0 % of samples, worst 8 %, clustered at session edges.

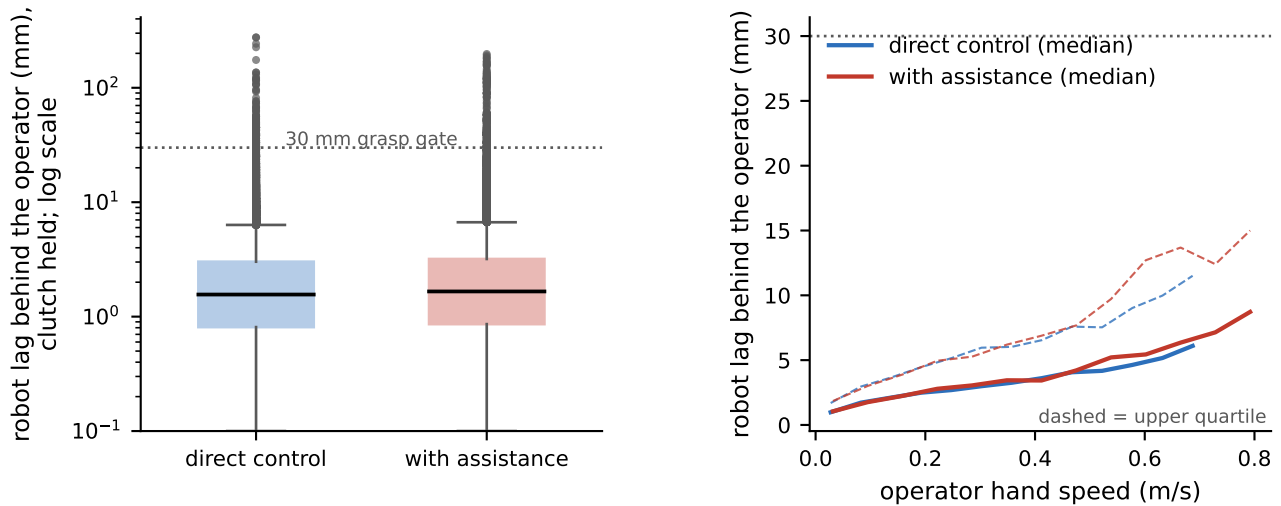


Figure O.4: Robot lag behind the operator while steering: median 1.6 mm direct, 1.7 mm with assistance; past the 30 mm capture gate in 0.5 % and 0.9 % of samples; lag grows roughly linearly with hand speed.

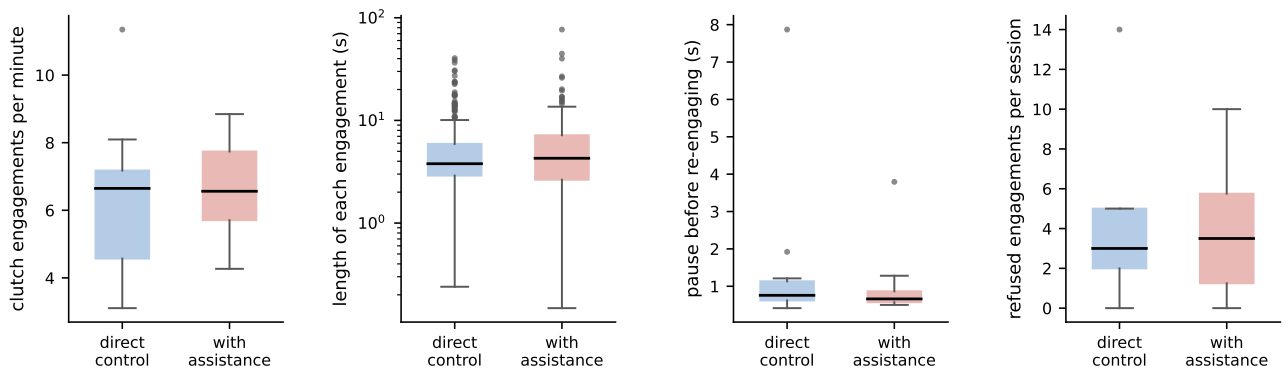
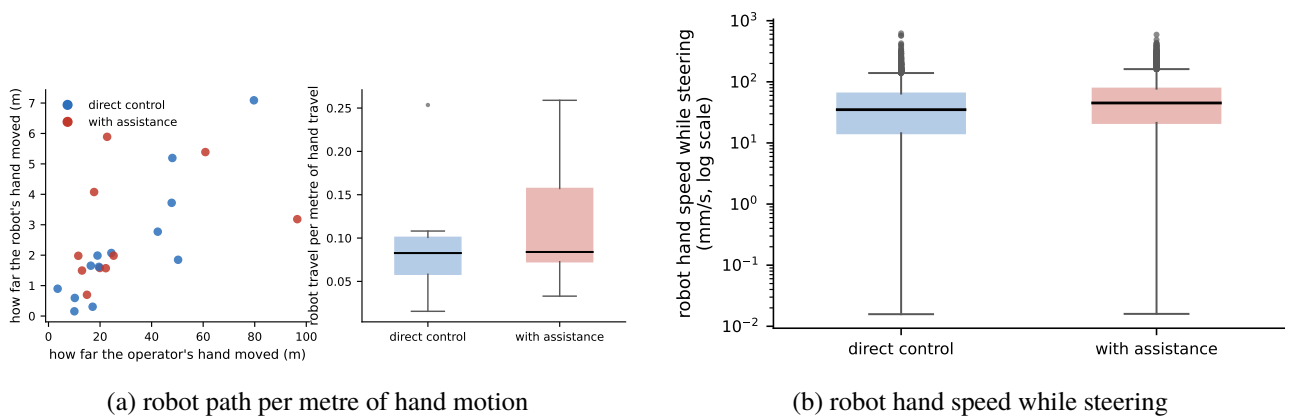


Figure O.5: Clutch use per session: engagements per minute (median 6.6 under both conditions), how long each engagement was held (3.8 s direct, 4.3 s assisted), the pause before re-engaging, and refused engagements (3.8 per session under both).



(a) robot path per metre of hand motion

(b) robot hand speed while steering

Figure O.6: Left: the operator's controller covers 10 to 50 m per session (raw motion, jitter included) while the robot's hand covers 0.2 to 7 m; the ratio is the same under both conditions. Right: the robot's hand moves at similar speeds under both conditions (medians 35 and 45 mm/s) – assistance lengthens the path, it does not speed the robot up.

O.2 Sim-to-real tracking, every session and every joint

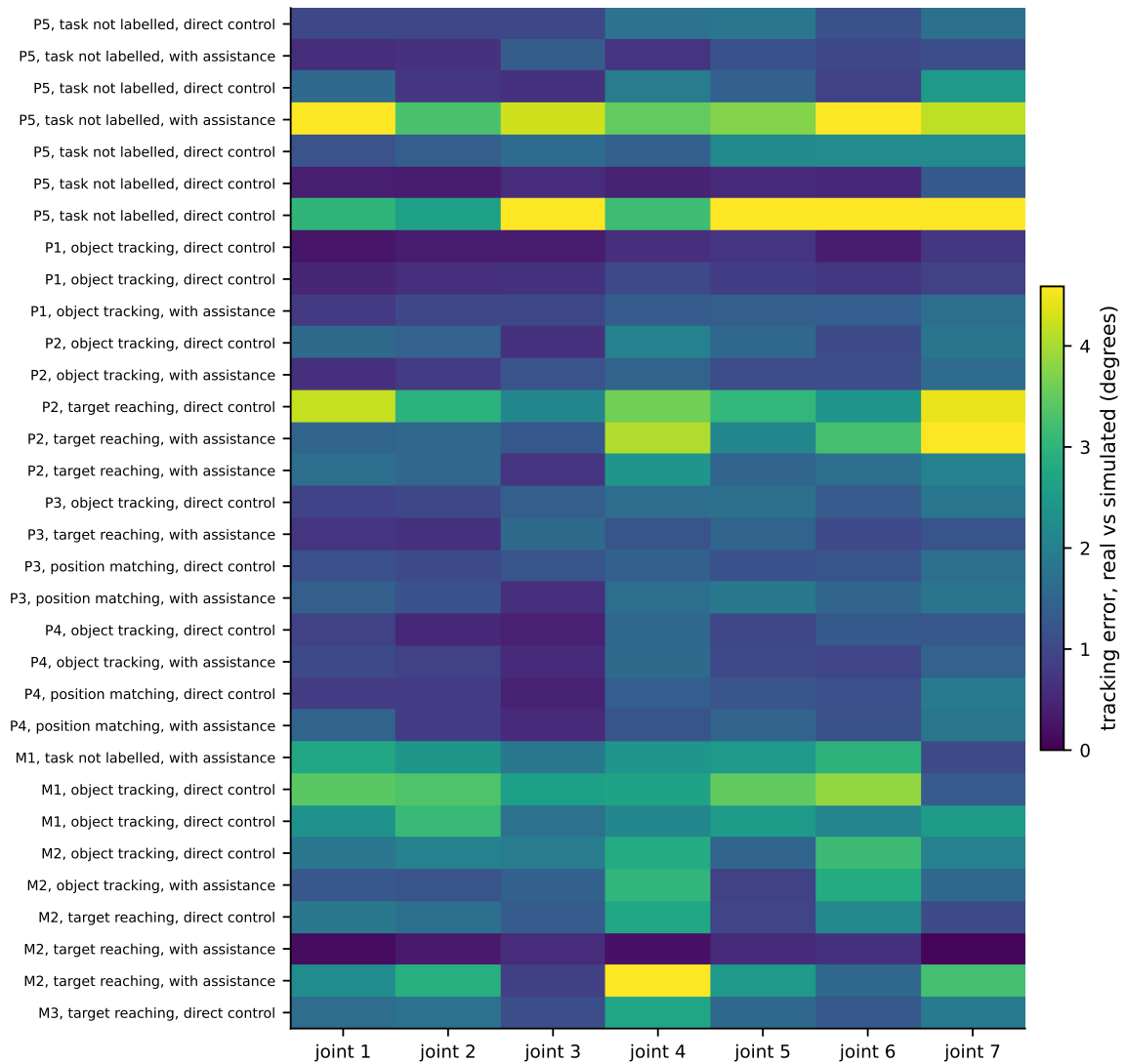


Figure O.7: How closely the real arm followed the simulated one, every session (rows) by joint (columns), both cohorts. Median per-joint error 1.44° ; the three M1 rows and two P5 rows light up on several joints at once, the signature of a bad input channel rather than a bad joint; every P1–P4 session stays under 4.6° on every joint.

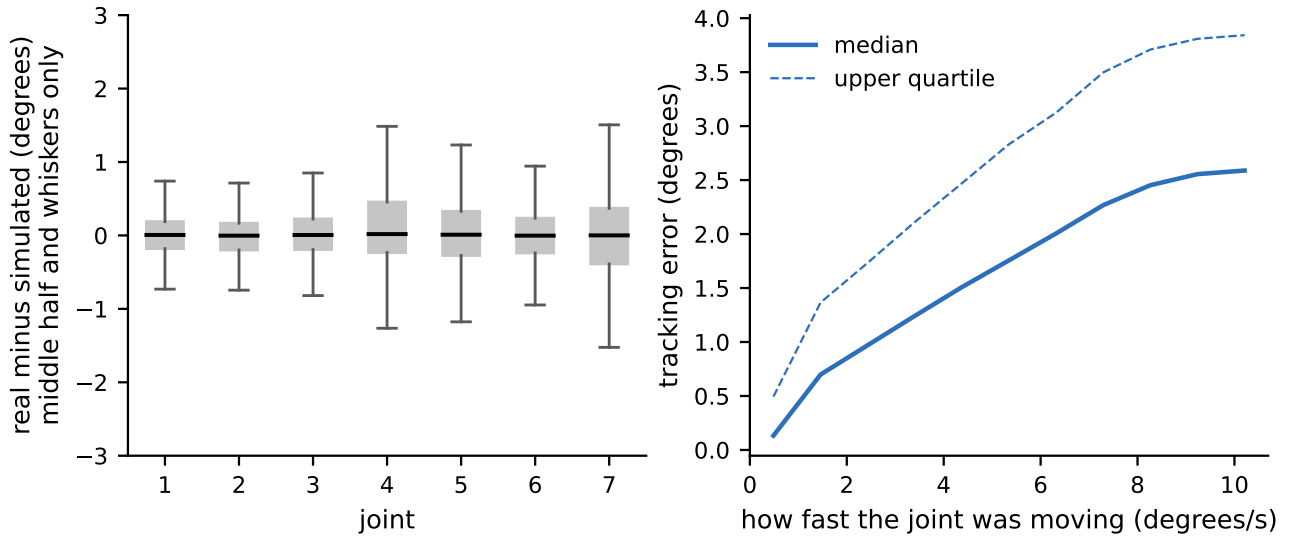


Figure O.8: Pooled over the VR cohort: the error is centred on zero for every joint and grows with joint speed (median 0.13° at rest, 2.59° in the fastest bin) – the real arm lags a moving command rather than sitting off a still one.

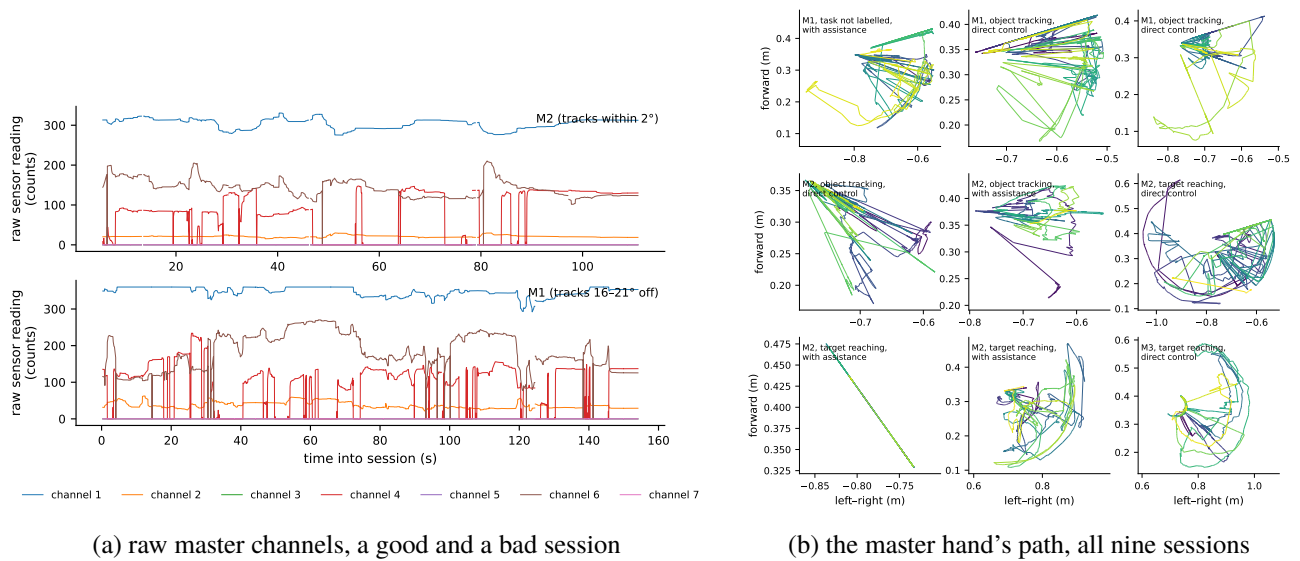


Figure O.9: Left: the seven raw sensor channels of one M2 session (tracking error 2.2°) beside one M1 session (21.2°); a channel that traces a trajectory is alive, one that sits flat or jumps between levels is the defect the health verdict names. Right: M1's hand paths are wide, jagged sweeps; M2's and M3's are compact.

O.3 The scripted task set, in pictures

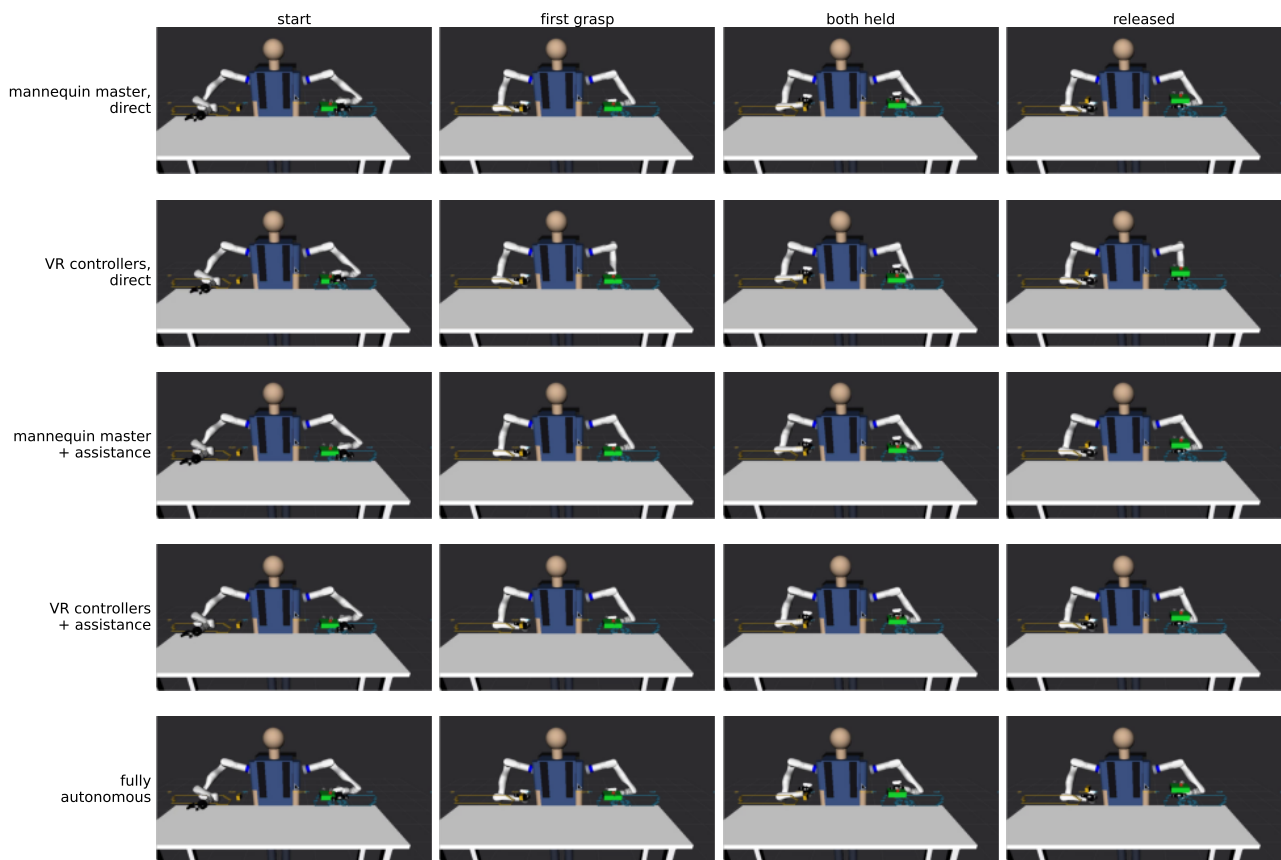


Figure O.10: The circuit-box-and-multimeter task under all five modes at four moments (simulation). Both arms lift the box and bring the meter to it in every mode; the runs differ in timing – grasps at 4.4 to 4.8 s in four modes, 37.0 s through the VR path – not in outcome.

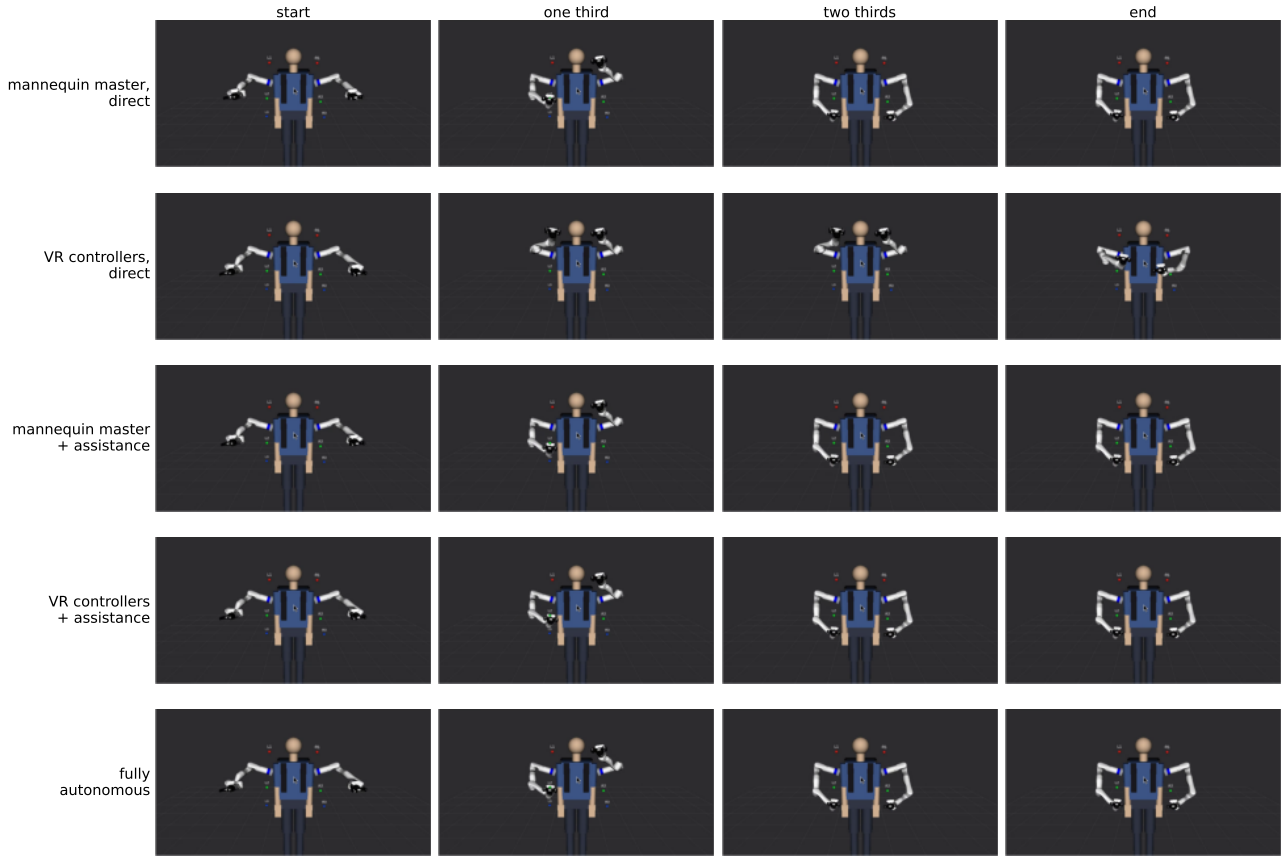
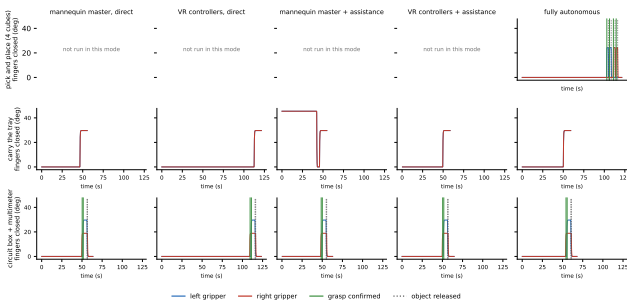
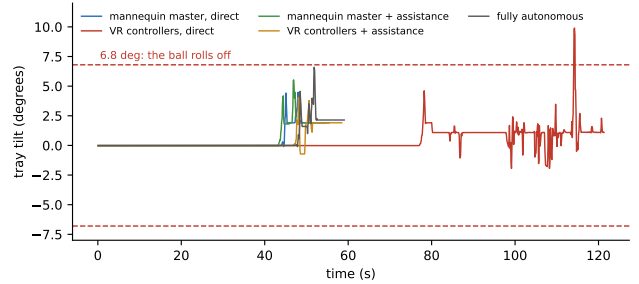


Figure O.11: Reaching three targets under all five modes, at the start, a third, two thirds and the end of the run (simulation). 16.5 s in four modes, 49.1 s through the VR path.



(a) finger angle through every grasping run



(b) tray tilt while carrying

Figure O.12: Left: in every mode the fingers close when the log says an object was grasped (24.2° on a 40 mm cube, 29.7° on the 30 mm objects) and open when it says released. Right: the tray stays within a degree of level in four modes; the VR-direct run spiked to 9.9° for 0.31 s, past the 6.8° roll-off angle.

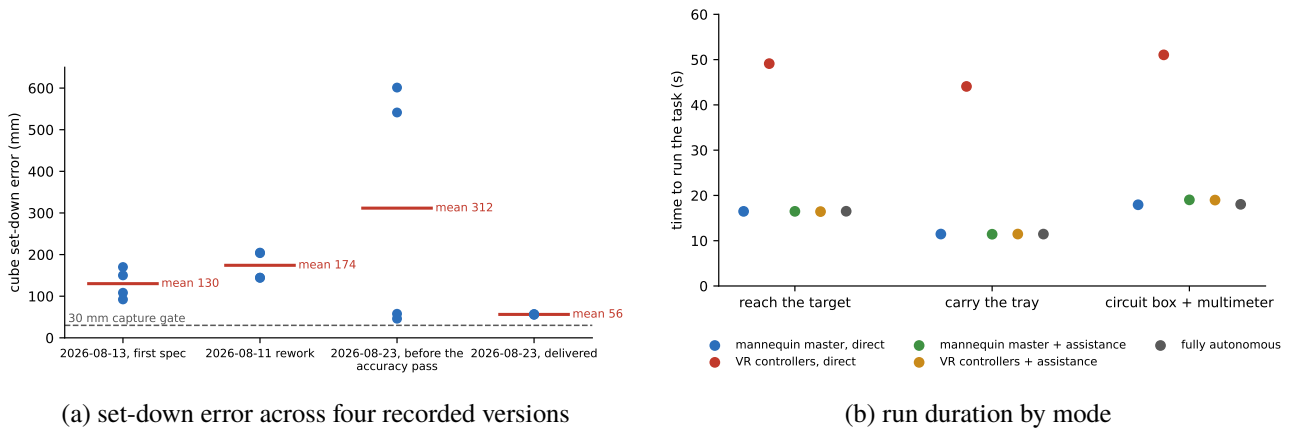


Figure O.13: Left: how far each cube was set down from its pad across the four recorded versions of the pick-and-place – means 130 mm, 174 mm and 312 mm, then 56 mm on every cube. Right: the same scripted task takes about three times longer through the VR-controller path (49 against 16.5 s for the reach task), which is rate-limited by its own filter.

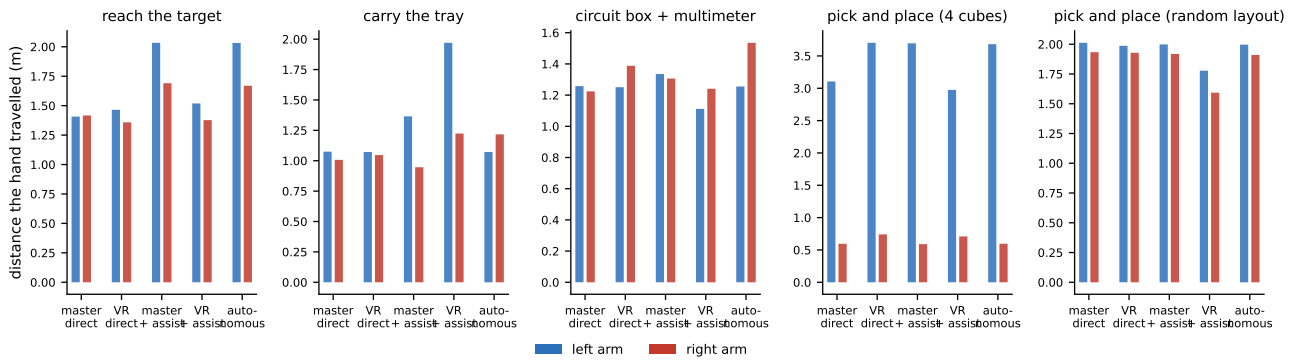


Figure O.14: With no person in the loop, the same commanded waypoints make the arm travel further under the assisted modes: reach task, left arm, 1.41 m direct against 2.03 m with mannequin-master assistance (45 %); 1.46 against 1.52 m for the VR pair (4 %). This is the no-operator baseline behind [Section 4.4](#).

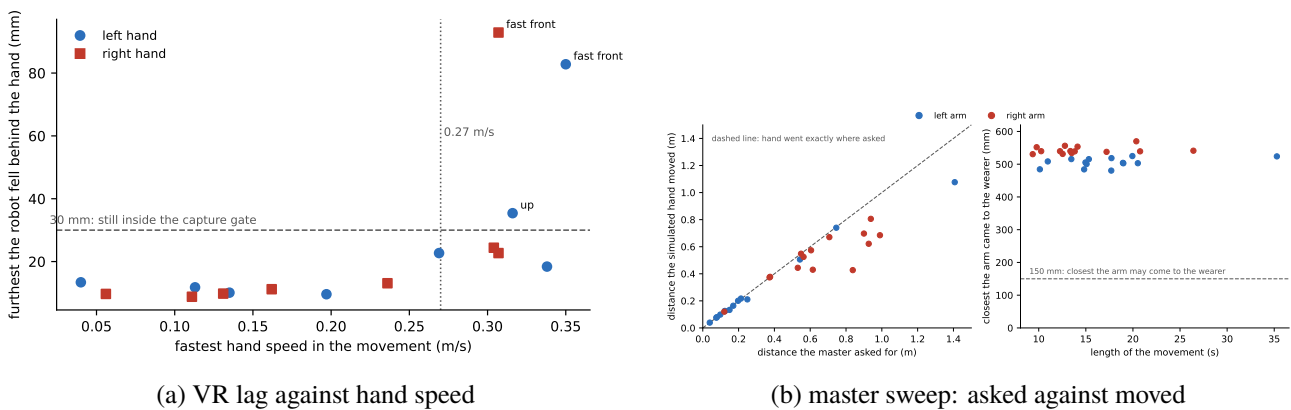


Figure O.15: Left: in a scripted VR protocol (simulation, 2026-08-26) the robot never fell more than 22.7 mm behind a hand moving under 0.27 m/s; a deliberately fast push at 0.35 m/s left it 83 to 93 mm behind, three times the capture gate. Right: twenty-eight set movements of the mannequin master: the left arm moved roughly what was asked, the right undershot on eight of fourteen; the hand never came within half a metre of the wearer.

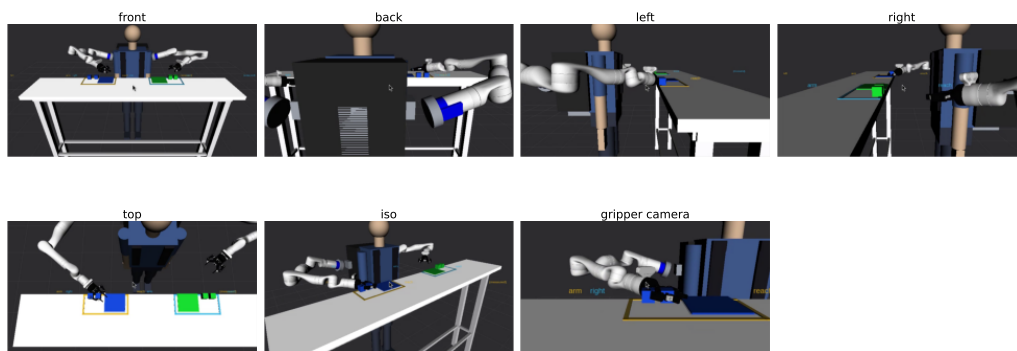


Figure O.16: The first grasp of the autonomous pick-and-place from the seven fixed viewpoints plus the gripper's own camera; every run was filmed this way so a grasp is judged from the pictures rather than asserted.

O.4 Every trajectory, every session

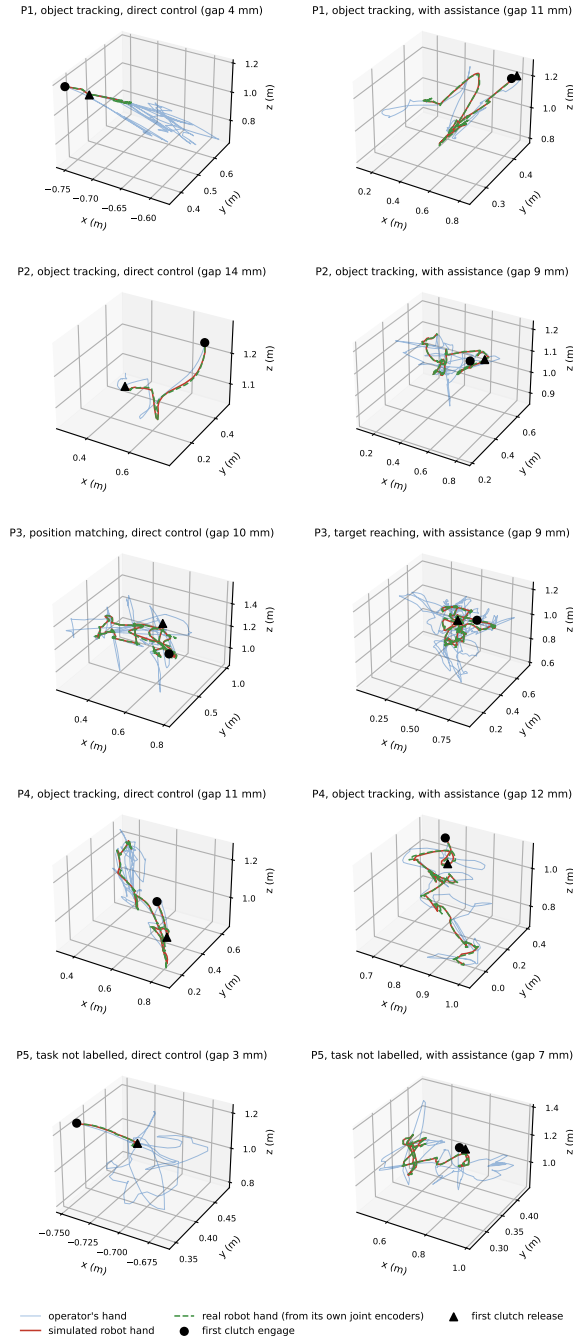
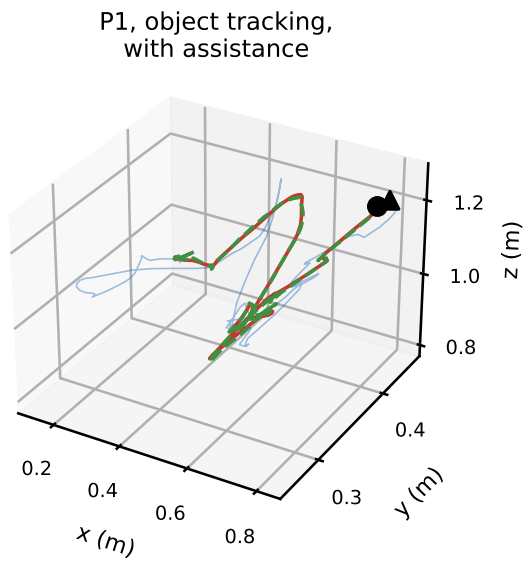
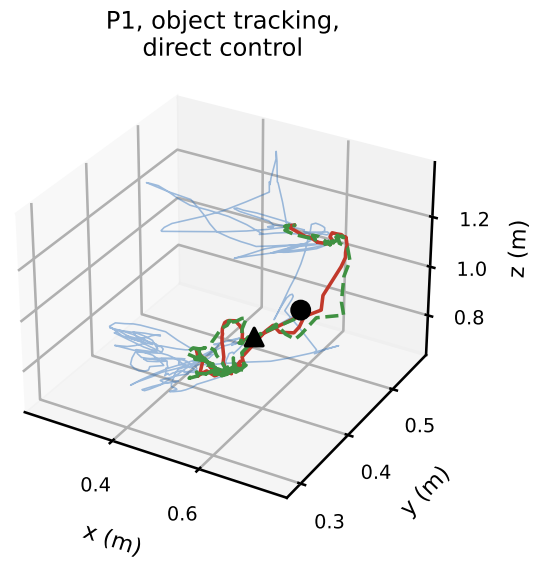
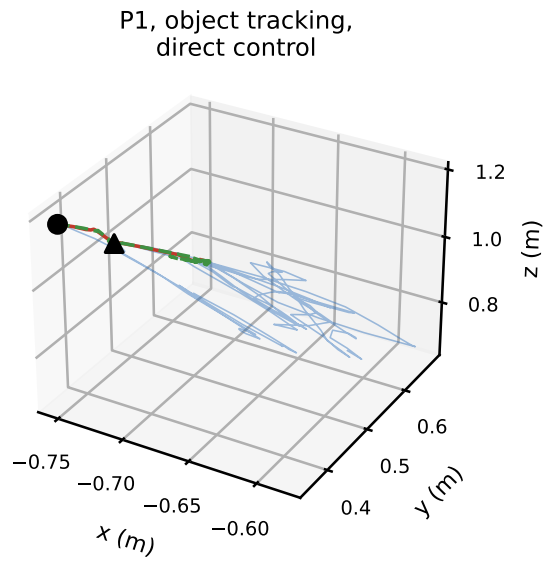


Figure O.17: Each of the five VR operators' cleanest direct (left) and assisted (right) session: the operator's hand (faint), the simulated robot's hand, and the real robot's hand computed from its own joint encoders; the first clutch press and release are marked, and each panel label carries the typical gap between simulated and real hand.



— simulated robot hand - - - real robot hand (from its own joint encoders) ● first clutch engage

Figure O.18: Every recorded session of P1: the operator's hand (faint, smoothed), the simulated robot's hand and the real robot's hand from its own encoders.

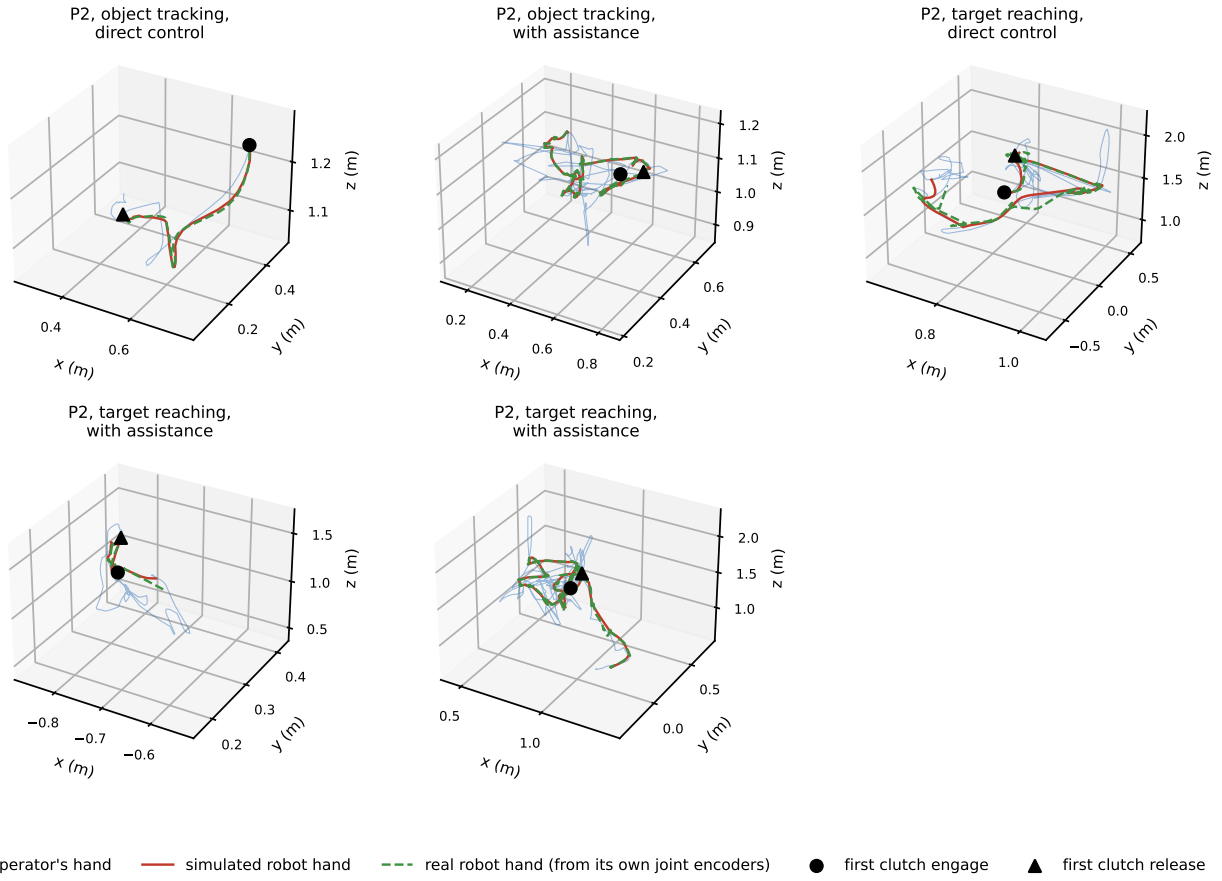


Figure O.19: Every recorded session of P2: the operator's hand (faint, smoothed), the simulated robot's hand and the real robot's hand from its own encoders.

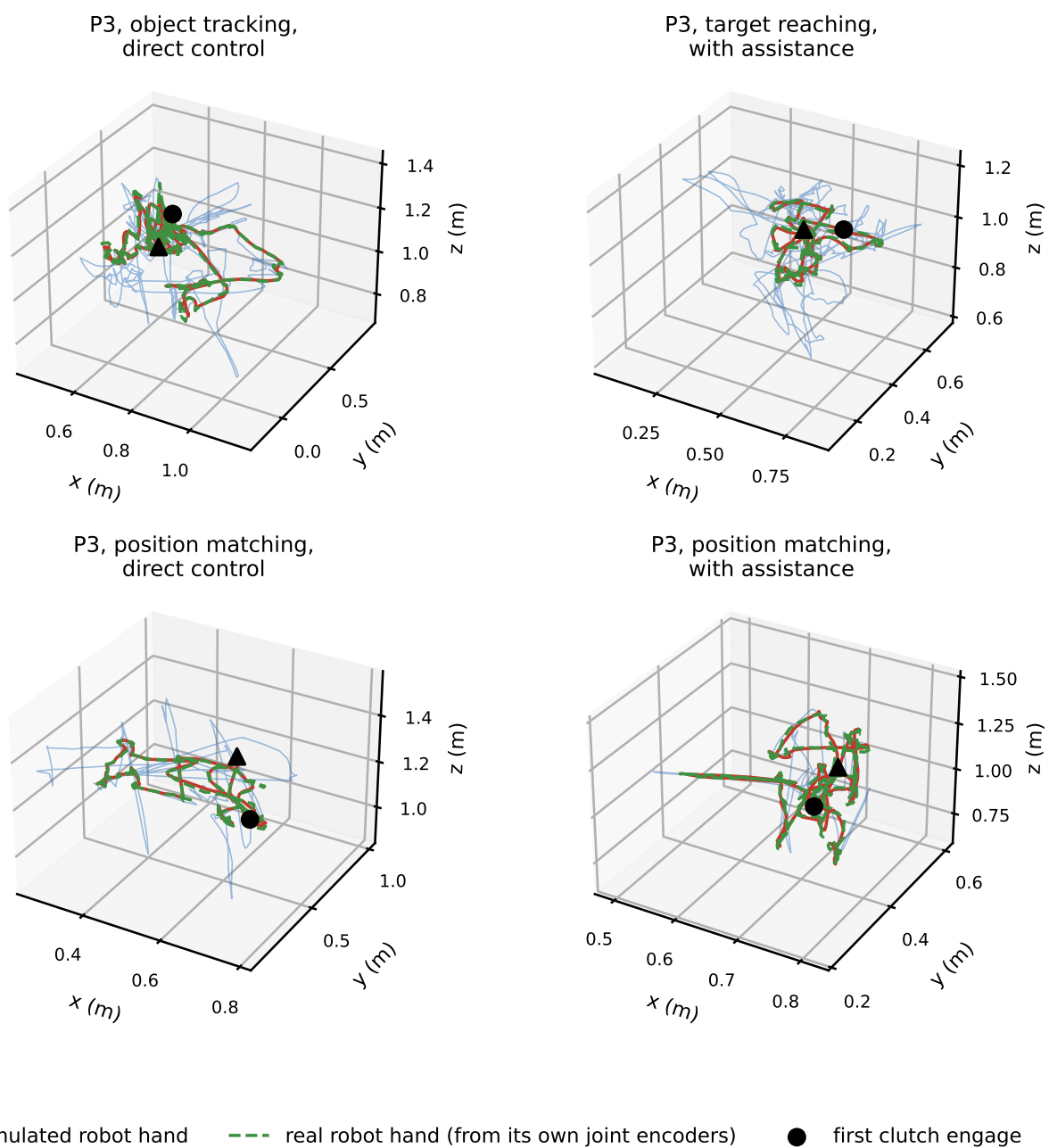


Figure O.20: Every recorded session of P3: the operator's hand (faint, smoothed), the simulated robot's hand and the real robot's hand from its own encoders.

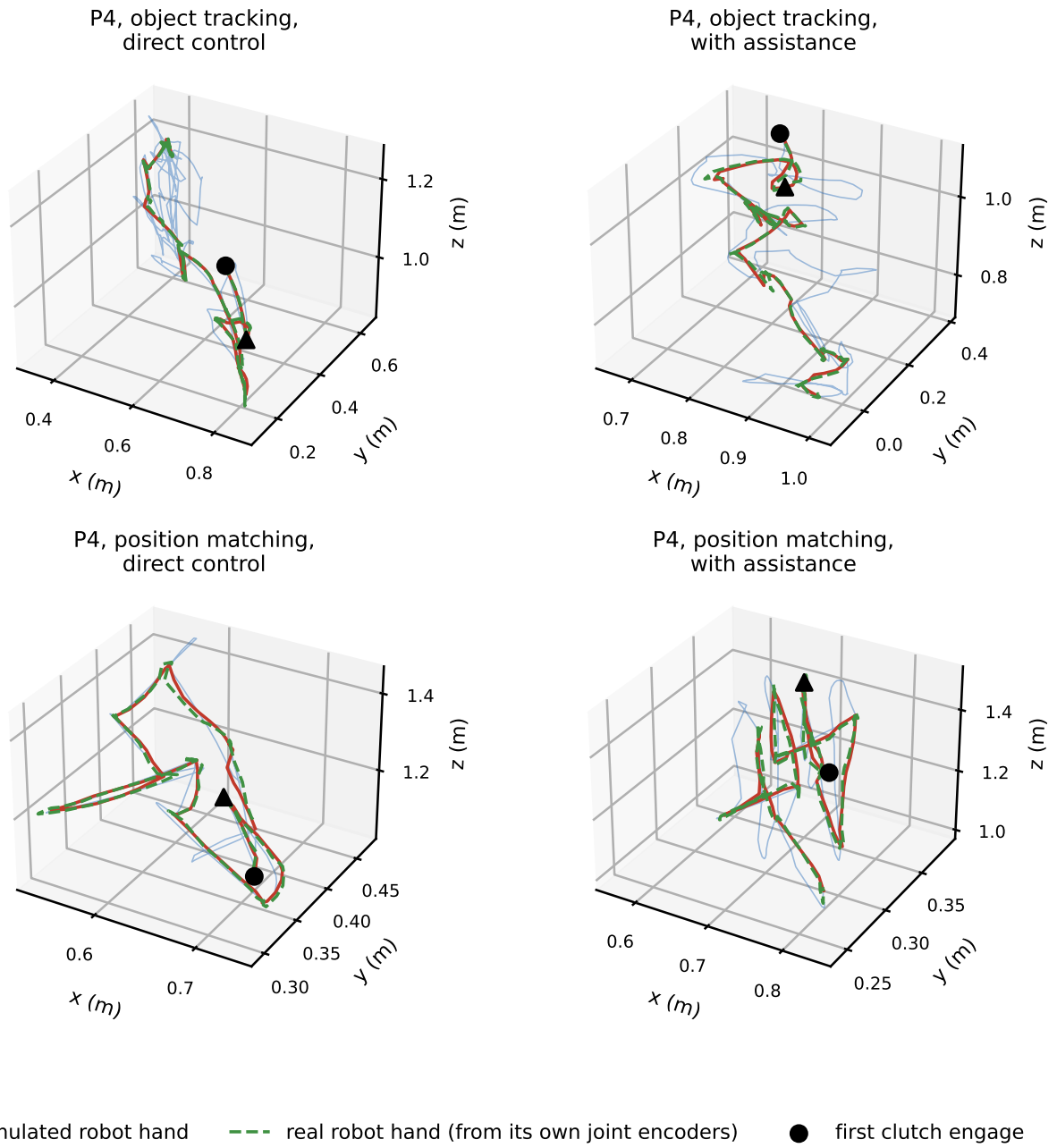


Figure O.21: Every recorded session of P4: the operator's hand (faint, smoothed), the simulated robot's hand and the real robot's hand from its own encoders.

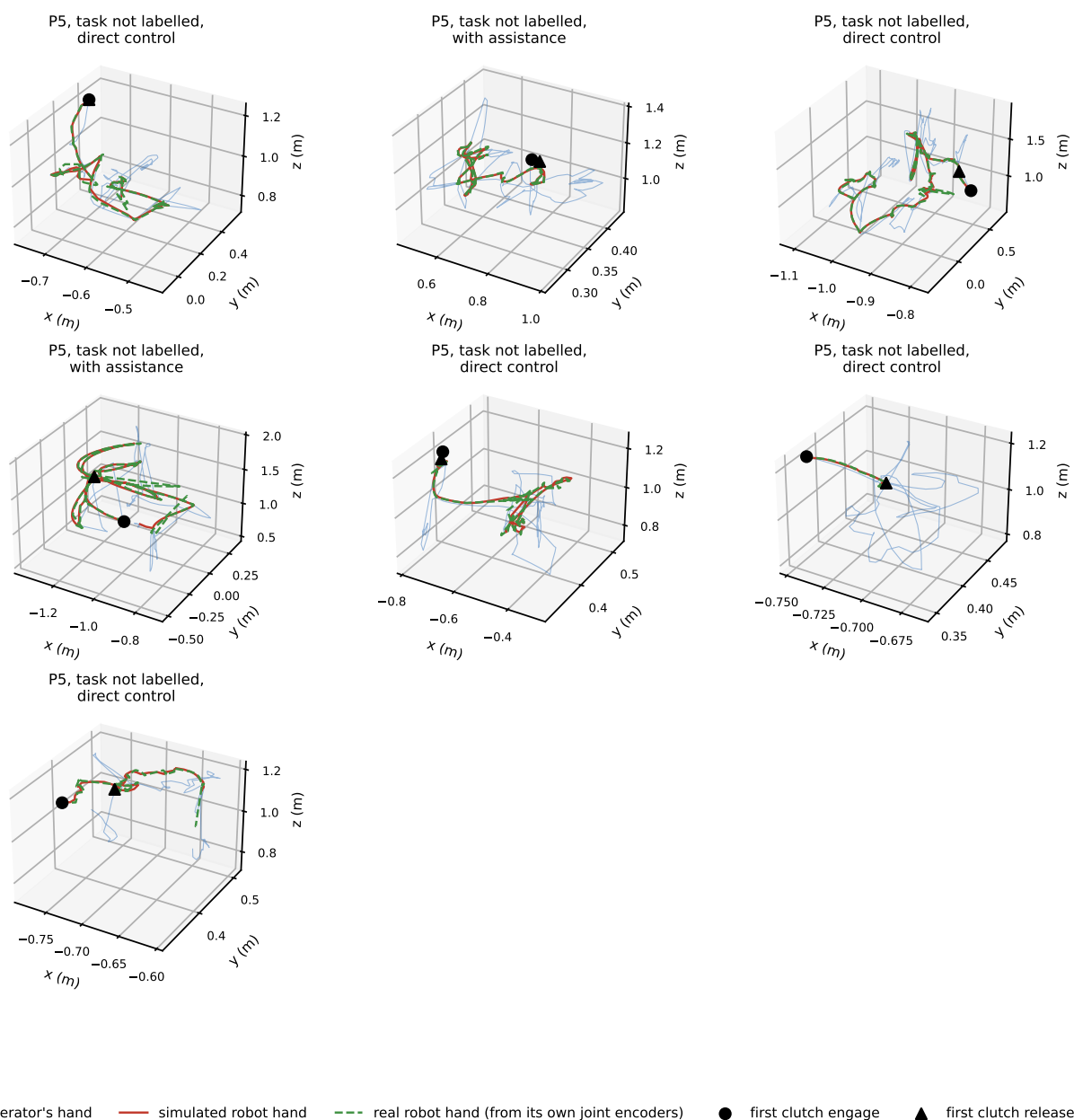
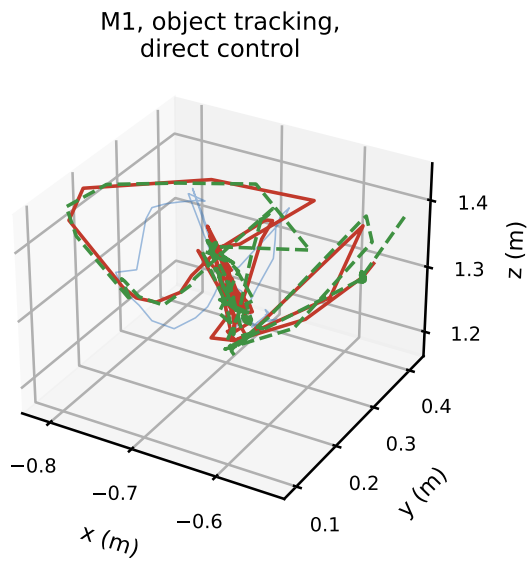
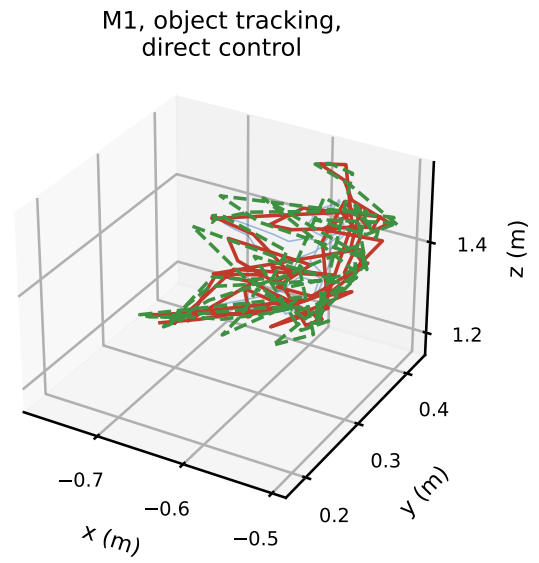
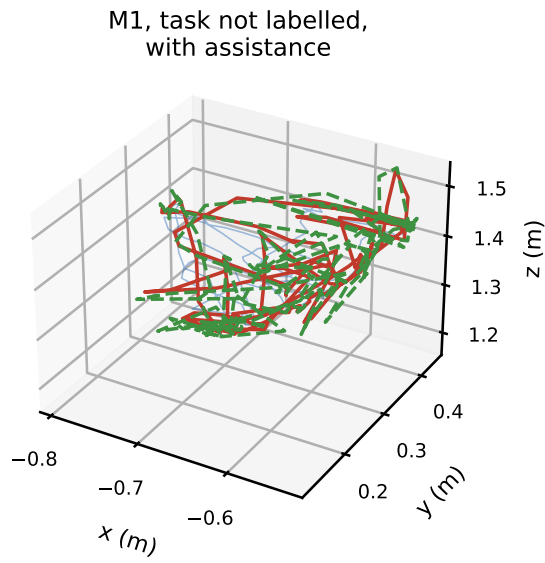


Figure O.22: Every recorded session of P5: the operator's hand (faint, smoothed), the simulated robot's hand and the real robot's hand from its own encoders.



— operator's hand — simulated robot hand - - - real robot hand (from its own joint encoders)

Figure O.23: Every recorded session of M1: the operator's hand (faint, smoothed), the simulated robot's hand and the real robot's hand from its own encoders.

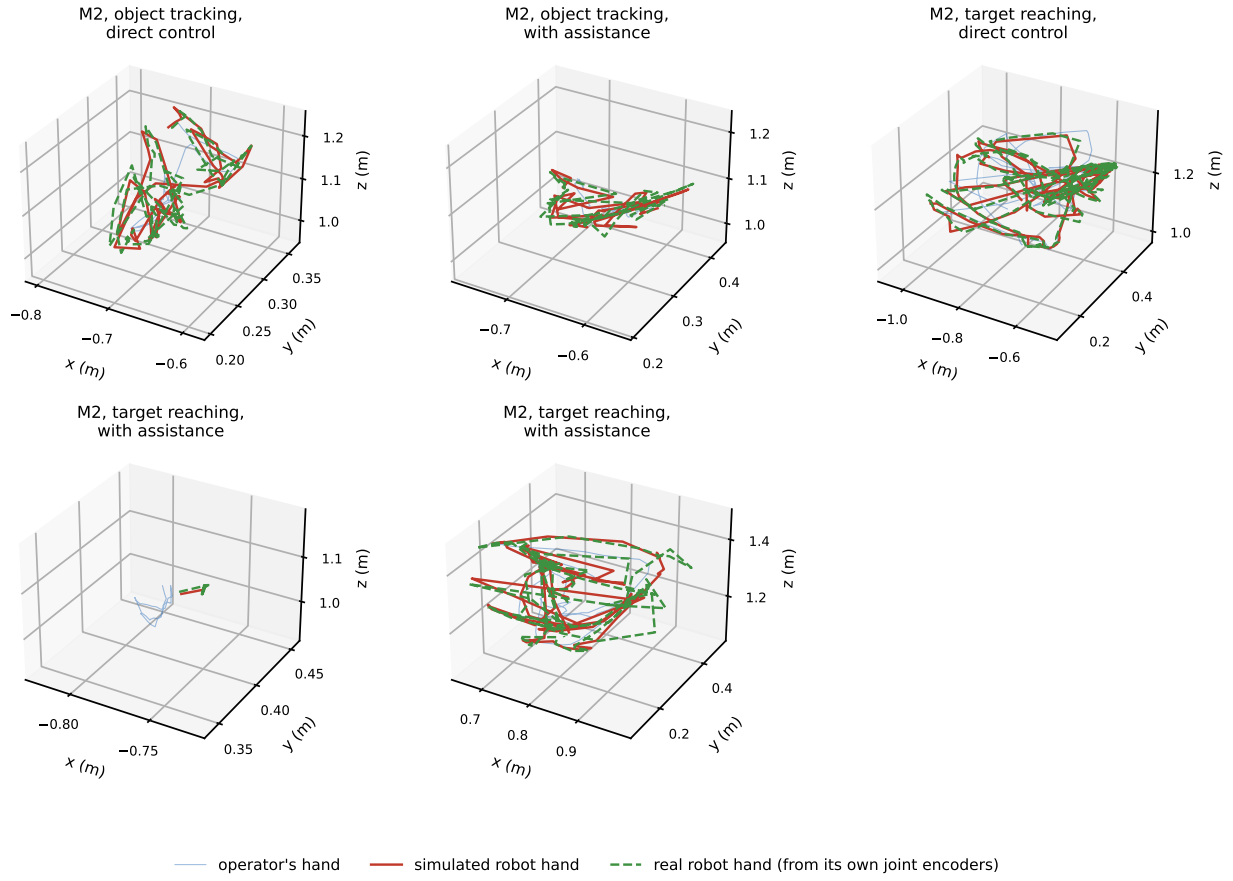
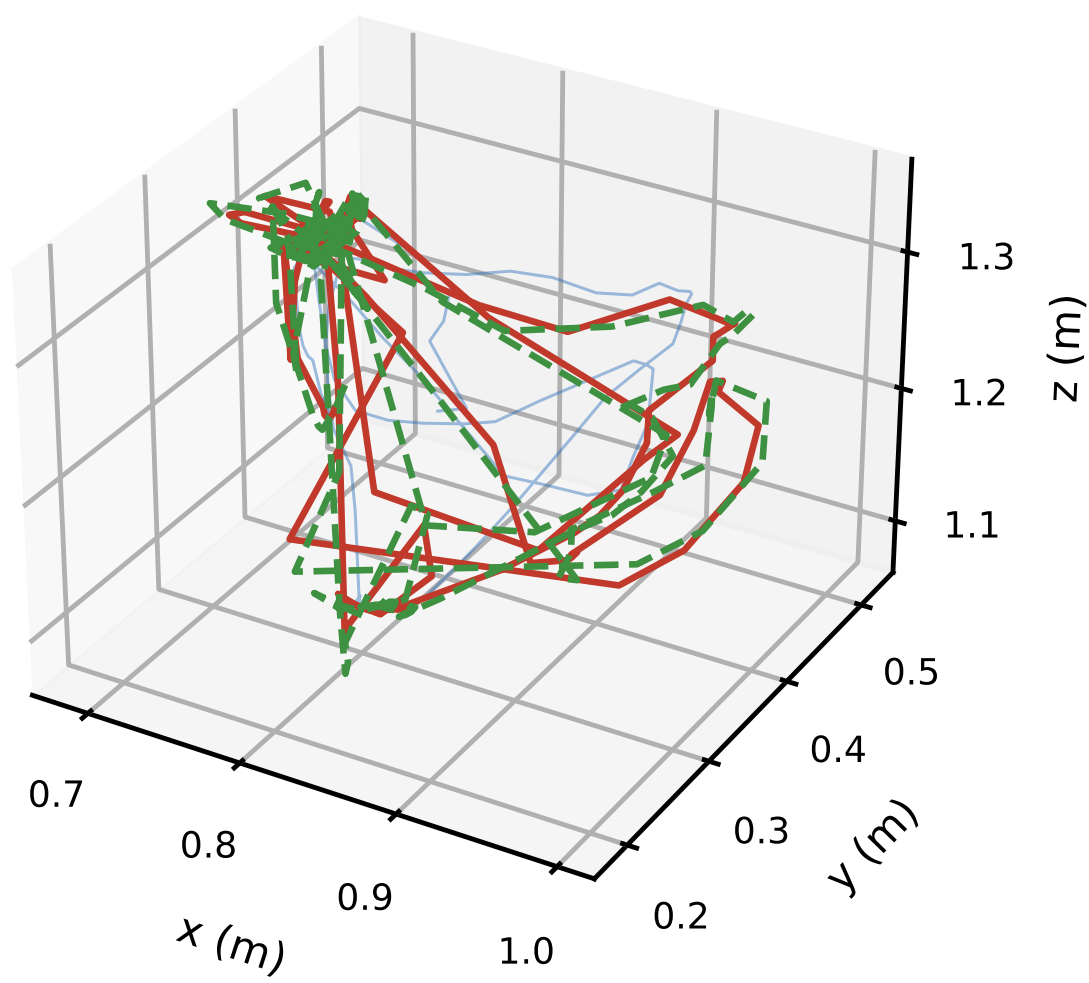


Figure O.24: Every recorded session of M2: the operator's hand (faint, smoothed), the simulated robot's hand and the real robot's hand from its own encoders.

M3, target reaching, direct control



— simulated robot hand - - - real robot hand (from

Figure O.25: Every recorded session of M3: the operator's hand (faint, smoothed), the simulated robot's hand and the real robot's hand from its own encoders.

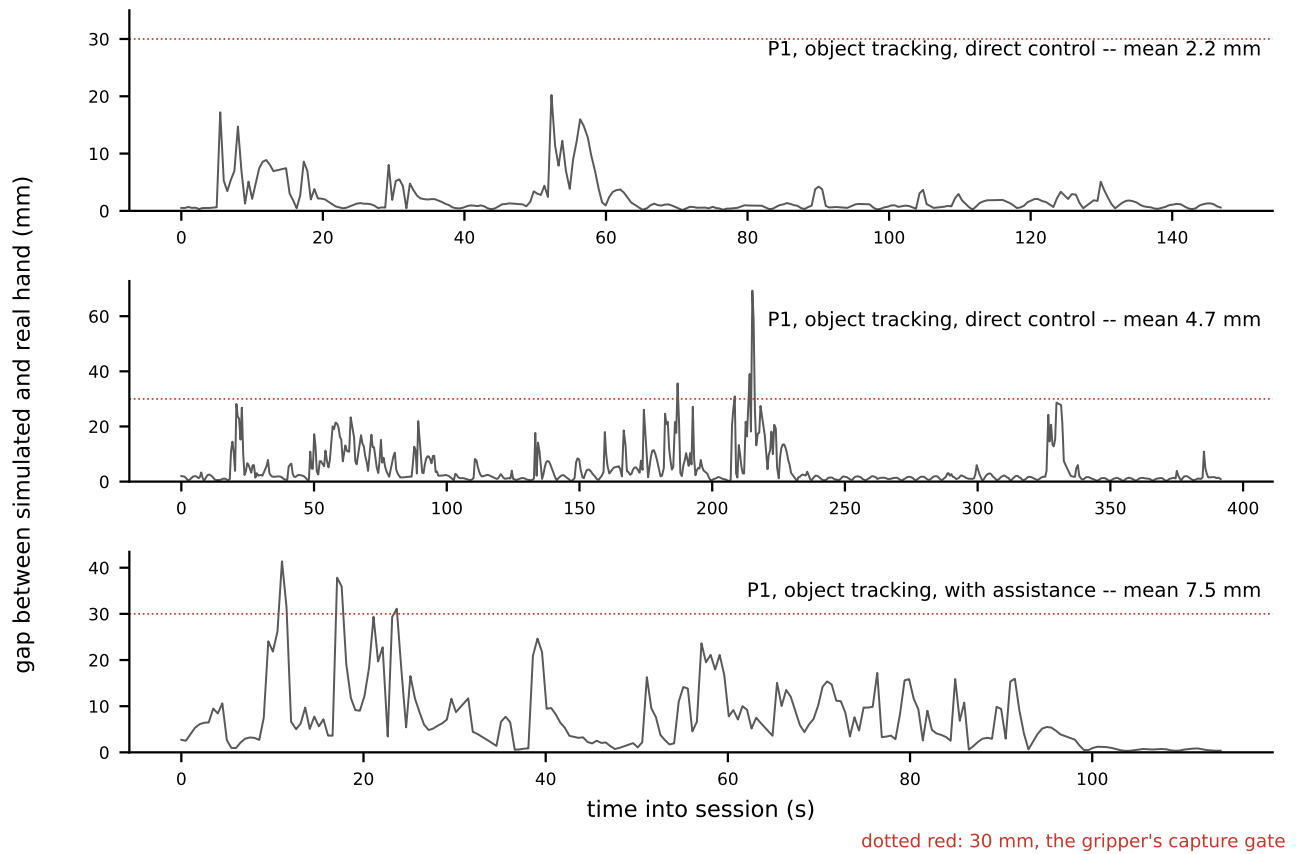
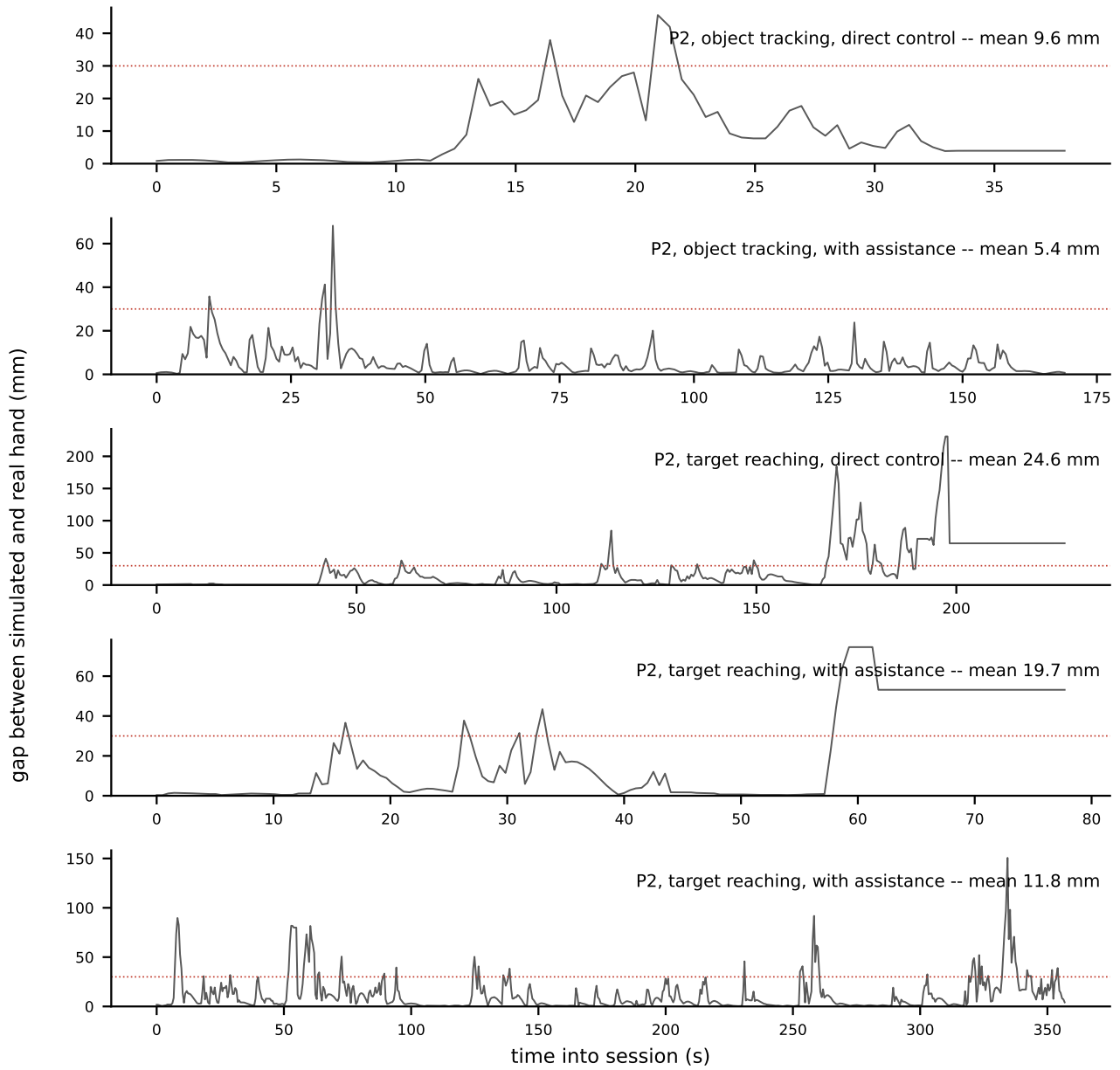


Figure O.26: The distance between the simulated and the real robot hand through each of P1's sessions, with the session mean in each label and the 30 mm grasp gate drawn. A flat tail at the end of a session is the real-arm columns freezing while the end-of-session emergency stop is held, not a standing offset.



dotted red: 30 mm, the gripper's capture gate

Figure O.27: The distance between the simulated and the real robot hand through each of P2's sessions, with the session mean in each label and the 30 mm grasp gate drawn. A flat tail at the end of a session is the real-arm columns freezing while the end-of-session emergency stop is held, not a standing offset.

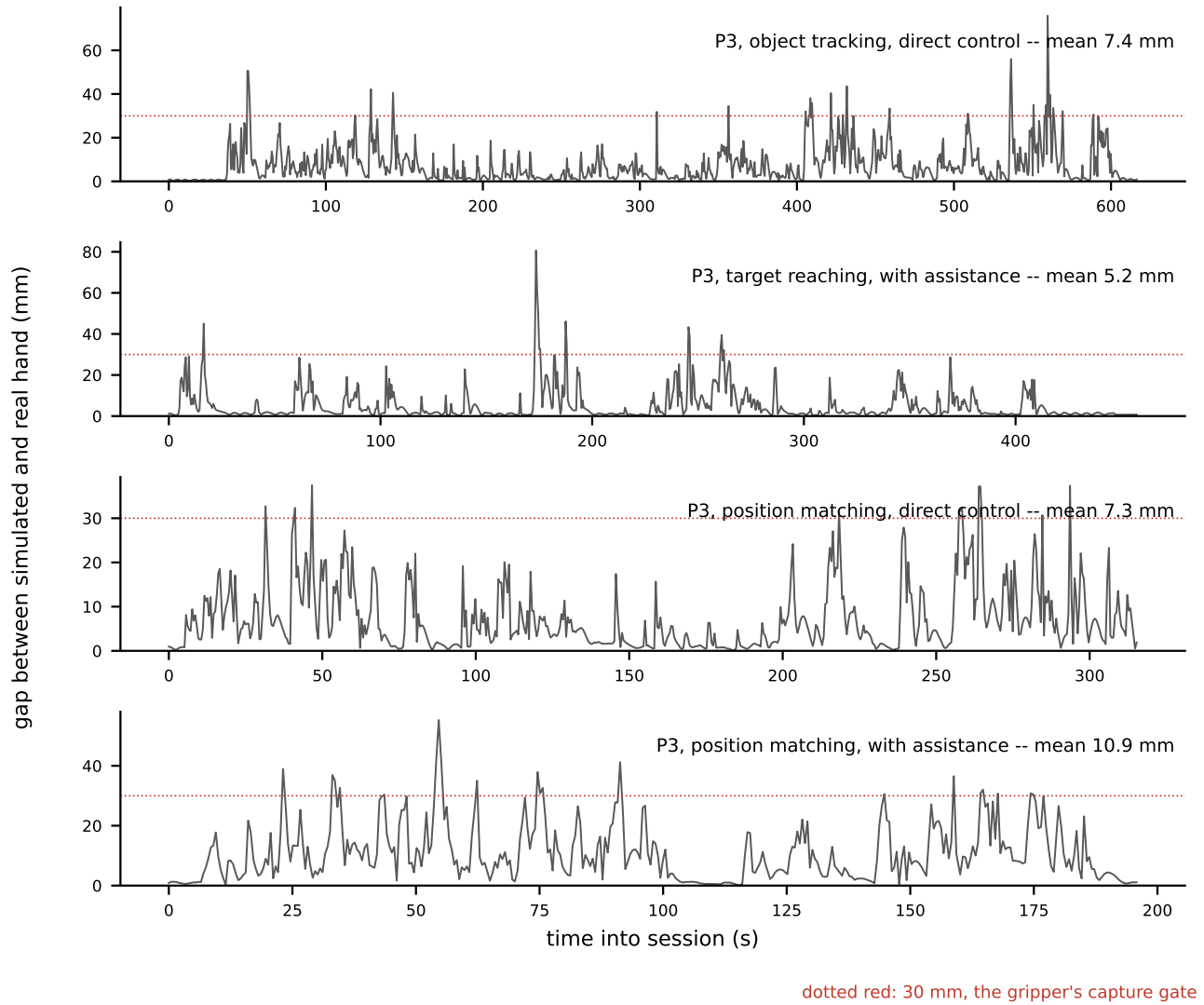
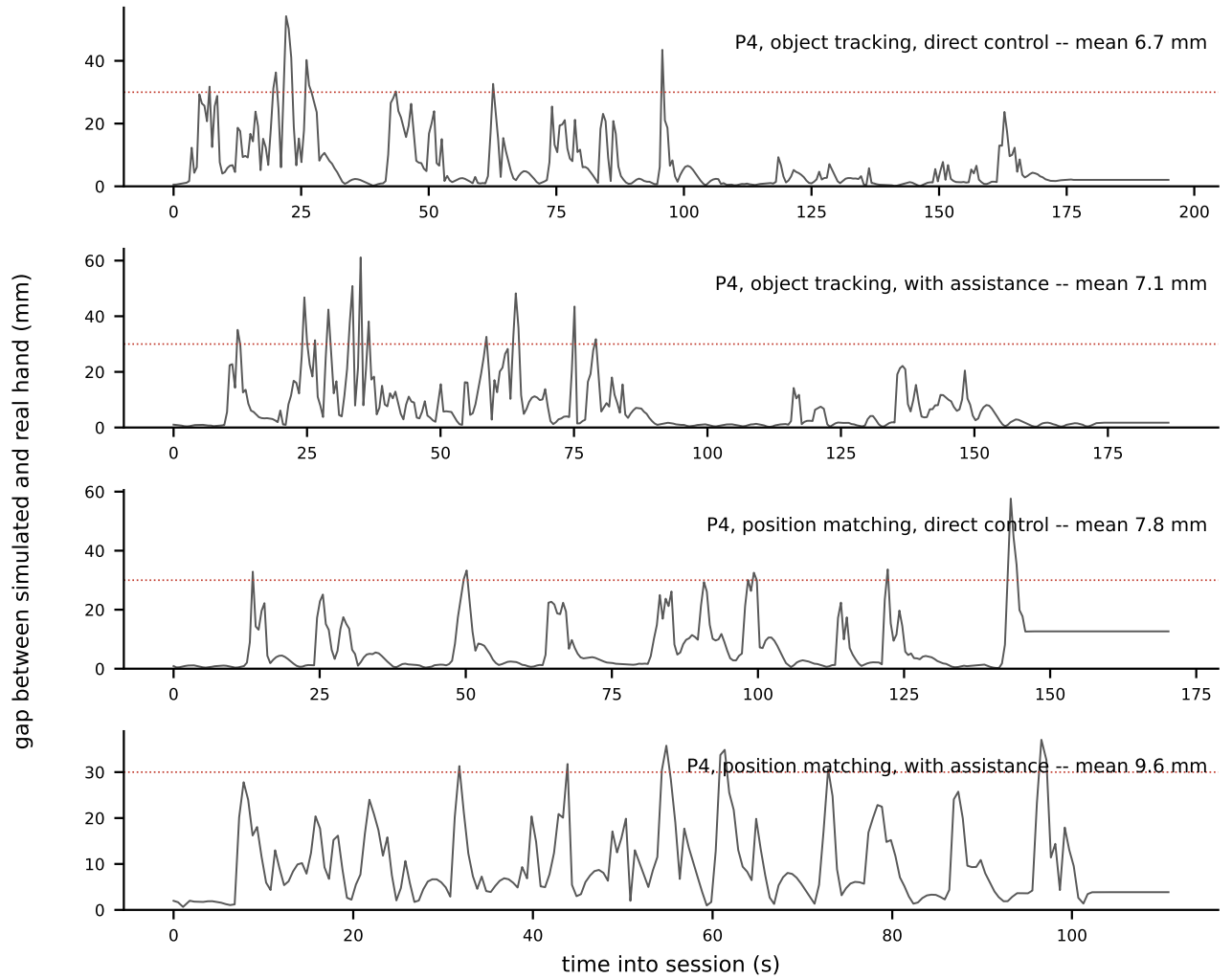
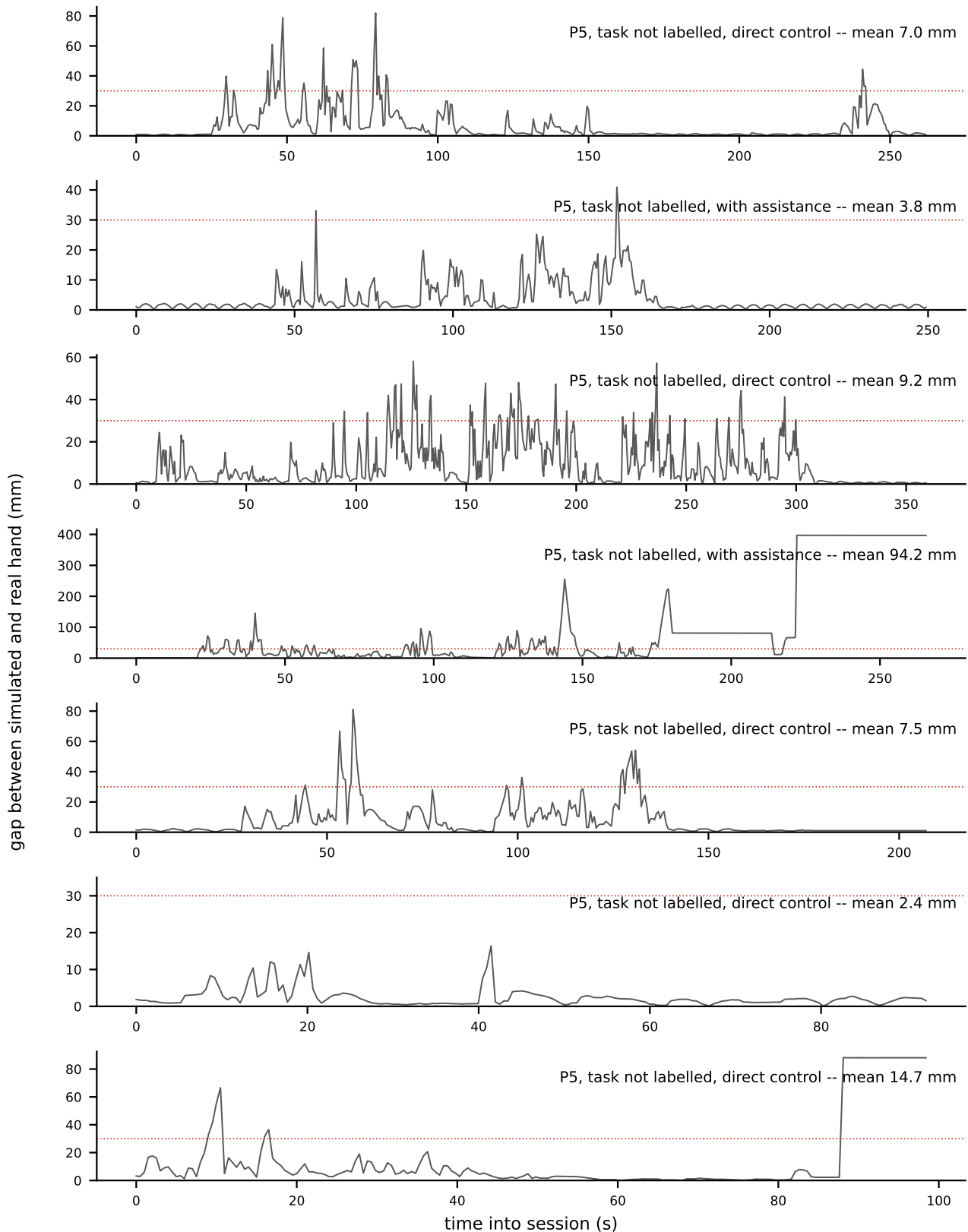


Figure O.28: The distance between the simulated and the real robot hand through each of P3's sessions, with the session mean in each label and the 30 mm grasp gate drawn. A flat tail at the end of a session is the real-arm columns freezing while the end-of-session emergency stop is held, not a standing offset.



dotted red: 30 mm, the gripper's capture gate

Figure O.29: The distance between the simulated and the real robot hand through each of P4's sessions, with the session mean in each label and the 30 mm grasp gate drawn. A flat tail at the end of a session is the real-arm columns freezing while the end-of-session emergency stop is held, not a standing offset.



dotted red: 30 mm, the gripper's capture gate

Figure O.30: The distance between the simulated and the real robot hand through each of P5's sessions, with the session mean in each label and the 30 mm grasp gate drawn. A flat tail at the end of a session is the real-arm columns freezing while the end-of-session emergency stop is held, not a standing offset.

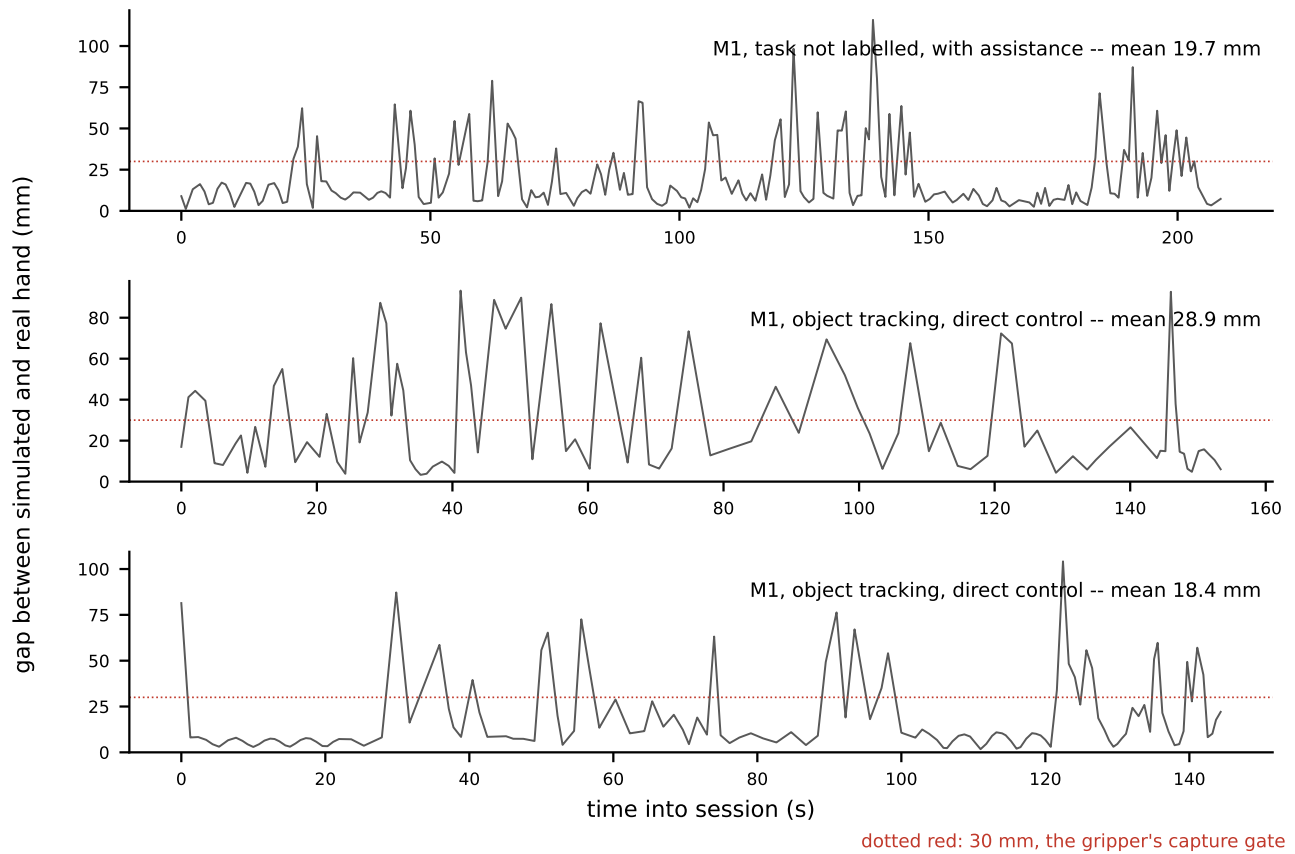
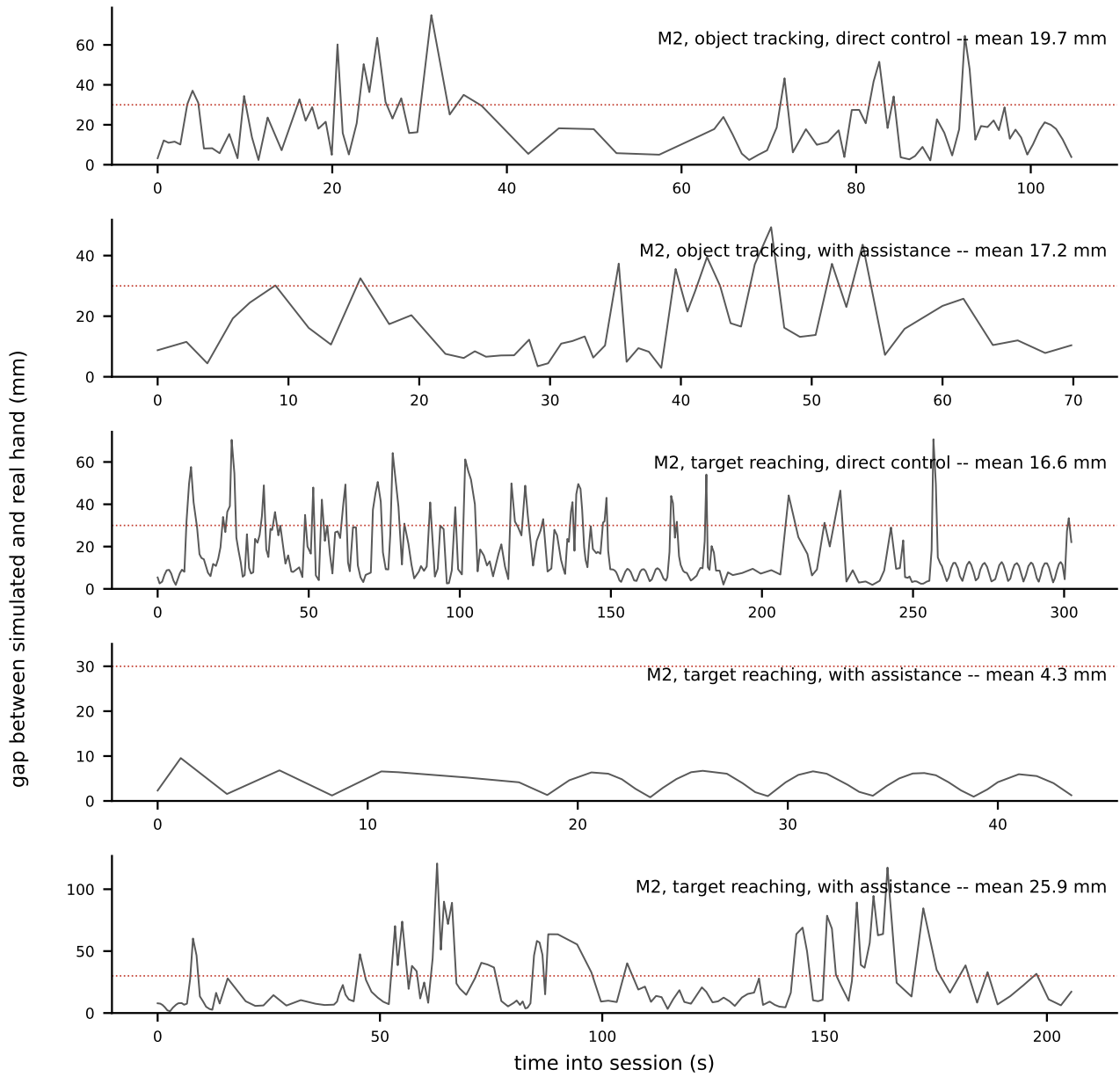


Figure O.31: The distance between the simulated and the real robot hand through each of M1's sessions, with the session mean in each label and the 30 mm grasp gate drawn. A flat tail at the end of a session is the real-arm columns freezing while the end-of-session emergency stop is held, not a standing offset.



dotted red: 30 mm, the gripper's capture gate

Figure O.32: The distance between the simulated and the real robot hand through each of M2's sessions, with the session mean in each label and the 30 mm grasp gate drawn. A flat tail at the end of a session is the real-arm columns freezing while the end-of-session emergency stop is held, not a standing offset.

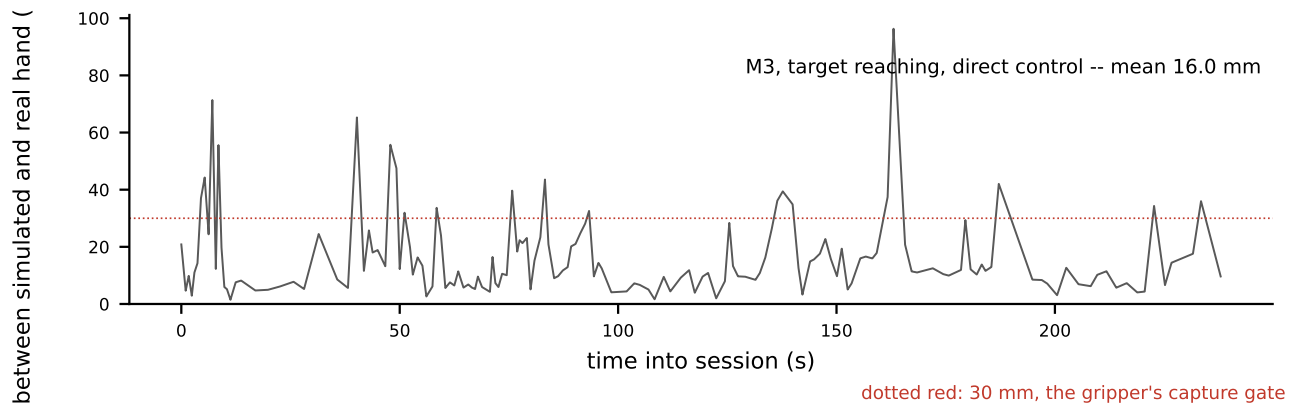


Figure O.33: The distance between the simulated and the real robot hand through each of M3's sessions, with the session mean in each label and the 30 mm grasp gate drawn. A flat tail at the end of a session is the real-arm columns freezing while the end-of-session emergency stop is held, not a standing offset.

Top and side views of every session, and each session's panel as its own file, are in `thesis_v3/figures/gallery/trajectories`

O.5 The cameras

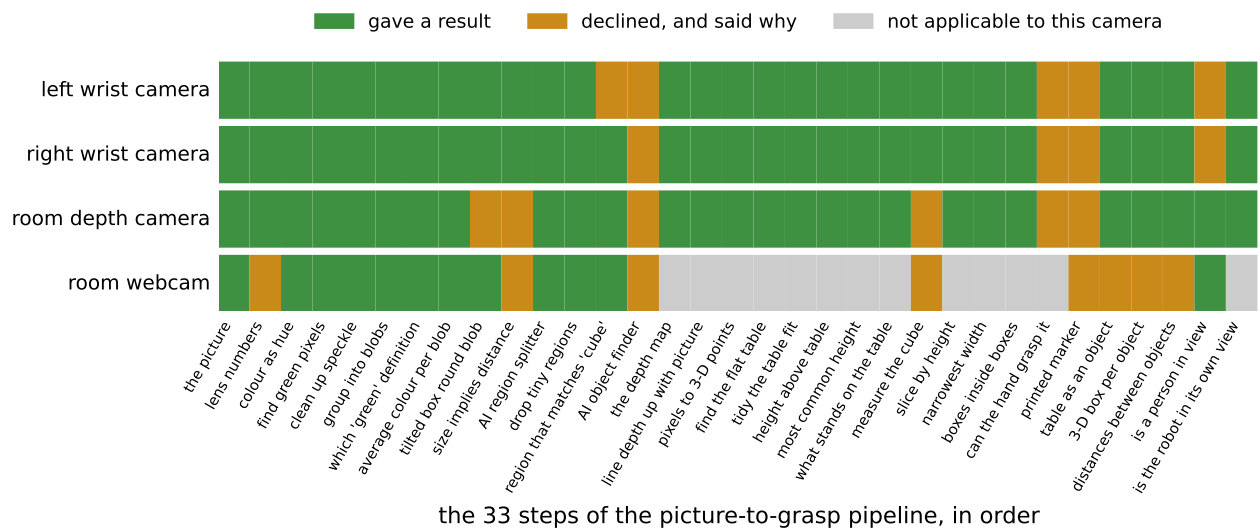


Figure O.34: Every stage of the picture-to-grasp pipeline on the four real cameras: 28, 29, 27 and 12 of 33 stages gave a result; every other outcome is a named refusal or a by-design absence, and none crashed.

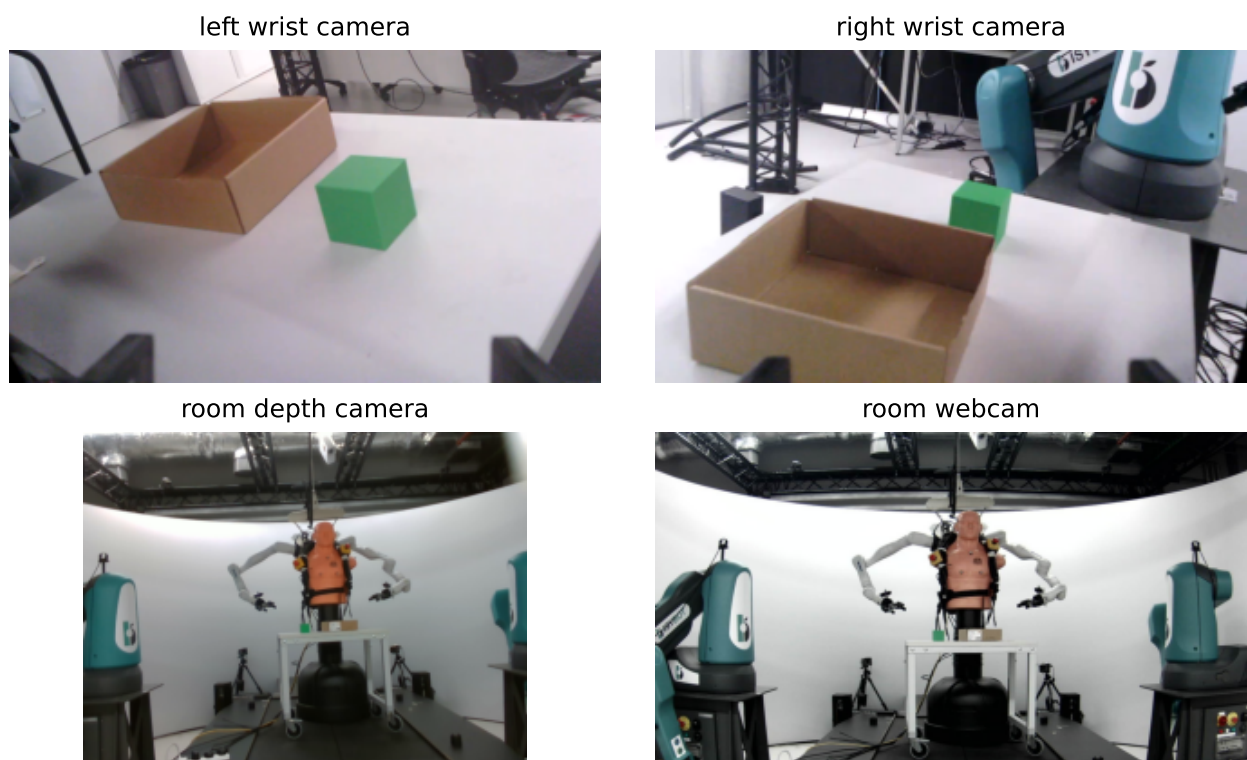


Figure O.35: The four cameras' views of the same moment: two ride on the hands and see the table; two stand across the room and see the whole rig.

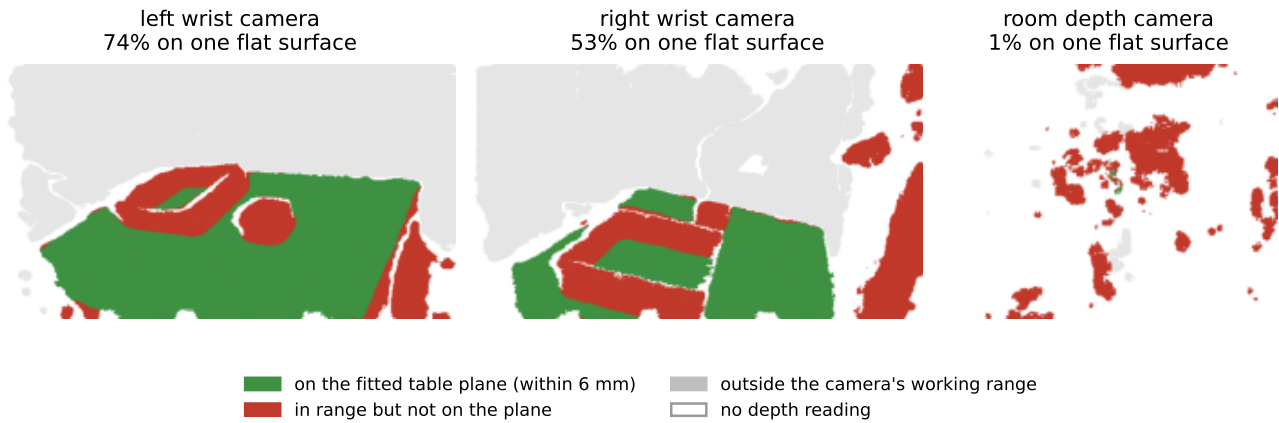


Figure O.36: The support plane the pipeline fitted, recomputed over each camera's own depth array with the stored plane and tolerance: 73.95 % of the left wrist camera's points lie on it, 52.79 % of the right's, and under 1 % of the room depth camera's (the pipeline's own log says 3.75 % for that camera; both mean no usable surface).

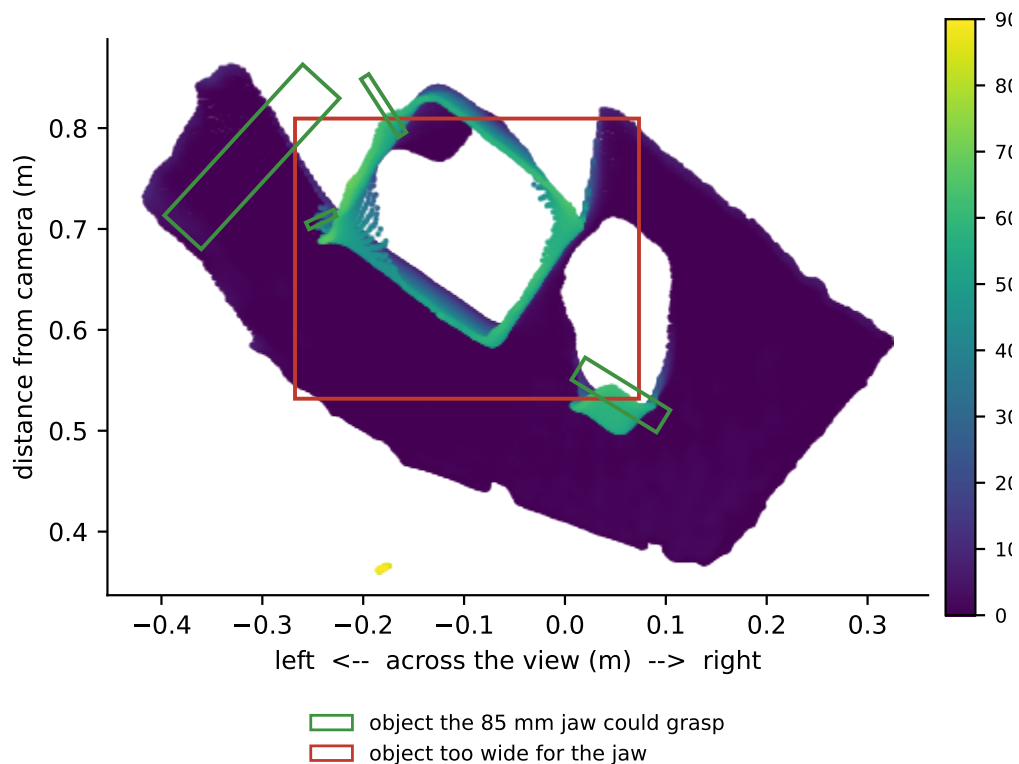


Figure O.37: The left wrist camera's point cloud from above, coloured by height, with the pipeline's oriented boxes: the box (red, too wide for the 85 mm jaw) and the cube (green, graspable) plus three small fragments; nearest object 0.539 m, furthest 0.843 m. This is the map the planner plans against.

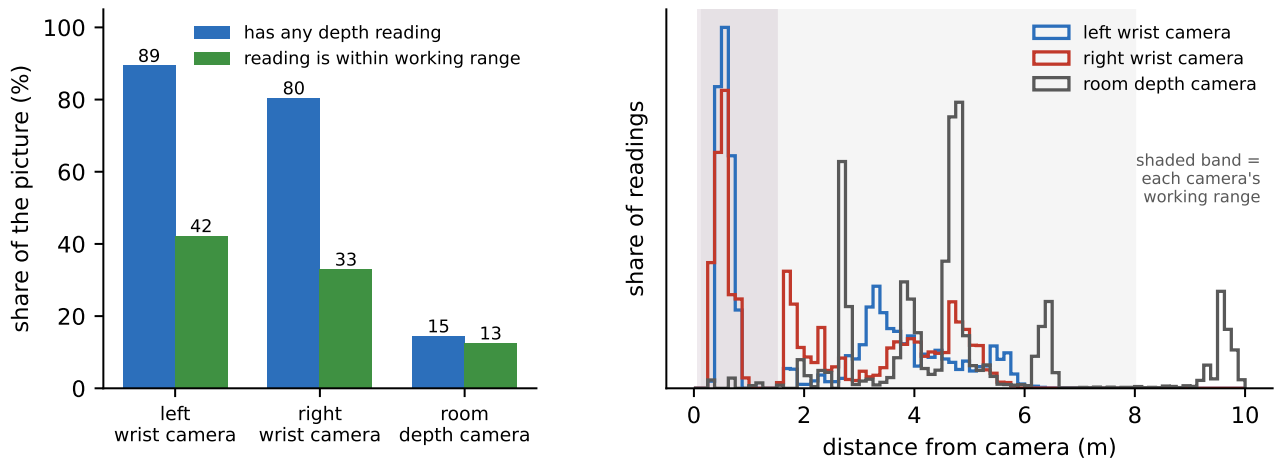


Figure O.38: Share of each picture with any depth reading (89 %, 80 % and 15 %) and within the working range (42 %, 33 % and 13 %); the room depth camera's returns lie mostly 2 to 6 m away – a different instrument from the wrist cameras, not merely a distant one.

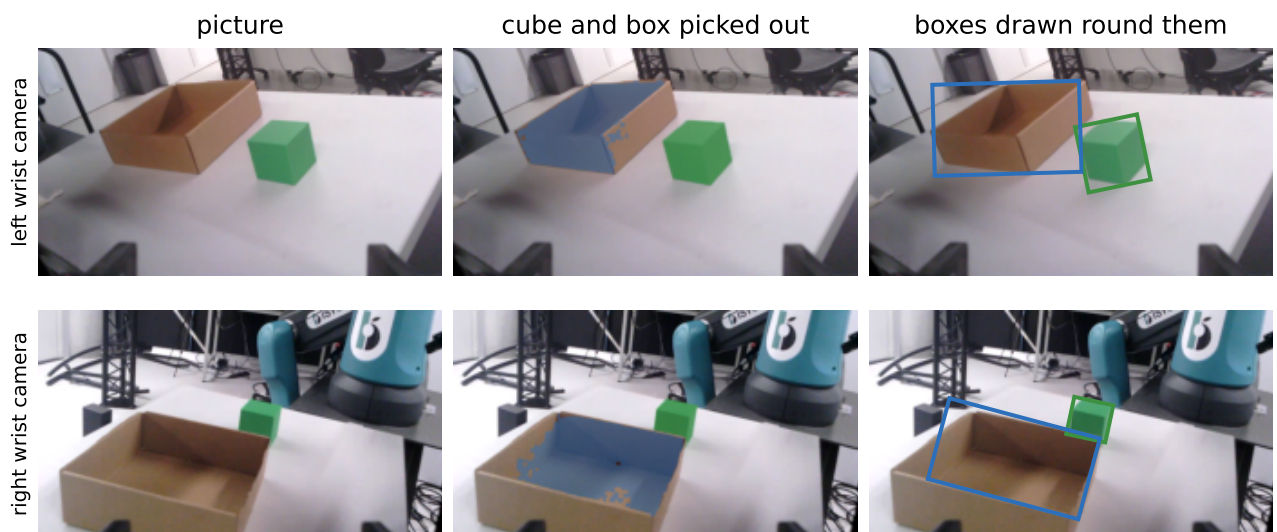


Figure O.39: The same scene from both hands: each wrist camera isolates the cube and the box.

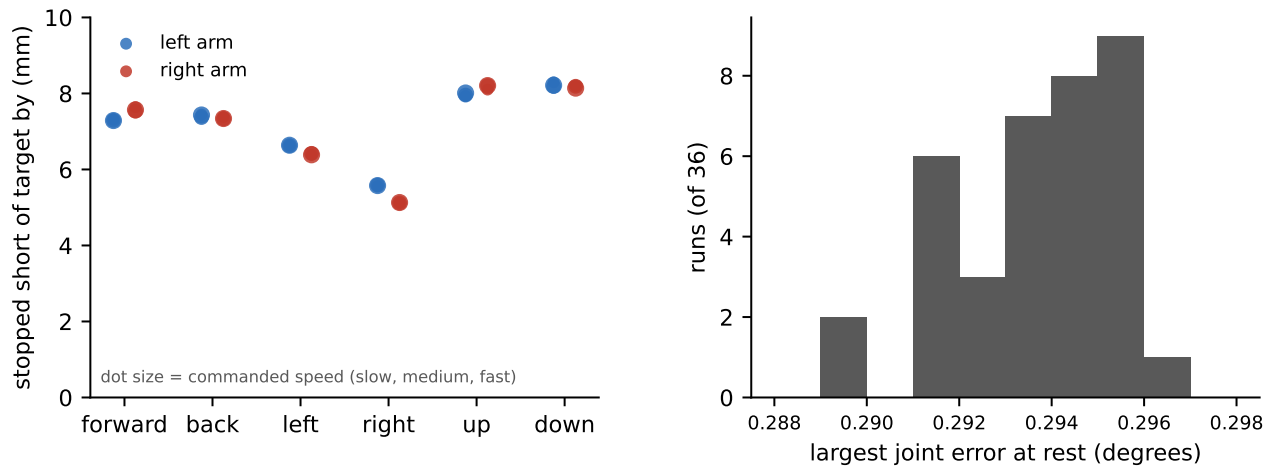


Figure O.40: The thirty-six calibration moves: every 100 mm move stopped 5.1 to 8.3 mm short, the same for both arms and all three speeds, with the worst joint's rest error in a 0.289 to 0.296° band (the pooled per-joint fit in [Section E.3](#) gives 0.305°).

O.6 Measured baselines, in pictures

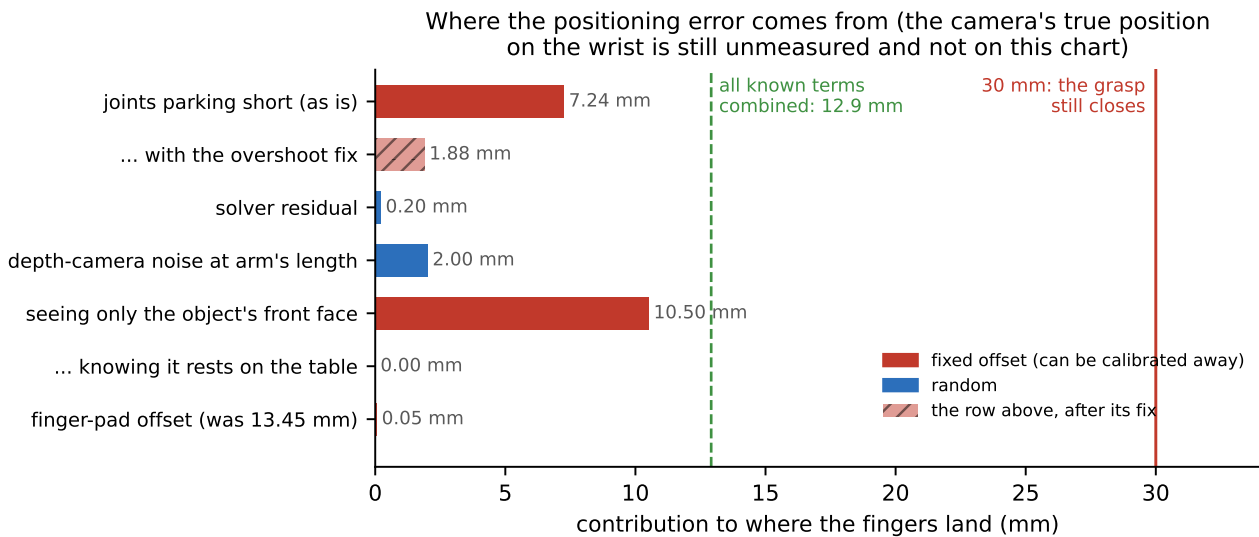


Figure O.41: Each known source of finger-landing error as a bar, fixed offsets distinguished from random terms and each fix shown against the row it fixes; combined 12.9 mm against the 30 mm grasp line. The camera's true position on the wrist is unmeasured and not on this chart.

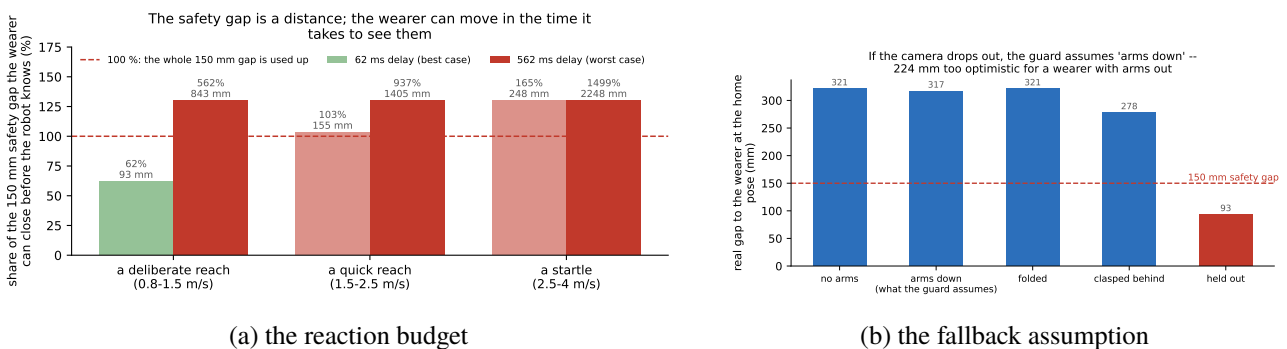


Figure O.42: Left: the share of the 150 mm safety gap a moving wearer closes before the robot knows, three limb-motion classes at the best (62 ms) and worst (562 ms) measured delay. Right: what the guard assumes when the wearer camera drops out (arms down, 317 mm) against a wearer with arms held out (93 mm).

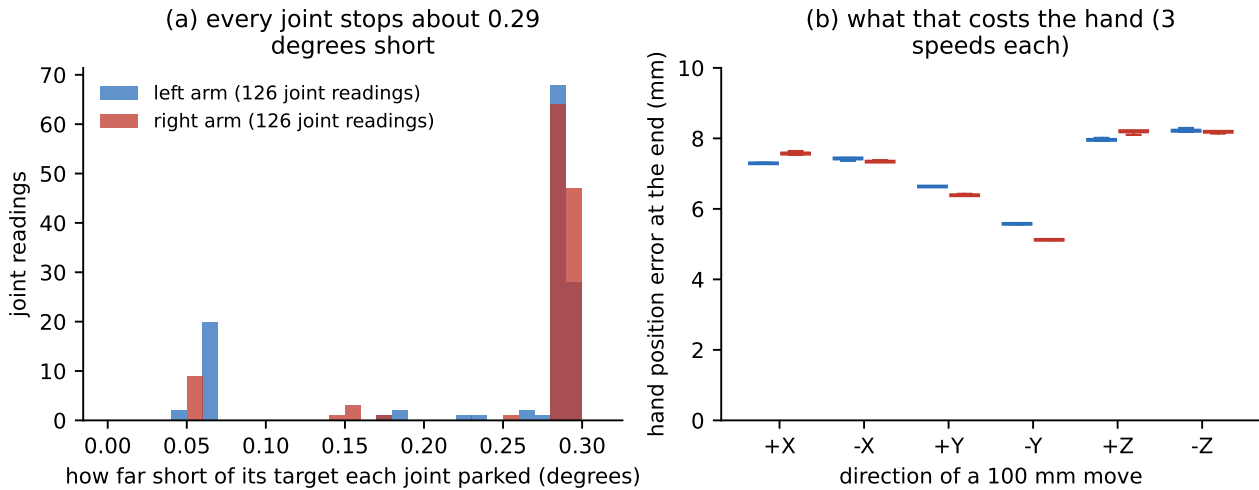


Figure O.43: Thirty-six real moves of 100 mm: every joint stops about 0.29° short (a), which costs the hand 5 to 8 mm by direction, the same at all three speeds (b).

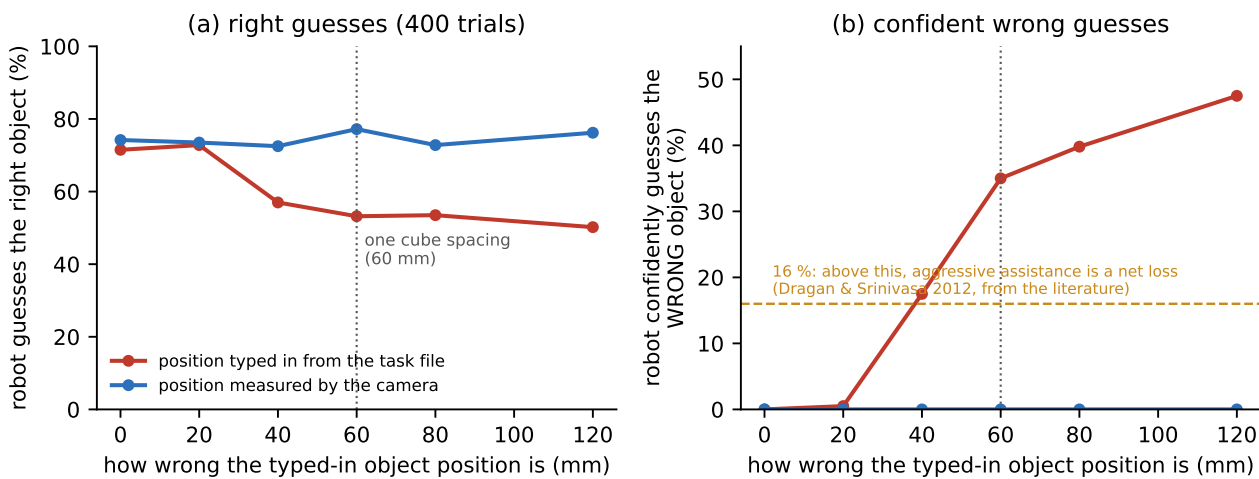


Figure O.44: Four hundred trials per point: from a typed-in object position the estimator's confident wrong guesses climb past the published 16 % break-even at 40 mm of error and reach 35 % at one cube spacing; from the camera's measured position they stay at zero.

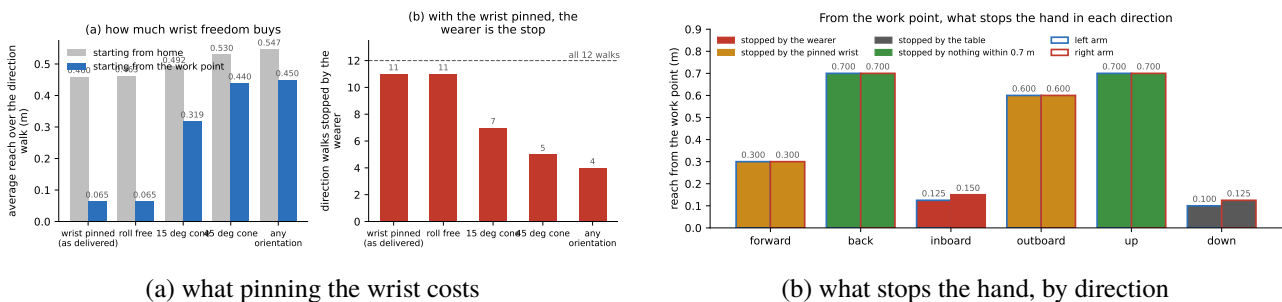


Figure O.45: Left: mean reach under each wrist policy from home and from the work point. Right: from the work point, what stops the hand in each of twelve directions – the wearer in eleven of twelve when the wrist is pinned.

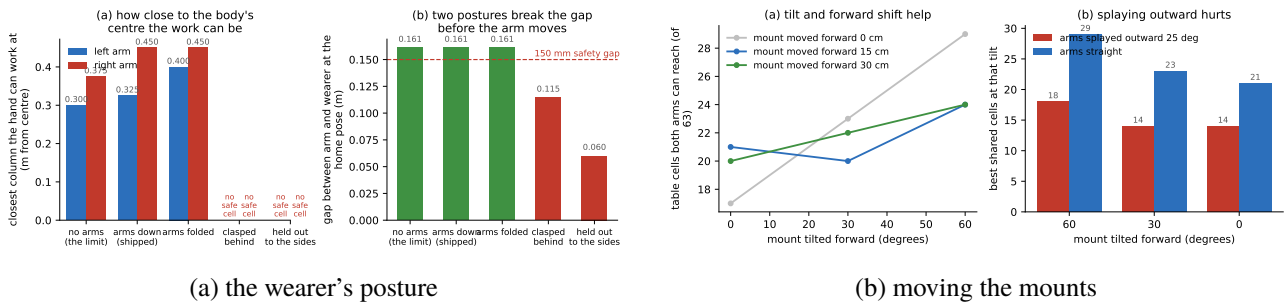


Figure O.46: Left: how the wearer's arm posture changes what the arms can reach, with the 150 mm gap drawn. Right: how moving the mounts changes what both arms can reach – as built the two share three of 63 cells; the buildable best is 22.

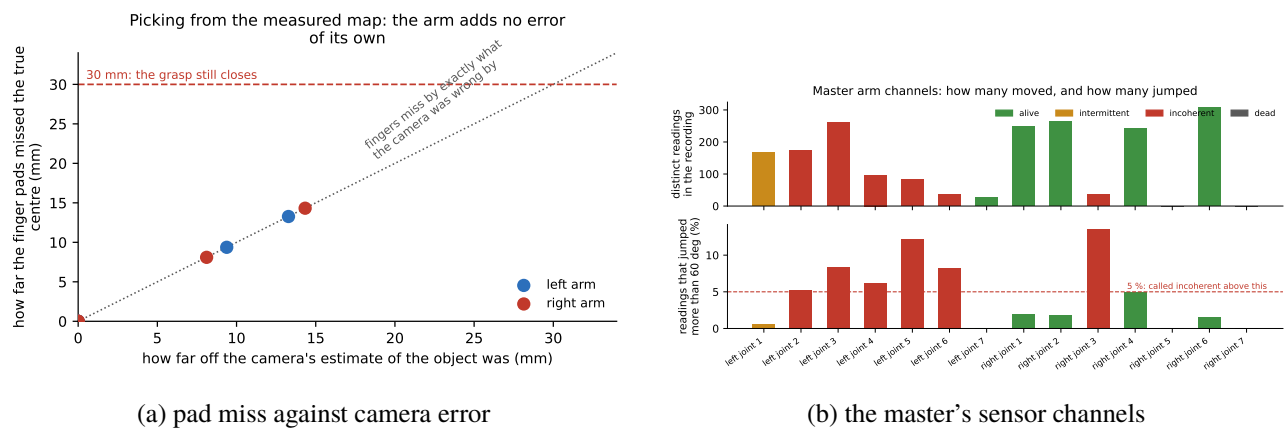


Figure O.47: Left: whatever the camera believes, the arm delivers the fingers there – the pad miss equals the belief error, so the residual is perception, not kinematics. Right: which of the fourteen master-arm sensors read cleanly, the verdict behind [Section H.2](#).

O.7 How the project was done

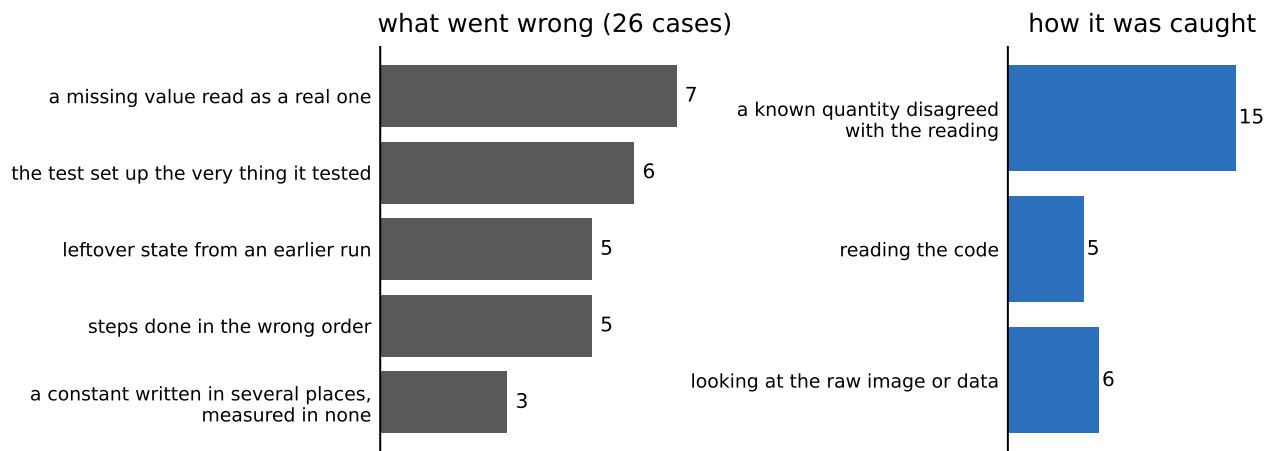


Figure O.48: What went wrong in the twenty-six cases where a measurement was wrong and the system was not, and how each was caught: fifteen by an independently known quantity disagreeing with the reading, five by reading the code, six by looking at the raw image or data.

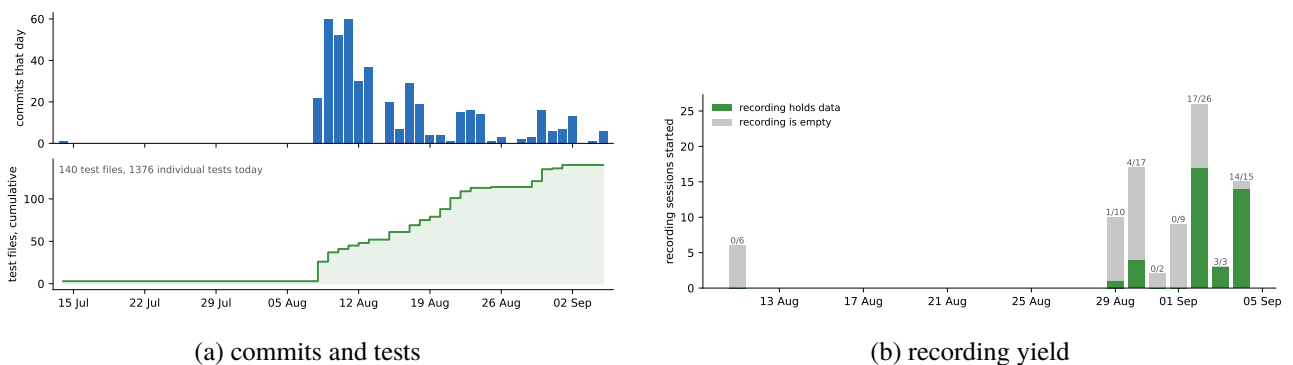


Figure O.49: Left: 449 commits over 27 active days and a test suite that grew with the code to 140 files and 1376 tests. Right: recording sessions started per day and how many held data – unreliable until 2 September (0/6, 1/10, 4/17, 0/2, 0/9), then 17/26, 3/3 and 14/15.

Appendix P

Ethical analysis

Section 2.5 states the approval and the substance. This appendix is the analysis behind it, organised by the department's six domains and the Engineering Council's four pillars (honesty and integrity; respect for life, law, the environment and the public good; accuracy and rigour; leadership and communication), and it names every stakeholder. Source: docs/research/03_ethics_and_safety.md, the document the approval application was adapted from.

P.1 Stakeholders

The **wearer**: bears 17 kg and the physical risk of motion they do not command; the party with the least control and the most at stake. The **operator**: bears the cognitive load and the responsibility for motion they cannot feel. The **experimenter**: calls breaks and stops, and must not let the schedule pressure either participant. **Fellow students and the laboratory**: share the arms, the room and the reputation of the group. **The College**: holds the approval, the data and the liability. **The wider public**: the eventual users of assistive limbs, and the people who will stand next to them. **The environment**: the hardware's energy and material cost.

P.2 A. Scientific integrity – is it true?

Every quantitative claim in the main body carries its evidence class (implemented, verified in simulation, measured on hardware, or measured in the pilot), and the report states that nothing below the pilot was validated on a loaded physical arm. Twenty-six occasions on which the measuring tool, not the system, was wrong are catalogued in [Chapter G](#) and summarised in [Section 4.3](#); each is reported as a defect found, not hidden. Numbers that disagreed with their source when this report was checked – the e-stop count, the estimator's wrong-guess rate at one cube pitch, the pinned-wrist reach ratio – were corrected and the correction is recorded in the repository log. The analysis for the full study was pre-registered and run on 20 000 constructed data sets before any participant data existed ([Chapter K](#)), so its false-positive rate is known (5.0 % against a nominal 5 %).

P.3 B. Scientific collegiality – is it fair?

The platform is shared laboratory equipment. Sessions were booked, the two-stack rule (never two control processes on one robot) protected other users' measurements as much as this project's, and every session leaves a manifest and a log. The report anonymises participants and credits prior work by citation; where this project

reproduces a published finding it says so, and where it contradicts one – the robot’s path being longer under assistance – it gives the measured mechanism rather than an appeal.

P.4 C. Protection of human subjects

Approval and consent. SETREC approval 7111986. Participants received an information sheet at least twenty-four hours in advance stating what the device is, that it moves near them, its weight if worn, the session length, the right to withdraw, what is recorded and how to withdraw data afterwards. Consent is taken separately for participation, for video (optional), for retention of anonymised data beyond the project, and for its use in publication. Withdrawal is possible at any time without reason, and data can be withdrawn up to publication through the participant code.

The two-person rule. A wearer and an operator who arrive together will not decline in front of each other, and the wearer consents to strictly more. Consent is therefore taken in separate rooms, on separate forms, and neither is told whether the other consented until both have. The wearer holds their own emergency stop on the watchdog path and presses it twice, watching the arms halt, before any recorded trial. No justification is ever requested for a stop. A session ended by a wearer is reported as data, not replaced with a fresh pair to make up the numbers, because that would select for tolerant wearers.

Risk register (the pilot’s operator-only sessions, and the planned worn study). Robot contacts the participant: collision-aware solving, a 150 mm clearance floor enforced from the robot’s true pose, the autonomy never permitted to move toward the wearer, the wearer’s own stop. Unexpected or fast motion: a participant speed cap stricter than the developer’s, re-asserted every two seconds so a stray parameter change cannot raise it; motion must be explicitly armed. Stop out of reach: the software refuses to arm until the reach check is confirmed. A sensor fails mid-trial: the trial is aborted and marked invalid, never silently continued, because a frozen command looks like a slow participant in the data. Physical strain from the master: blocks of at most eight minutes, two-minute rests, a session of at most forty-five minutes, a Borg fatigue probe between blocks. Strain from wearing the rig: twenty minutes maximum, two people to don and doff, exclusion for any back, neck or shoulder problem. Psychological risk: participants are told the *system* is under evaluation and that cancelling an autonomous action is normal use.

What is not yet safe, stated rather than glossed. The arms’ proximal links interfere with the wearer’s own upper arms in the collision model; this is set by the mount position and no planner can avoid it. Worn operation therefore was not sought for approval; the pilot ran with a mannequin wearer. The clearance check that runs on the physical arm samples link origins, not link geometry ([Chapter N](#)) – the one item this report classes as a safety defect rather than an incompleteness.

P.5 D. Animal welfare

No animals are involved.

P.6 E. Institutional integrity – is it wise?

Data handling: participant codes only; no name, email, date of birth, address or telephone number appears in any file, enforced by `write_manifest()` raising if one is passed; demographics as age bands. The code-to-person mapping exists only to allow withdrawal, is held on encrypted College storage outside the repository, and is

destroyed at publication. Video, if consented to, frames the rig and hands, not the face. Anonymised data are retained under institutional policy; the repository is not a data store. Legal basis: public-task processing under UK GDPR and the Data Protection Act 2018, with consent for the optional elements. No external sponsor or conflict of interest exists; the arms are laboratory property and the findings belong to the College.

P.7 F. Social responsibility

The device this work points toward is an assistive limb worn by one person and, in this configuration, directed by another. The result that assistance shortens the operator's task while lengthening the robot's path, with the wearer's trust and agency unmeasured, is exactly the shape of harm that professional ethics exists to notice: a benefit to the party who chooses and a cost to the party who does not. That is why H3 and H4 – that the wearer's trust and agency fall under assistance – are the hypotheses the full study exists to test, and why the report refuses to call the pilot evidence about the wearer. Sustainability and cost: the platform re-uses existing laboratory arms and a consumer headset; the 4 GiB video-memory limit was met by choosing lighter methods rather than buying hardware; the software is open in the laboratory repository so the work can be continued rather than repeated.

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